HYDRAULICS &
PNEUMATICS Workshops
User's Guide

Famic Technologies Inc.

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Hydraulics and Pneumatics Workshops User's Guide for Automation Studio™.

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Famic Technologies Inc.
9999, blvd. Cavendish, suite #350, St-Laurent, QC, Canada, H4M 2X5
Tel.: 514-748-8050, Fax: 514-748-7169, Url: www.automationstudio.com

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1 Introduction

Automation Studio™ Hydraulics has been specifically tailored for hydraulic system engineering purposes. It is intuitive, simple, and provides all the required functions of the most demanding users. Automation Studio™ Hydraulic provides tools to implement and document your schematics, to generate bills of material, and to animate all applications which include industrial systems.

As for Hydraulics, Automation Studio™ Pneumatics is a complete standalone design environment specially designed for pneumatic systems engineering purposes.

The *Hydraulics and Pneumatics Workshops User's Guide* contains the necessary information to use Automation Studio™ fluid power workshops efficiently. This guide represents topics directly related to the creation and simulation of diagrams using the Hydraulics and Pneumatics workshops. All standard functions, common to all workshops, are described in *Automation Studio™ User's Guide*.

This guide is divided into the following sections:

Chapter *2 Creation of a Fluid Project*: In order to properly design a system, one should learn the standards used for symbols and diagrams, and the ways to modify them in Automation Studio™.

Chapter *3 Systems Simulation*: In order to properly test the systems, one should have the appropriate knowledge on simulated and simplified events, such as charge loss or fluid thermal variation.

Chapter *4 Component Description*: This section explains the use of components in Automation Studio™ and does not pretend to explain real component behaviour.

Chapter *5 Component Builders*: This section explains how to design and configure complex components, such as pumps, directional valves, sequencers, reservoirs or gear boxes.

Chapter *6 Additional Information*: Many visual options (line's color for low and high pressure, etc.) and physical characteristics (high pressure limit, ambient temperature, etc.) can be configured in Automation Studio™. This section provides the information to accomplish this.

2 Creation of a Fluid Project

It is possible to create hydraulics and/or pneumatics projects, without using a document or project template. Also, it is not compulsory to configure the edition and simulation options for the project. This approach is acceptable when working on small projects; however, if your projects are to evolve, if you create your projects based on your company/school standards, or if you want to share your projects with your colleagues, clients or students, it becomes essential to configure your projects and documents correctly. Once configured, your projects/documents can be saved as templates.

For the documentation concerning the creation and management of templates, refer to the *Automation Studio™ User's Guide*.

This guide explains how to configure the edition and simulation options for your pneumatics and hydraulics circuits. A correctly configured document will help you save time during both the edition and the simulation of your circuit.

2.1 Configuring Installations and Circuits

The standards ISO 1219-2 specify that a hydraulics or pneumatics system is to be subdivided into installations, which are themselves further subdivided into circuits. Initially, the standards were only applied when constructing ISO identifiers.

Each component (hydraulic, pneumatic, and electric) has an internal ID, which is built according to the standards: an ID is a concatenation of component properties.

Internal Code = "Installation Number" + "-" + "Circuit Number" + "ISO Component Code" + "Auto Counter".

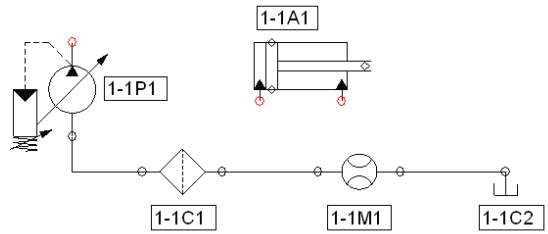


Figure 2-1: Example of Some Internal IDs (ISO)

The ISO Component Code is defined as follows:

- Pumps and Compressors: P;
- Actuators: A;
- Driving Motors: M;
- Sensors: S;
- Valves: V;
- Other devices: Z (L for Lines, C for Fluid Conditioners, M for Measuring Instruments);
- For other technologies, the norms IEC 61131, JIC are followed. The internal code for a component is constructed in the same way as for hydraulic/pneumatic components. Thus guaranteeing that each component has a unique internal code. This code is used most notably to identify variables.

Currently, the management of installations and circuits follows the recommendations of ISO 1219-2, without the norms limited constraints.

An installation or functional group is an autonomous system: excavator, tractor, wastewater treatment plant, pumping station, etc.

A circuit is a subcomponent in the installation, and plays a specific part: cooling circuit, lifting circuit, etc.

2.1.1 Managing Installations and Circuits

Installations and Circuits are managed in the dialog box found under “Project Properties” → “Fluid System” → “Management”.

In this guide, when a circuit is referred to, it is understood that the circuit is part of an installation. In the project properties, a circuit is identified by the pair (Installation, Circuit).

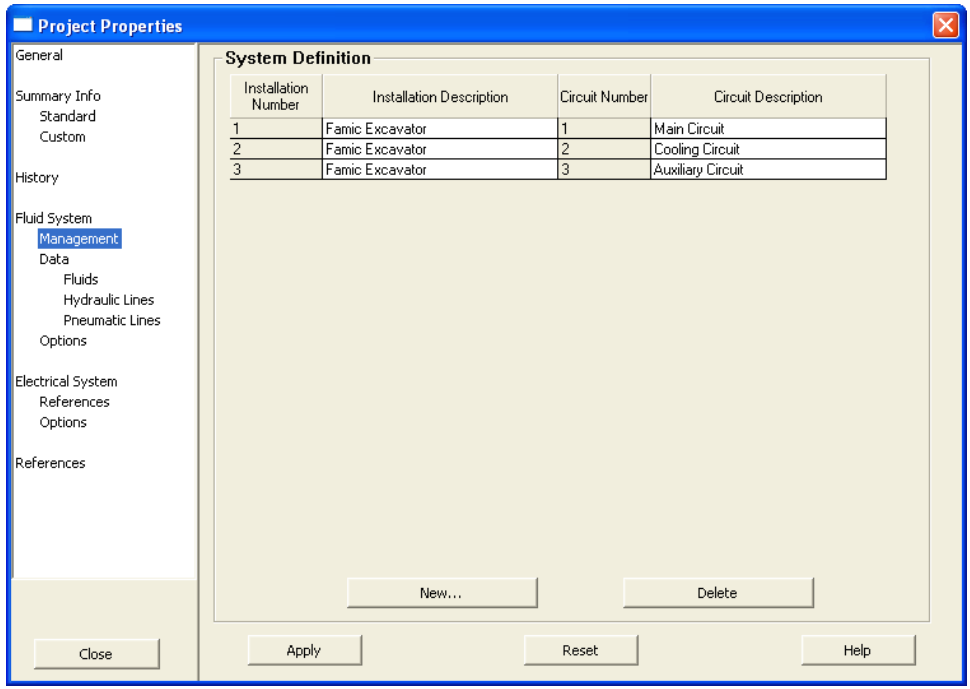


Figure 2-2: "Project Properties" → "Fluid System" → "Management" Dialog Box

Field	Description
Installation Number	Assigns a number to an installation. Standard ISO 1219-2 specifies that the code is composed of numerals, and starts with 1. This installation number is used when the circuit contains an installation.
Installation Description	Describes an installation. An installation can have at most one description.
Circuit Number	Assigns a number to a circuit. Standard ISO 1219-2 specifies that the code is composed of numerals. The number is incremented with each circuit.
Circuit Description	Describes a circuit. A circuit can have at most one description.
New…	Allows the creation of a new pair (Installation, Circuit), by adding a new line in the dialog box.

Field	Description
Delete	Allows the suppression of an existing pair (Installation, Circuit). This is possible only if the project has no component that belongs to this circuit.

☞ While names can be given to installations and circuits, it is preferable not to do so, since the ISO standard and the component variable names already use them.

(Installation, Circuit) Pair Assignment

To assign specific pair (installation and circuit) to a component, after you configured properly installationss in the « Project Properties » → « Fluid System » → « Management », there are different ways to proceed:

- When a component is inserted on a diagram, the selected subdivision in the « View » toolbar is automatically associated to the component;

- When a component is selected, it is possible to modify installation and circuit pair by choosing the desired pair in the « View » toolbar.

Figure 2-3: "View" Toolbar, (Installation, Circuit) Assignment

2.2 Other Management Tools

Automation Studio™ gives the possibility to create, modify and reuse properties that are useful for a simulation, without having to redefine them over and over. These properties describe the fluids as well as the lines used in your circuits. If you decide so, these properties can be kept when you install a new version.

These tools are accessed by the "Tools" menu, and are used to manage:

- Hydraulic fluids;
- Line materials;
- Types of Lines.

2.2.1 Fluid Management

For every circuit it is possible to define the type of oil used. If you cannot find the type you want, you can define your own new type. Fluid properties are available in the "Project Properties" dialog box.

☞ WARNING: if you use different oil types in different circuits, then components of one circuit cannot be connected to components from another circuit. The editor assumes the oil type is not compatible.

☞ It is possible to use the same oil for different circuits. Notice that the oil properties remain identical (Thermal properties, Bulk Modulus and Pressure, air Percent, Minimum/Maximum Temperatures).

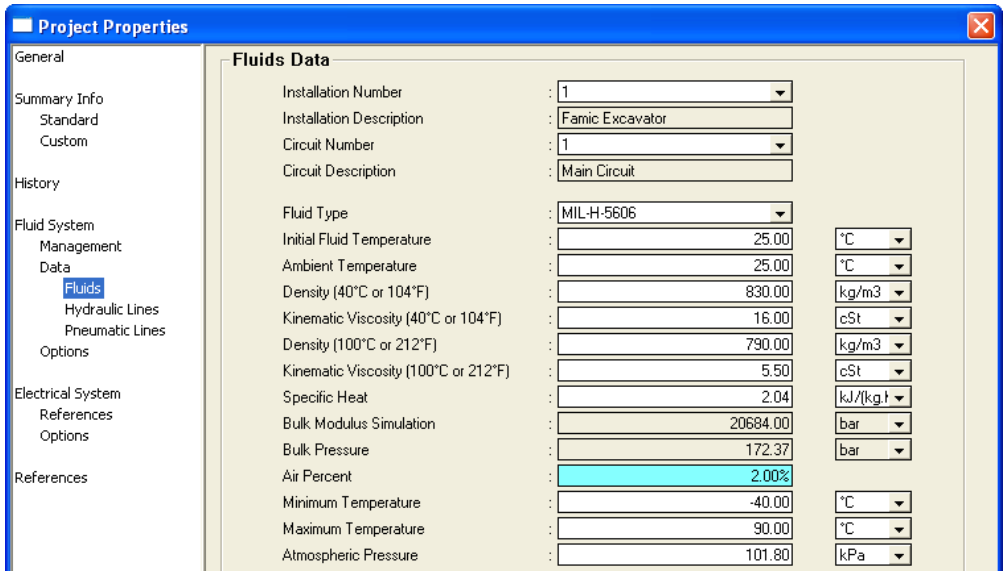


Figure 2-4: "Project Properties" → "Fluid System" → "Data" → "Fluids" Dialog Box

For the first four fields in the above figure, please refer to *Figure 2-2: "Project Properties" → "Fluid System" → "Management" Dialog Box.* The other fields are described below:

Field	Description
Installation Number	Specifies the installation for which this oil type is defined.
Installation Description	Shows the description corresponding to the installation number.
Circuit Number	Specifies the circuit for which this oil type is defined.
Circuit Description	Shows the description corresponding to the circuit number.
Fluid Type	Chooses among frequently used fluids. It is possible to define your own fluid type. Refer to section *Creating and Modifying Fluids.*
Initial Fluid Temperature	Specifies the fluid temperature when the simulation begins.
Ambient Temperature	Specifies the ambient temperature. This temperature is usually assumed to be constant during a simulation run; to make this temperature variable, use an ambient temperature generator (see section *3.2.1 Ambient Environment*).
Kinematic Viscosity (40ºC or 104ºF)	Specifies the kinematic viscosity at the given temperature and the atmospheric pressure.
Density (100ºC or 212ºF)	Specifies the ratio mass over volume of fluid at the given temperature.
Kinematic Viscosity (100ºC or 212ºF)	Specifies the kinematic viscosity at the given temperature and the atmospheric pressure.
Specific Heat	Specifies the specific heat: amount of heat per unit mass required raising the temperature by one degree Celsius.

Field	Description
Atmospheric Pressure	Specifies the atmospheric pressure involved in various simulation calculations.

☞ If you want a constant kinematic viscosity, enter the same values for the kinematic viscosity for both temperatures, 40ºC and 100ºC (104ºF and 212ºF).

☞ In order to please the mobile and industrial hydraulics and pneumatics market, the number of pipe types was increased with the addition of Load Sense and Return pipes, along with five other Custom types.

Creating and Modifying Fluids

The fluids you can use are not limited to the ones offered by default. You can create your own fluids and reuse them. Fluid properties can be used in other version of Automation Studio.

These operations are accessible through the menu item "Tools" → "Fluid Management…". This function opens the following dialog box:

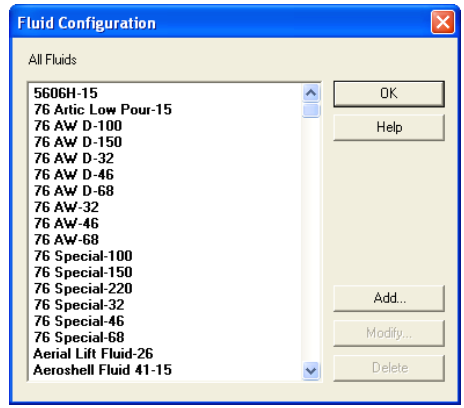


Figure 2-5: "Fluid Configuration" Dialog Box

By choosing the corresponding button, it is possible to "Add…", "Modify…" or "Delete" a fluid.

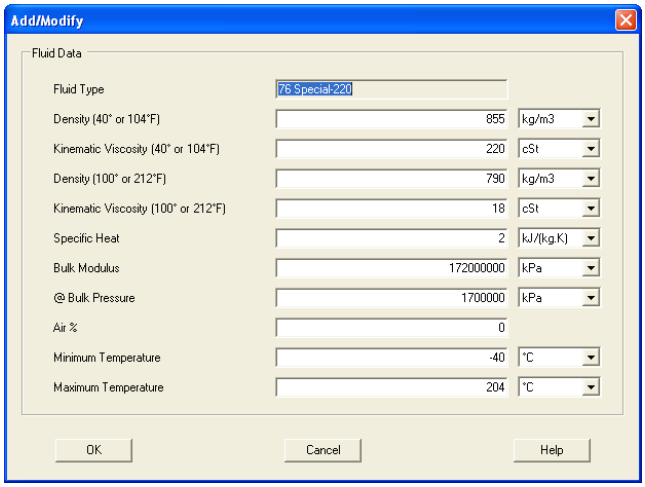


Figure 2-6: "Add/Modify" Dialog Box for Fluids

The fields in the "Add/Modify" dialog box are explained below:

Field	Description
Fluid Type	Specifies the name or identifier of the fluid type to add or modify.
Density (40 °C or 104 °F)	Specifies the ratio mass over volume of fluid at the given temperature.
Kinematic Viscosity (40 °C or 104 °F)	Specifies the kinematic viscosity at the given temperature.
Density (100 °C or 212 °F)	Specifies the ratio mass over volume of fluid at the given temperature.
Kinematic Viscosity (100 °C or 212 °F)	Specifies the kinematic viscosity at the given temperature.
Specific Heat	Specifies the specific heat: amount of heat per unit mass required raising the temperature by one degree Celsius.

Field	Description
Bulk Modulus	This module B = P/(ΔV/Vo), describes the incompressibility of solid bodies and fluids. For an equal pressure P, if B2 > B1 then fluid 2 is less compressible than fluid 1.
Pressure Bulk	The Bulk Modulus is measured at a given pressure, called Bulk Pressure.
Air %	Specifies the percentage of air in the fluid.
Minimum Temperature	Specifies the minimum temperature at which the fluid can be used.
Maximum Temperature	Specifies the maximum temperature at which the fluid can be used.

2.2.2 Line Material Management

The hydraulics simulator uses the information on the line material to evaluate the pressure drop. You can modify or add new materials to the default list offered by Automation Studio™.

Creating and Modifying Materials

These operations are accessible through the menu item "Tools" → "Material Management…". This function opens the following dialog box:

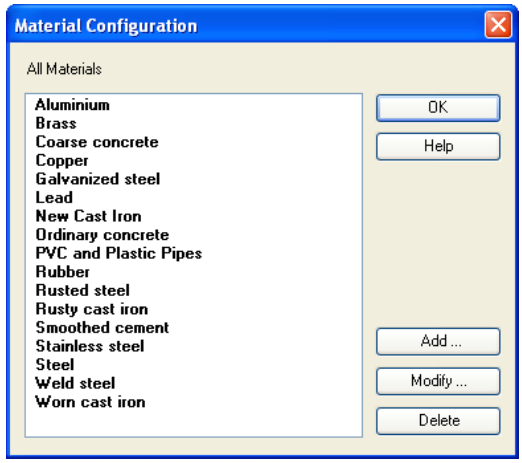


Figure 2-7: "Material Configuration" Dialog Box

By choosing the corresponding button, it is possible to "Add…", "Modify…" or "Delete" a material.

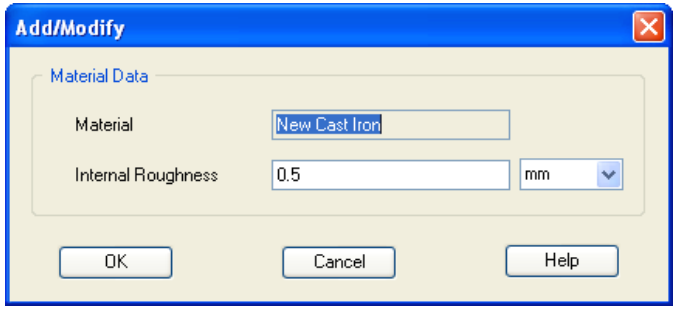


Figure 2-8: "Add/Modify" Dialog Box for Materials

Field	Description
Material	Specifies the name for the material to add or modify.
Internal Roughness	Specifies the default internal roughness for a material.

2.2.3 Line Management

It is easier to refer to a line by its name than to remember its diameter. For such reasons, it is possible to manage a list of lines.

For every circuit, it is possible to define types of lines, their properties, materials, and their default appearance. This information is accessible through the "Project Properties" dialog box, as illustrated below.

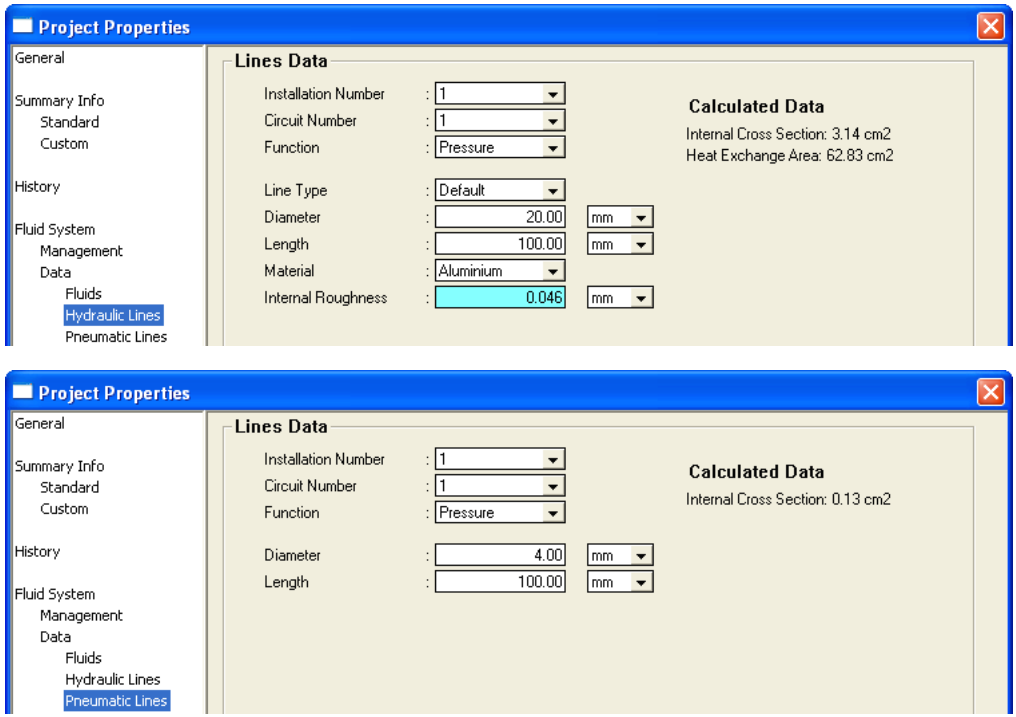


Figure 2-9: "Project Properties" → "Fluid System" → "Data"→ "Lines" Dialog Box

Field	Description
Installation Number	Specifies the installation for which this line type is defined.
Circuit Number	Specifies the circuit for which this line type is defined.

Field	Description
Function	Specifies the line function according to specific properties of this line.
Line Type	Specifies the line type.
Diameter	Specifies the line internal diameter.
Length	Specifies the line length.
Material	Specifies the line's material.
Internal Roughness	Specifies the line internal roughness.
Line Appearance	Chooses the line appearance in the editing mode. By clicking on "…", you can choose the desired format.

From this dialog box it is possible to reload the line properties of a circuit.

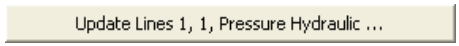


Figure 2-10: Button to Reload a Circuit or Installation

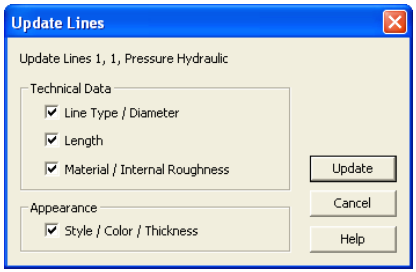


Figure 2-11: Options to Reload a Circuit or Installation

☞ When reloaded, flexible lines are assigned the circuit pressure line properties.

Creating and Modifying Lines

These operations are accessible through the menu item "Tools" → "Lines Type Management

…". This function opens the following dialog box:

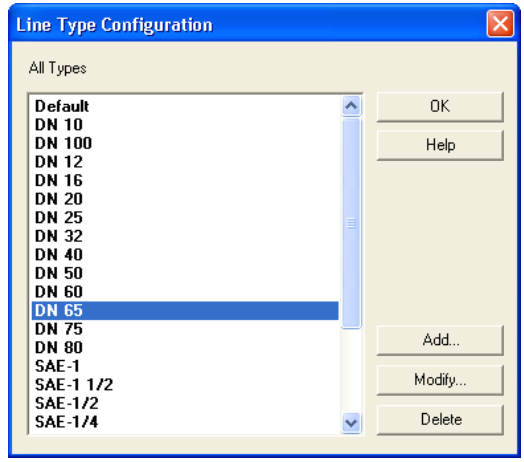


Figure 2-12: "Line Type Configuration" Dialog Box

By choosing the corresponding button, it is possible to "Add… ", "Modify…" or "Delete" a line type.

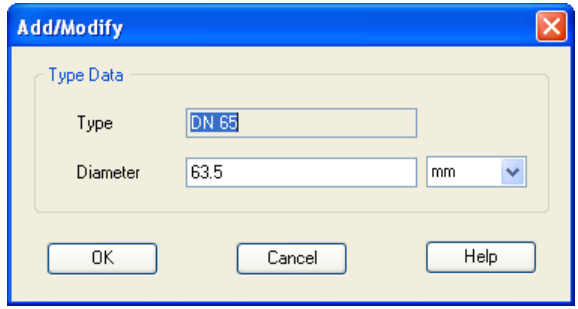


Figure 2-13: "Add/Modify" Dialog Box for Line Types

Field	Description
Type	Specifies the name or identifier for the line type to add or modify.
Diameter	Specifies the default diameter for this line type.

3 Systems Simulation

Circuits' simulation gives access to physical values such as pressure, flow, fluid temperature at measuring points, as well as to several kinematics and dynamics values for valve actuators. Measuring instruments and plotter make it possible to follow the evolution of these values and to analyze them.

Note that before testing a circuit, you must be acquainted with the ambient system physical characteristics, as well as the fluid properties used to transmit the hydraulic or pneumatic power.

3.1 Fluid Simulator

Automation Studio uses a combination of methods that reproduce the behaviour of dynamic and static interconnected components. Dynamics models are reproduced using either differentials equations or non-linear models based on manufacturer's specifications.

The simulator main goal is to deliver a global behaviour of a circuit as close as possible to reality, in real time if possible. Very short transitions phenomenon (less than 10ms) are resolved by approximation. In order to reproduce physical behaviour at very small scale of a particular component, others tools may be more appropriate.

It is otherwise possible, upon request, to adapt and model components for Automation Studio™ at smaller scale as if it was a system.

3.2 System Environment

For fluid power, the simulator must know the values associated with two types of data to initiate calculation:

1. The initial properties of the ambient environment: pressure, temperature (hydraulic only);
2. The initial properties of each fluid: initial oil temperature, viscosity, density and Bulk modulus (hydraulic only).

We will further describe some of the following points:

- The Ambient environment;
- The Fluid;
- The concept of temperature.

3.2.1 Ambient Environment

The ambient environment is the circuit's exterior environment. To access the ambient environment options, go to "File" → "Project Properties…" → "Fluid System" and into "Options" branch as on the figure below:

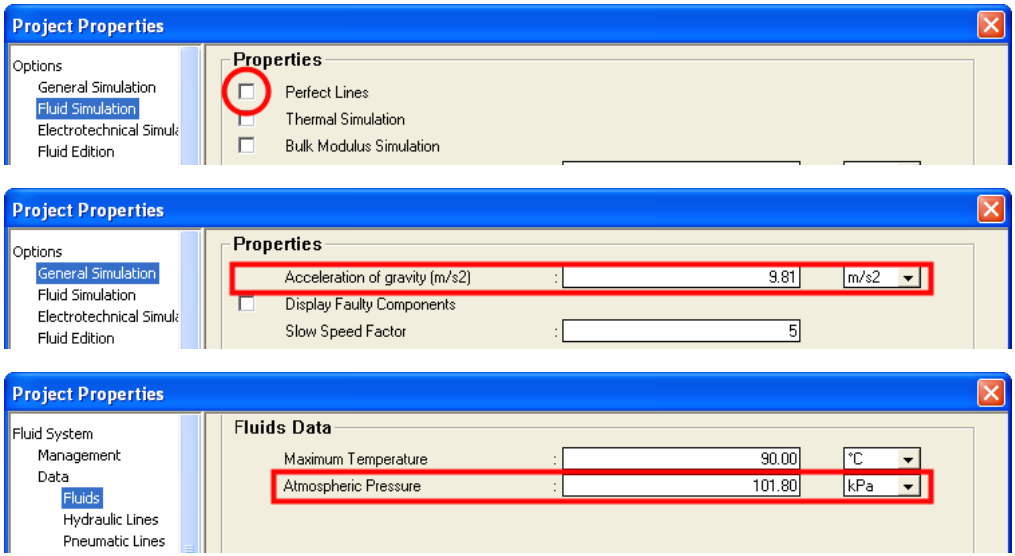


Figure 3-1: Ambient Environment Options and Pressure Drop

Field	Description
Perfect Lines	When selected, this option allows the simulation of a perfect pipe with no load drop by friction, hence without heat transfer.
Acceleration of gravity (kPa)	Allows specifying the acceleration that results from a gravitational force for the simulation.
Atmospheric Pressure	Allows specifying the atmospheric pressure for the simulation of the system. This pressure is characterized per circuit.

It is defined by:

- External pressure;
- Ambient temperature;
- Gravitational field.

External Pressure

The external pressure can be atmospheric, and can vary from 1 to 2 bars. The value can be modified from the software fluid simulation options, as shown in *Figure 3-1: Ambient Environment Options and Pressure Drop*.

To work with a higher external pressure, you must use a tank. Refer to section *4.3.3 Fluid Storage* for further information.

In the "Flow Lines and Connections" branch, there are components that allow to connect or to seal fluid lines.

Ambient Temperature

The default ambient temperature is 25ºC, and is defined independently for every circuit. Refer back to section *2.1.1 Managing Installations and Circuits*. The ambient temperature is accessible in the project properties.

The ambient temperature is constant. However, one can use an "Ambient Temperature Generator" component (in "Sensors", under the "Hydraulic" section of the main library) to make the ambient temperature vary with time. These components will only act on the ambient temperature for the circuit to which it belongs.

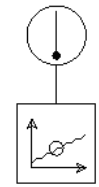


Figure 3-2: Ambient Temperature Generator

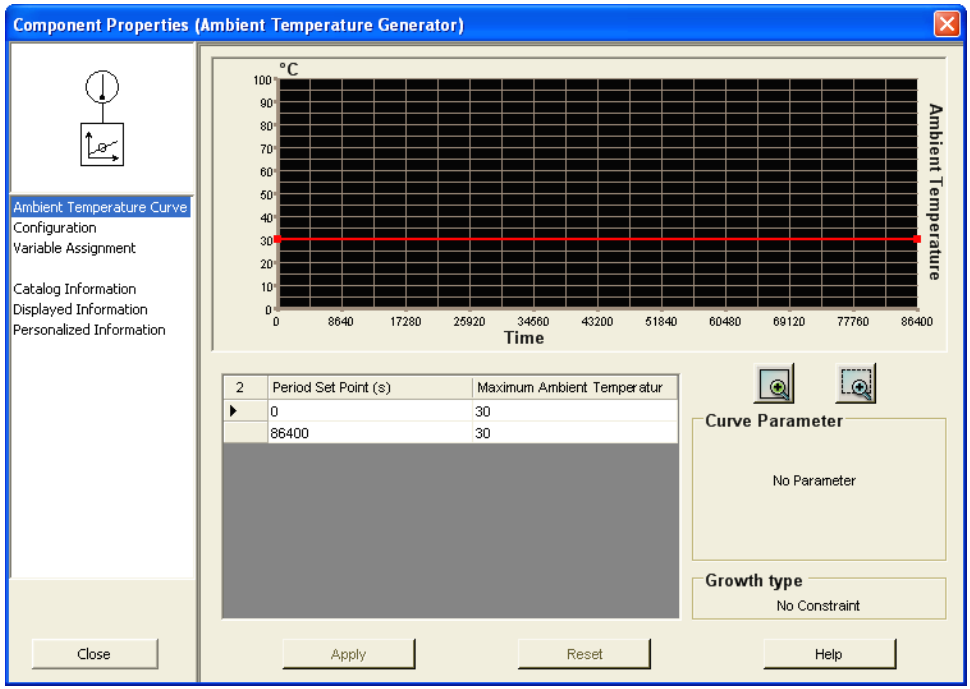


Figure 3-3: Ambient Temperature Curve

Gravitational Field

In general the gravitational field is not used by the simulator, except in the following circumstances:

- Modelisation of loads on actuators: Mass + Angled Cylinders;
- Modelisation of a hydrostatic line; situations with non-negligible height differences (Mining, Petroleum Industries);
- Some stocking components can simulate hydrostatic pressure.

☞ The hydrostatic pressure of oil is generally negligible in the usual industrial and mobile applications.

3.2.2 The Fluid

In pneumatic applications, the fluid is considered as compressible, while it is considered incompressible in hydraulic applications. The behaviour of the air is determined by the atmospheric pressure; hydraulic applications also consider the initial temperature, which assigns the kinematics density and viscosity. Depending on the temperature's evolution, the viscosity and density values are adjusted based on the fluid characteristics, for every circuit (independently), in the project properties.

☞ Note that the viscosity grows with pressure. The viscosity index doubles when reaching 400 times the atmospheric initial pressure (approx. 400 bars).

The "Project Properties" → "Fluid System" → "Data" → "Fluids" dialog box allows setting the project's fluid environment parameters, as well as the ambient temperature.

The fluid properties are only taken into account if the thermal option is activated.

☞ If the "Thermal Simulation" option is unchecked, then the considered fluid is the oil that is defined for each circuit; its viscosity will be calculated by interpolation, functions of the initial temperature value of the specified oil in the circuit properties.

☞ The air percent in the oil is only considered when the Bulk Calculation is activated from "Project Properties" → "Fluid System" → "Options".

3.3 Major Simulation Options

Only the options influencing the simulation are covered in the chapter. The other options are described at the end of this guide.

The options configuration from the dialog box "Project Properties"→ "Fluid System"→"Options" is global to the whole project. Therefore, if you choose not to simulate the load loss in the pipes, then all the pipes in the simulated diagrams of a project will be considered as perfect lines. This can explain the difference in the calculations obtained between two simulation sessions.

3.3.1 Fluid Dashboard

The fluid dashboard is a floating window that allows to quickly visualize and modify simulated phenomena in hydraulic and pneumatic systems. It is shown in the following figure.

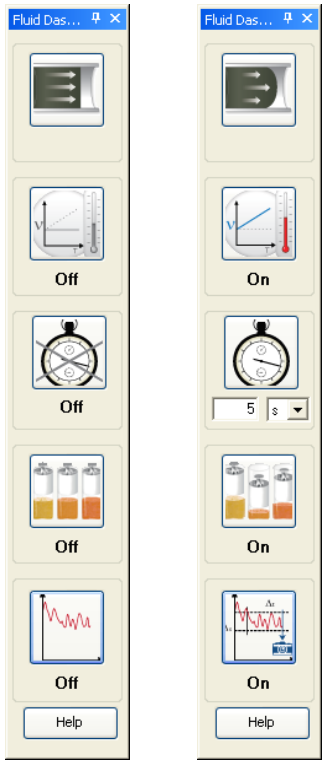


Figure 3-4: "On" and "Off" States of the "Fluid Dashboard"

Option	
	Pressure drops
	Thermal simulation

Option	
	Snapshot delay
	Bulk modulus simulation
	Steady State Evaluation

When the "Fluid system" options are modified, the "Fluid Dashboard" is automatically updated. Also, modifications in the "Fluid Dashboard" automatically modify corresponding options of the project.

To open and use "Fluid Dashboard":

1. Choose "Window" → "Display" → "Fluid Dashboard";

or

1. Click on the button in the "Project" toolbar;

   The "Fluid Dashboard" window appears.

2. Modify Boolean values by directly clicking on them and modify numerical options by typing the value and choosing the unit.

   Simulation options are updated.

3.3.2 Common Hydraulic and Pneumatic Options

Pressure Drops

For all lines, pressure drop is proportional (linear function) to the line length in hydraulics as well as in pneumatics. However, the calibration results from empirical data supposing that the

lines length is long enough (several meters) for the pressure drops across connectors to be insignificant. The pressure drop depends on the type of flow (turbulent or not).

The pressure drops effects can be excluded in the simulation, for pedagogical or performance purpose, by selecting the menu "File" → "Project Properties" → "Fluid System" → "Options" → "Perfect Lines".

When the pressure drop is not considered, the simulator does not display the arrows indicating the flow direction.

3.3.3 Hydraulic Simulation Options

Thermal Simulation: To access the thermal simulation options, go to ""File" → "Project Properties" → "Options", and check the "Thermal Simulation" box.

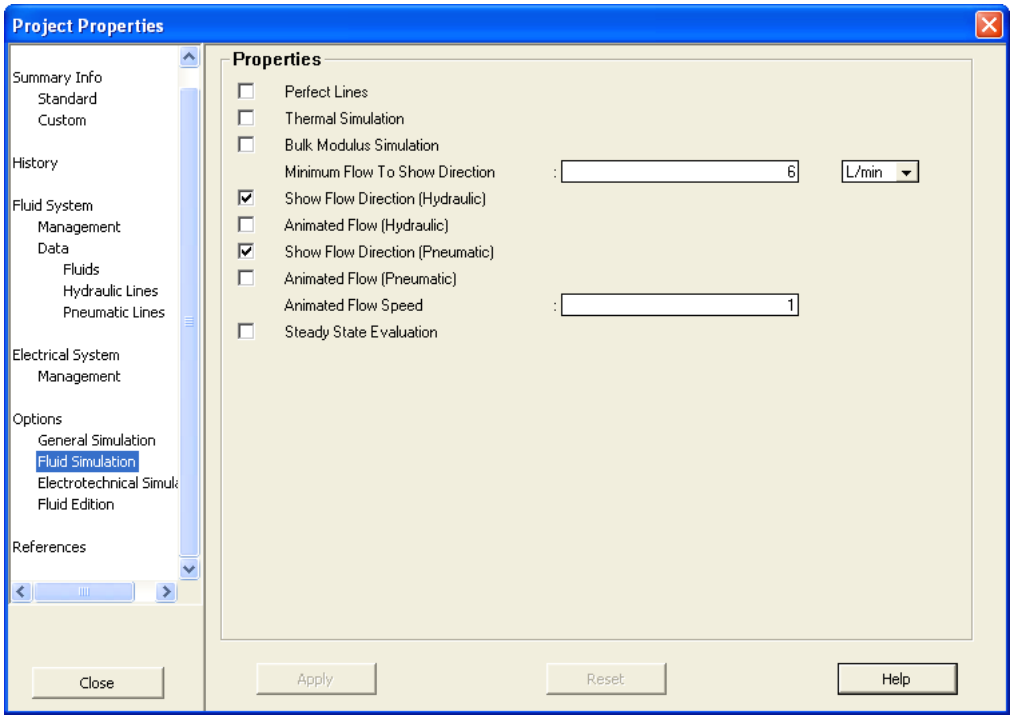


Figure 3-5: "Thermal Simulation Options"

Field	Function
Thermal Simulation	Allows you to take into consideration the fluid heating in contact with the pipes' inner surface.
Bulk Modulus Simulation	Allows considering incompressibility modulus of each hydraulic fluid used in the project.

☞ If you don't activate the "Thermal Simulation" option, then the fluid is assumed to be the oil defined for the installation/circuit, and its temperature is interpolated from the initial temperature value (in the circuit properties).

☞ Kinematic Viscosity is a function of both pressure and temperature. However if the "Thermal Simulation" option is not active, the kinematic viscosity is assumed constant for the installation/circuit.

If your circuit is spread over several documents, you will have to use library links.

In the Library Explorer, go to "Hydraulic" → "Flow Lines and Connections" → "Jump-To Label". You must be careful to name them adequately under the configuration root in the component's properties.

Consideration of the Bulk Modulus: In order to obtain a better precision of calculations in simulation, you can choose the option "Bulk Modulus Simulation". Refer to *Figure 2-6: "Add/Modify" Dialog Box for Fluids* for details. If you activate that option, the simulation will take into account the Bulk Modulus and the air percent in each used oil in the circuits.

This option can slow down the simulation time.

3.3.4 Simulation Accuracy and Optimisation

Some options are available in Project Properties to achieve better performance, or to simplify computations. For a large system, simulation can be slow. Some options are available if you're only interested in obtaining a quick qualitative behaviour of the system (but the results will be less accurate).

Depending on the size of your system, and which processor is used, simulation can be slowed-down seriously. To improve speed, deactivate the "Thermal Simulation" and the "Bulk Modulus" options, and choose Perfect Lines.

☞ Only the Thermal Simulation is active by default.

Minimize Document Windows: minimizes the display. This option may seem useless, but if you build a circuit along with a logical controller you may want to reach a certain level in his simulation scenario before beginning to analyse it.

3.3.4.1 Thermal Simulation

Each hydraulic component carries, in his advanced properties, a thermal transfer index, a transfer surface, as well as the volume used by the fluid at any time. The volume can be either automatically calculated (for lines and cylinders for instance) or set. The flow in the circuitry warms up the oil and the external environment according to the temperature difference between the ambient environment and the oil, the thermal transfer index, the transfer surface and the oil volume.

☞ Thermal simulation only requires one storage tank in the circuit with an infinite volume.

How does Automation Studio™ calculate the heat exchange?

First Step

For each component, a certain amount of energy in the form of heat is transferred to the fluid. This heat transfer is mostly due to friction as the fluid passes through the component; the energy loss is converted to heat and transferred to the fluid. Since ΔP is a function of flow, viscosity, flow profile and the geometry of the component, this parameter will be considered for the heat transfer calculation. The other parameter introduced is the fluid specific heat that allows defining the amount of energy needed to increase the temperature of the fluid. Therefore, with these parameters, Automation Studio™ uses the following first equation for heat transfer:

$$\Delta T = \frac{\Delta P \cdot \Delta t}{\rho \cdot C_s}$$

Figure 3–6: Temperature Increase in a Fluid

Where:

ΔT is the calculated temperature variation of the fluid;

ΔP is the pressure differential between the inlet and outlet of the component (P2-P1) (Pa);

ρ is the fluid density at temperature T (K);

Δt is the fluid's specific heat value (kJ/kg·K);

C_s is the calculated time interval (s) (for AS = 1 second).

Second Step

Since components transfer energy to the fluid, there is also heat dissipation from the fluid to the surrounding environment. Some components, such as valves, do not permit a great amount of energy to pass from the fluid to the atmosphere; others, such as coolers, are specifically designed for that purpose.

To quantify the capacity of components to dissipate the heat from the fluid, the "Thermal conductance" parameter in W/m2·K is introduced. This parameter, multiplied by the heat exchange area allows Automation Studio™ to calculate the amount of heat passed from the fluid to the environment. The temperature difference between the fluid and the environment (ambient) will also affect the dissipation rate, and thus the appearance of the term T_A in the equation. Finally, the total amount of energy in the fluid is determined by using the specific heat and volume of the fluid. Therefore, by simplifying some parameters, the thermal dissipation is calculated by using:

$$\Theta_t = \frac{(T_0 + \Delta T - T_A) \cdot C_t \cdot S_e \cdot \Delta t}{\rho \cdot C_s \cdot V_{Tot}}$$

Figure 3–7: Thermal Dissipation

where,

Θ_t is the total thermal dissipation;

T_0 is the temperature of the fluid at the beginning of the interval t;

ΔT is the calculated temperature variation of the fluid;

T_A is the ambient temperature;

C_t is the Thermal Conductance;

S_e is the heat exchange area;

ρ is the fluid density at temperature T (K);

C_S is the fluid's specific heat value (kJ/kg·K);

V_{Tot} is the total amount of fluid having passed through the component during the time interval t.

The specific heat is not known for all of the oils listed in Automation Studio™. When unknown, an average value is used.

Third Step

Once Automation Studio™ has calculated all the heat transfer in the component, the difference is calculated therefore determining the total amount of heat fluctuation of the fluid:

$$T_t = (T_0 + \Delta T) - \Theta_t$$

Figure 3–8: New Temperature

Thermal equilibrium is reached and the temperature remains stable when the total of all heat transfers equals 0:

$$\sum \Delta T = \sum \Theta_t$$

Figure 3–9: Global Thermal Results

Independently of the oil temperature, the simulator always considers the fluid as being liquid. Boiling or freezing phenomena are not shown. You must, by your own, use the default component displayed functions to monitor the oil limit temperatures.

3.3.4.2 Simulation without Thermal Transfer

If pressure drops are simulated, then the flow of the fluid against the line's sides and components will warm up the fluid. By default, Automation Studio™ does not take into account these parameters in its calculations. If you want to change this setting, go to "Tools"

menu → "Options…" → "Diagram Simulation" → "Fluid Simulation" → "Hydraulic Workshop", as shown in *Figure 3-5: "Thermal Simulation Options"*.

3.3.5 Layer Simulation

To validate some operational functionalities of subsystems, it is possible, directly from the "Layers" manager, to specify for each diagram which layers will be simulated. For details, refer to the *Automation Studio™ User's Guide*.

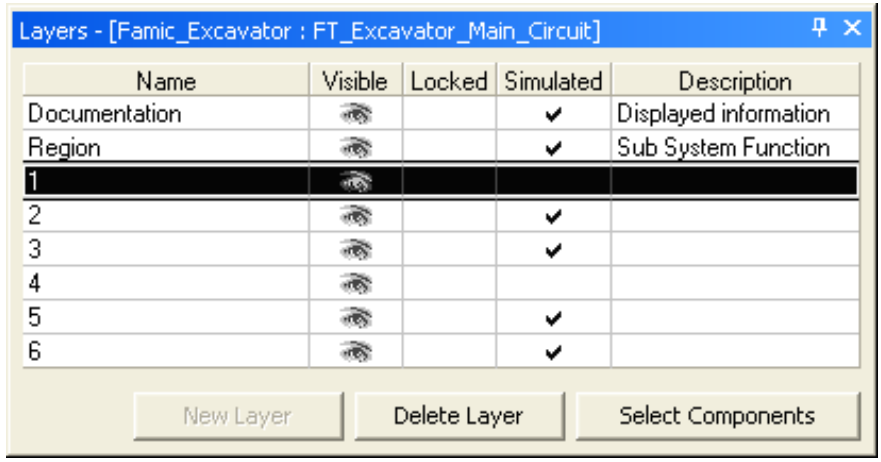


Figure 3-10: Layer Simulation

3.4 Advanced Analysis Tools

Advanced analysis tools allow you to perform thorough systems analyses, which are divided into two parallel and distinct parts.

- The first part is composed of the "Steady State" and "Snapshot" tools. You can trigger a data simulation snapshot according to many criteria. These values can be used to restart a simulation with saved initial conditions. This data can also be exported to be included in a complete simulation analysis report.

- The second part is the "Measuring Instruments" tool. You can insert dynamic measuring instruments on desired components and connection ports. Each instrument record variable evolution in simulation mode. After simulation, the measured values can be saved. These values can be plot or export for analysis.

For details on creation and management of the advanced analysis tools, refer to the *User's Guide*. This section contains only information on Steady State configuration and on measuring instruments that own to *Hydraulic and Pneumatic Workshops*.

3.4.1 Steady State

This feature allows you to define global or local stability criteria of your system or of the hydraulic and pneumatic components, in order to trigger a snapshot automatically with variables calculated by the simulator.

Steady State Setting

You can configure Steady state parameters in the "Steady State settings" dialog box shown in the following figure.

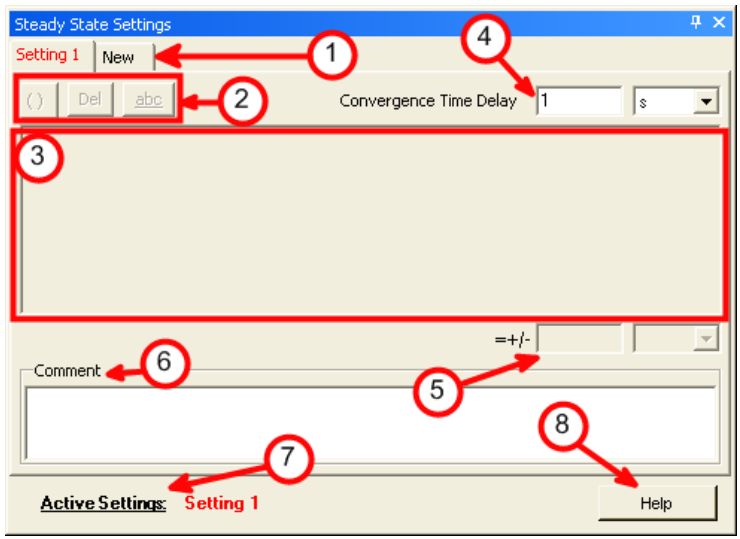


Figure 3-11: "Steady State Settings" Dialog Box

It includes the following elements.

	Description
1	These tabs contain a set of steady state conditions.

	Description
	This button allows you to group multiple consecutive conditions.
2	This button allows you to remove one or more selected conditions.
	This button allows you to visualize the component related to the selected condition.
3	This zone allows you to define steady state condition with various component properties.
4	Allows you to specify convergence time delay for which a condition is considered true (steady state) according to the value intervals "=+/-" of each property.
5	Allows you to specify the maximum variations compared to an equilibrium value, for a condition to be considered as true (Steady State).
6	Allows you to enter a comment for the steady state condition of the active tab.
7	Allows you to visualize which condition is active.

To open "Steady State Settings" dialog box:

1. Choose "Windows" → "Display" → "Steady State Settings" or click on the button of the "Simulation" toolbar.

   The "Steady State Settings" dialog box appears, as shown in the *Figure 3-11: "Steady State Settings" Dialog Box.*

For a new project or a project that doesn't have steady state condition, there are two tabs: "Setting 1" and "New".

To add a new tab:

1. Click on “New” tab. A new tab is created on the left side of the existing tab

There is no limit on the number of steady state conditions for a project. But only one set will be considered in a simulation session.

Also, the shortcut menu of a tab, other than the “New” tab, contains configuration options shown in the following figure.

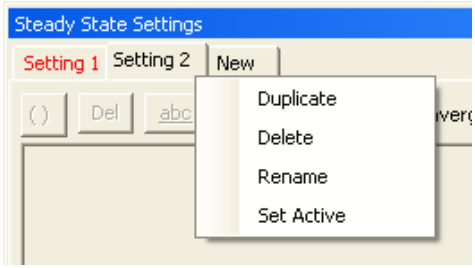


Figure 3-12: “Steady State Settings” Shortcut Menu

Command	Description
Duplicate	Allows you to copy all properties associated with a steady state condition in a new tab.
Delete	Allows you to delete selected tab.
Rename	Allows you to rename selected tab.
Set Active	Allows you to activate selected tab.

Edition of Steady State Conditions

To add steady state properties:

1. After you have opened the tab you want to add properties to, drag and drop a component on the “Steady State Settings” window;

   For example, for a fixed displacement pump, a dialog box with the properties that can be plotted appears.

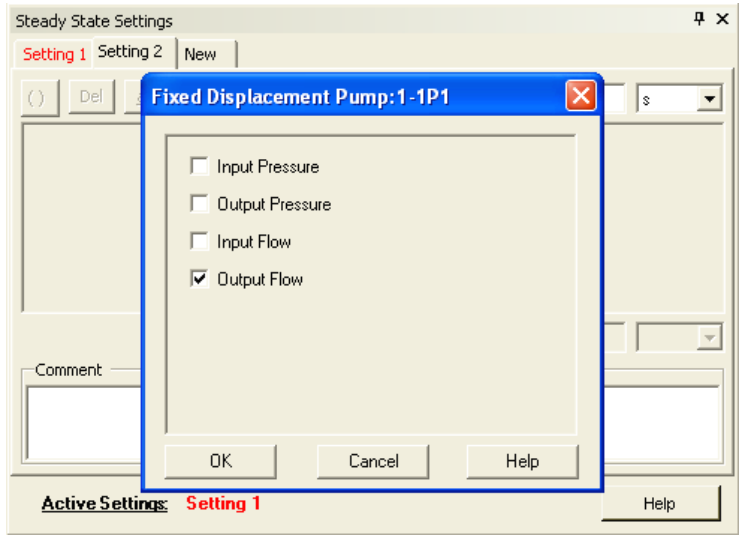


Figure 3-13: Properties Insertion of the Steady State Condition

2. Check all properties you want to consider for the steady state;

☞ If there is no property available for the steady state evaluation, the following dialog box appears.

Figure 3-14: Component without Available Property

3. Click on "OK".

The chosen properties are added and can be edited, as it is shown in the following figure and this is valid for three properties available of a fixed displacement pump.

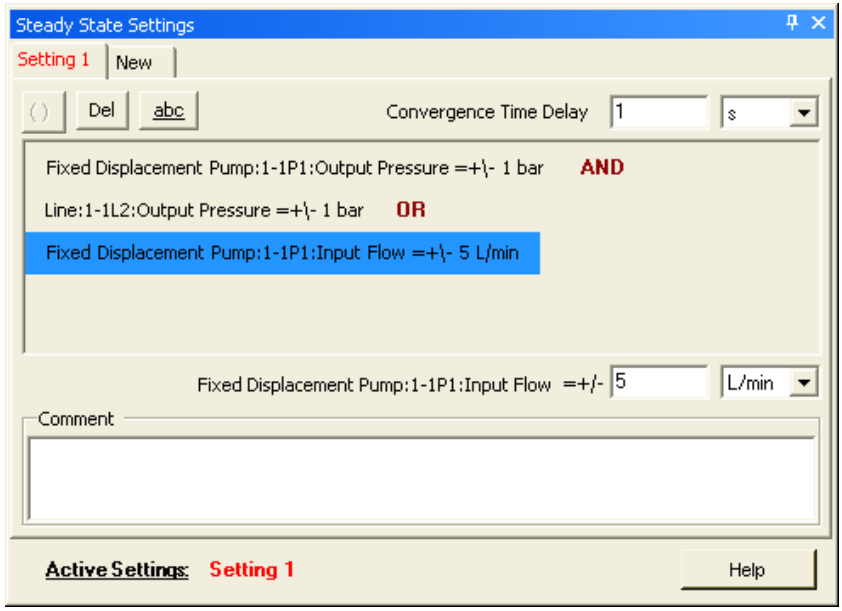


Figure 3-15: "Steady State Settings" with Added Properties

To edit properties of the steady state:

1. Select a property to edit;

   The property is displayed in the bottom of the dialog box and there is a field to edit its maximum variation value and a drop list to choose property units from.

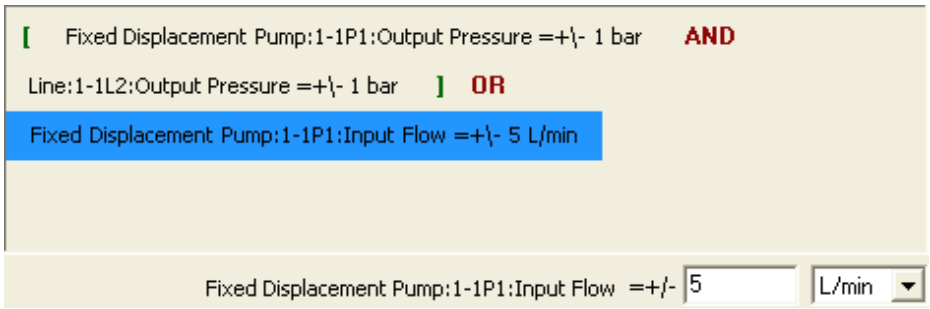


Figure 3-16: Selection of a Property

2. Type and choose the maximum variation and its unit.

☞ The maximum amplitude value is equal to twice the specified variation.

☞ You can select multiple properties or groups of properties. This activates the ( ) button and you will be able to add parenthesis in order to group some consecutive properties.

☞ By selecting only one component, you can use the abc to visualize the component associated with the selected property in the corresponding diagram. You can also use the "Go Back" button to return to the initial view.

3. Select logical operator "AND" or "OR" that you want to modify;

   A drop (pop-up) list appears in the bottom left area and allows you to choose the desired operator for the steady state conditions to consider (analyse).

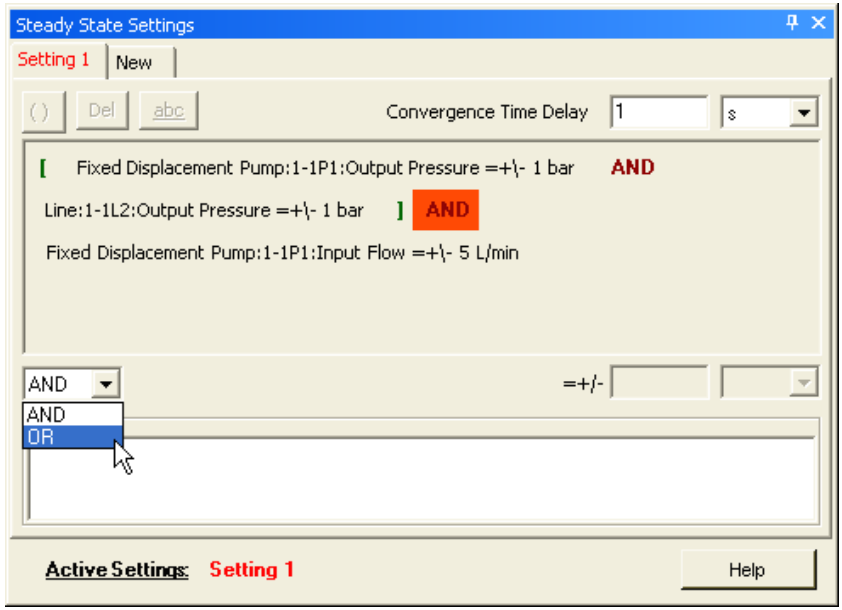


Figure 3-17: Logical Operator Selection

To delete one or more conditions, or parenthesis:

1. Select what you want to delete;

2. Click on the Del button.

3.4.2 Dynamic Measuring Instruments

Dynamic measuring instruments are specialized components satellites that allow to obtain useful information on the variables used in a simulation. In order to avoid inserting useless components in real schematics, these instruments, accessible by a dedicated tool bar, can be inserted in a circuit during simulation.

The main purpose of these dynamic measuring instruments is to display and record a given property. It is then possible for you to plot evolution curves of variables and to build a complete analysis report of the simulation. Unit and refreshment period can be specified.

The following figure shows an example of utilization of measuring instruments.

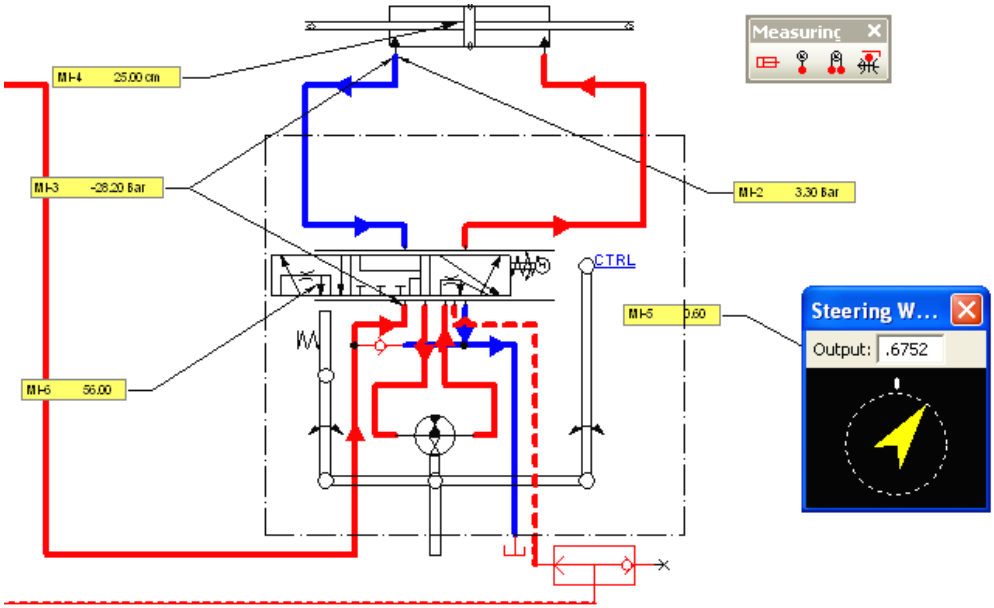


Figure 3-18: Example of measuring instruments in simulation mode

3.4.2.1 Measuring Instruments Toolbar

The toolbar contains options which allow you to insert dynamic measuring instruments on a diagram.

Figure 3-19: "Dynamic Measuring Instruments" Toolbar

The "Dynamic Measuring Instruments" Toolbar contains the following buttons.

	Command	Description
	Component Dynamic Measuring Instrument	Allows you to insert a measuring instrument directly on a component.
	Connector Dynamic Measuring Instrument	Allows you to insert a measuring instrument directly on a connector.
	Differential Dynamic Measuring Instrument	Allows you to insert a measuring instrument between two connectors.
	Power Dynamic Measuring instrument	Allows you to insert a measuring instrument on a fluid line or on a connector.

3.4.2.2 Creating Dynamic Measurement Instruments

Component Measuring Instrument

To insert measuring instrument on a component:

1. Choose "Insert" → "Measuring Instruments" → "Component Dynamic Measuring Instrument" or click on the button in the "Dynamic Measuring Instruments" toolbar;

   The cursor takes this shape .

2. Click on a component for which you want to measure a property;

The selected element turns blue.

3. Move the cursor to the location where you want the measuring instrument arrow tip to be;

4. Click and, by holding mouse button, drag and drop to determine its location on the diagram;

   The dialog box for measuring instrument configuration appears.

5. Modify the property to measure as needed (if necessary);

☞ For this instrument, the available properties are the same as the ones that can be plotted.

6. Click on “OK”.

   The measuring instrument takes the following shape.

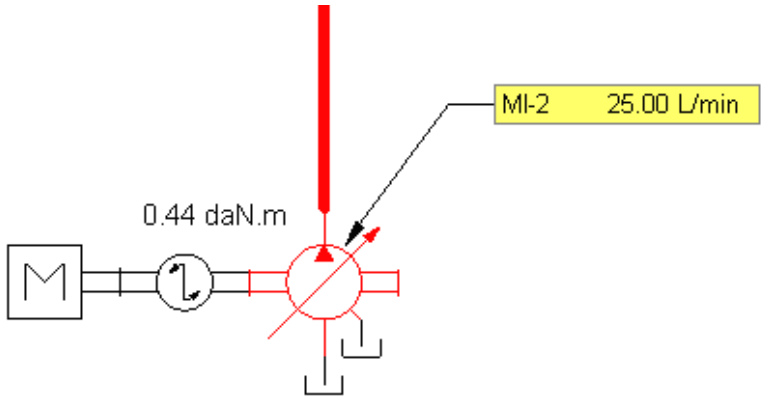


Figure 3-20: Component Dynamic Measurement Instrument Creation

Node Dynamic Measuring Instrument

To insert a measuring instrument on a connection port:

1. Choose “Insert” → “Measuring Instruments” → “Node Dynamic Measuring Instrument”

or click on the button in the "Dynamic Measuring Instruments" toolbar;

The cursor takes this shape .

2. Choose a connector for which you want to measure a property by moving the cursor over the desired connector;

   The cursor takes this shape .

3. Click and, by holding the mouse button, drag and drop to determine its location on the diagram;

   The dialog box for measuring instrument configuration appears.

4. Modify the property to measure as needed;

For this instrument, the available properties are pressure, temperature, density and cinematic viscosity of a fluid.

5. Click on "OK".

   The measuring instrument takes the following shape.

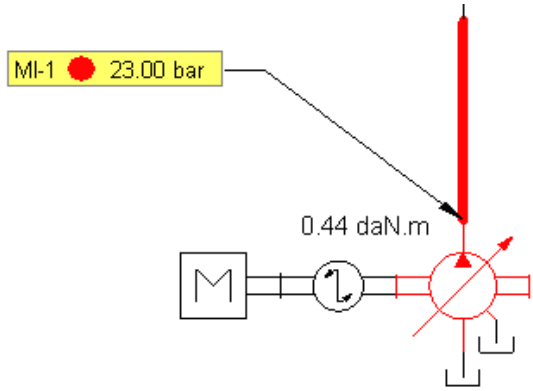


Figure 3-21: Node Dynamic Measuring Instrument Creation

Differential Dynamic Measuring Instrument

To insert measuring instrument between two connection ports:

1. Choose "Insert" → "Measuring Instruments" → "Differential Dynamic Measuring Instrument" or click on the button in the "Dynamic Measuring Instruments" toolbar;

   The cursor takes this shape .

2. Choose a connector for which you want to measure a property, by moving the cursor on the desired connector;

   The cursor takes this shape .

3. Click and, by holding the mouse button, drag and drop to determine its location on the diagram;

   The instrument location is fixed.

4. Click on a second connector for which you want to measure differential property;

   The dialog box for measuring instrument configuration appears.

5. Modify the property to measure as needed;

☞ For this instrument, the available property is the differential pressure.

6. Click on “OK”.

   The measuring instrument takes the following shape.

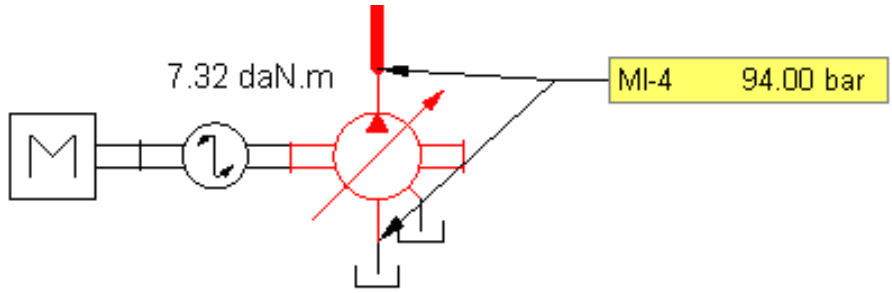


Figure 3-22: Differential Dynamic Measuring Instrument Creation

Power Dynamic Measuring Instrument

This method requires a line selection in order to specify a flow, and one or two pressure measurements to calculate power loss (Q x (P2- P1)) or available power (Q x P2)).

To insert a power measuring instrument:

1. Choose “Insert” → “Measuring Instruments” → “Power Dynamic Measuring Instrument” or click on the button in the “Dynamic Measuring Instruments” toolbar;

   The cursor takes this shape .

2. In order to identify flow (Q) and pressure (P1) needed for calculation, click on a line for which you want to measure the flow, by moving the cursor on it;

   The line turns blue only in simulation mode.

3. Click again on the line to define where you want the arrow tip to be, by holding the mouse button, then drag and drop to determine the instrument location on the diagram;

   The instrument location is now fixed.

4. Choose a connector to define the required pressure (P2) for power calculation, by clicking on the desired connection port;

   The cursor takes this shape .

5. Click on this connector;

   The dialog box for measuring instrument configuration appears.

6. Modify the property to measure as needed;

For this instrument, the available properties are power and power loss.

7. Click on “OK”.

   The measuring instrument takes the following shape.

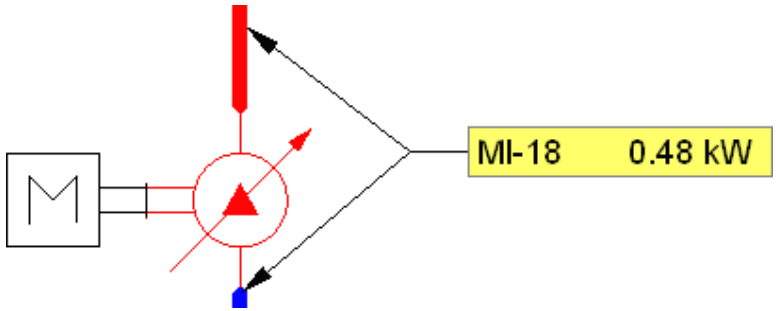


Figure 3-23: Power Dynamic Measuring Instrument Creation

4 Component Description

4.1 Introduction

Here is a few important information about components in order to avoid error messages during simulation.

This section contains the required information to fully understand the components' behaviours during simulation.

When simulating a project created with Automation Studio™, each circuit is made of components linked to each other by lines. Power is provided through the fluid. Each component has a mathematical model and for some of them, an additional graphic model.

This section aims at providing the general guidelines to use fluid components in the software.

To find out the role of each component property or the behaviour of Automation Studio™ components that are not listed in this guide, please refer to the contextual help using the F1shortcut key.

4.2 Standards and Specifications

Automation Studio™ can be used to design hydraulic/pneumatic circuits according to ISO 1219 – 1 (1991) standards. For the displayed information of the components, the ISO 1219-2 standards are respected.

4.3 Main Library Structure for the Fluid Workshops

The hydraulic and pneumatic workshops are similarly structured as the main library. To get access to the components of a specific workshop in the library explorer window, an appropriate license is required.

At the root of the workshop, you can find the components used most often for simple circuits. At the root of each component family of a workshop, there are components with separate mathematical model.

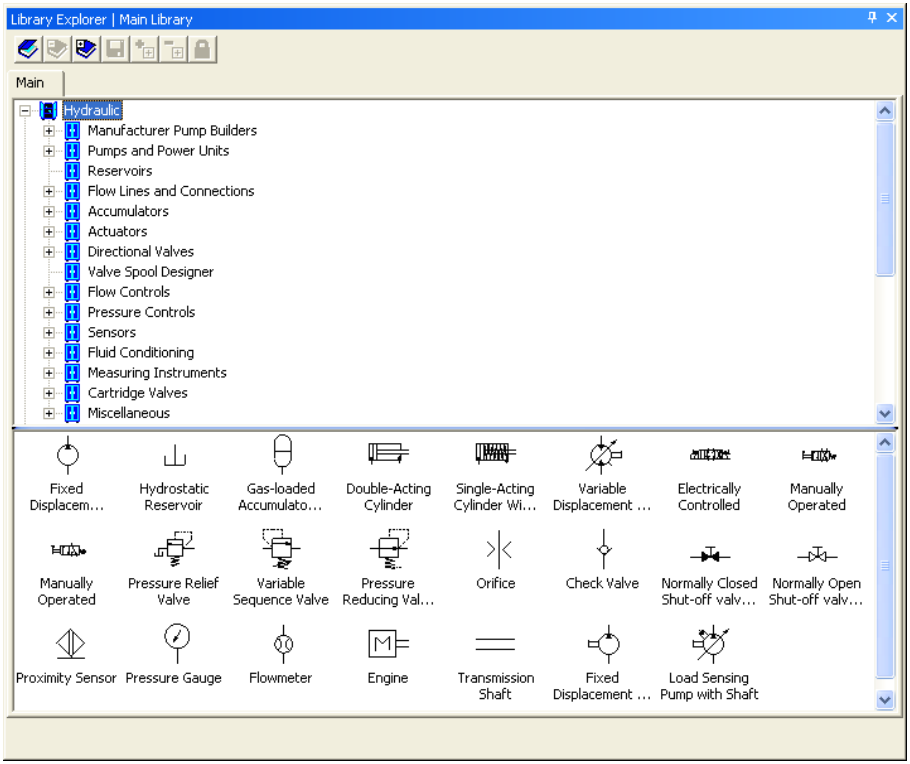


Figure 4–1: Example of "Hydraulic" Library Structure

The pneumatic library has some difference, for example, it includes sequencers, timers, and logic valves but it does not have poppet valves.

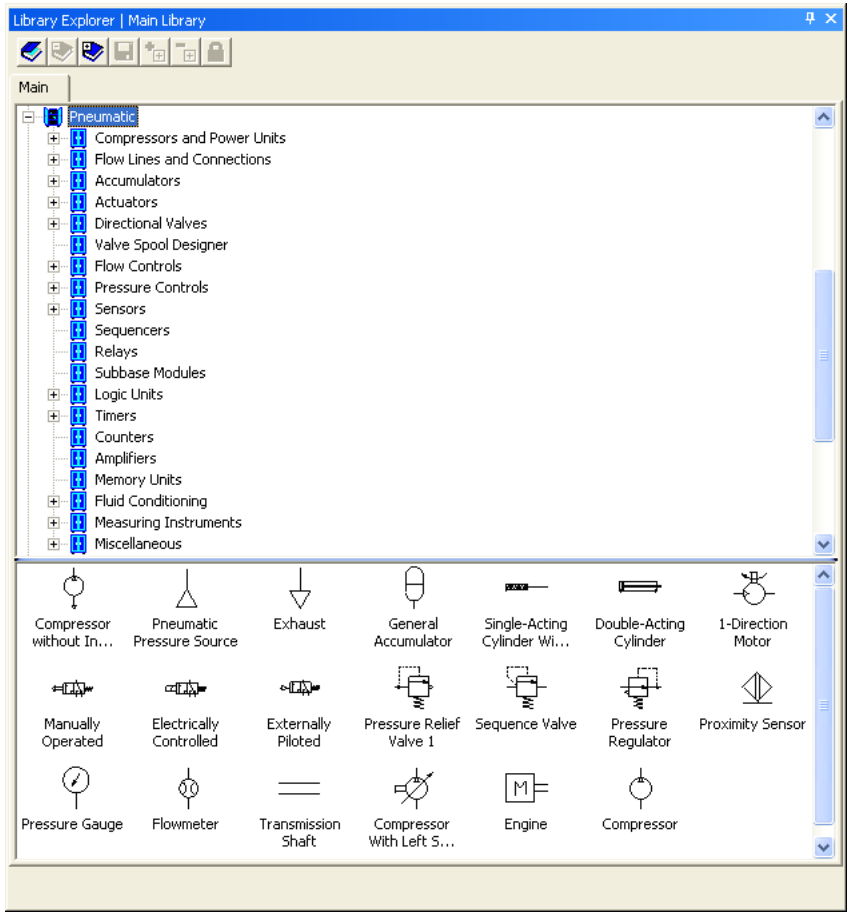


Figure 4-2: Example of "Pneumatic" Library Structure

The proportional hydraulic and pneumatic workshop does only include components with continuous or infinite displacement, transducers and proportional control devices.

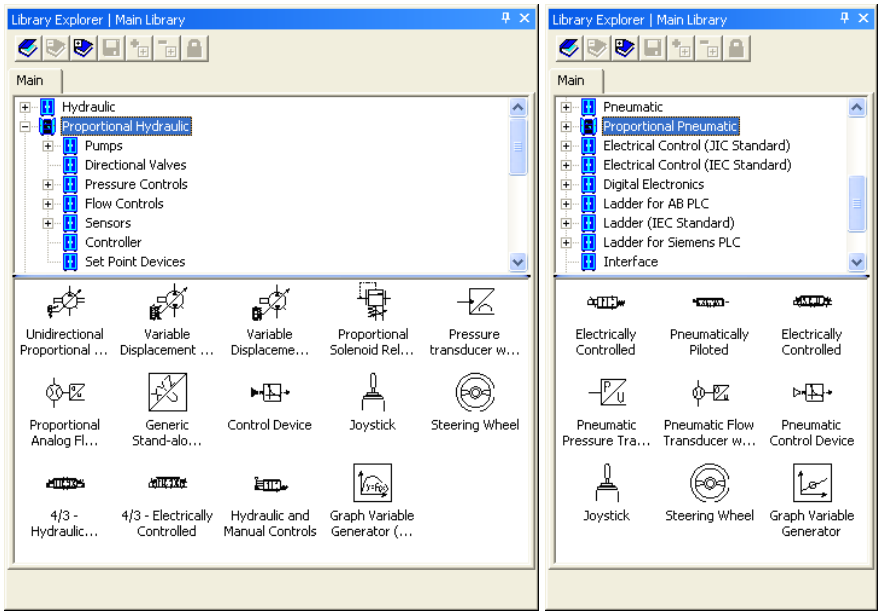


Figure 4-3: Example of "Proportional" Library Structure

You can draw your own symbols, based on a collection of elements recommended by standards ISO 1219-1.

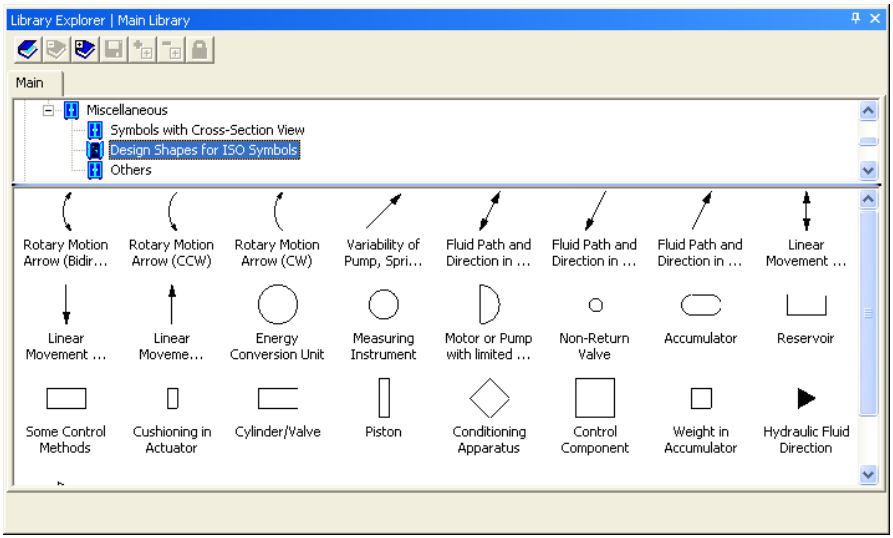


Figure 4–4: Symbol Elements ISO 1219

In order to satisfy all requests for new symbols, we have developed a new feature in the main library, without overcharging it. This feature simplifies research of symbols and differentiation of simulation models.

This feature can be visualized in the component zone. For all symbols in the library which are represented on a blue rectangle (master symbols) the user can be accessed by double clicking to a floating window containing similar symbols. Depending on your preferences, you can replace the master from the similar components after selecting the new representative and click on "Replace" in the similar symbol window.

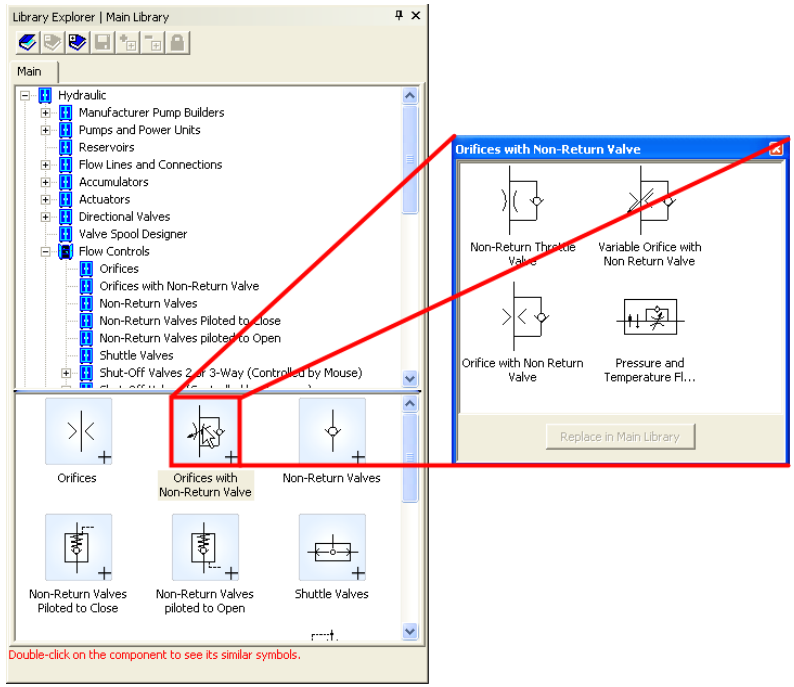


Figure 4-5: "Library Explorer" and Similar Symbols Windows

- A similar symbol has the same simulation behaviour than the master component.

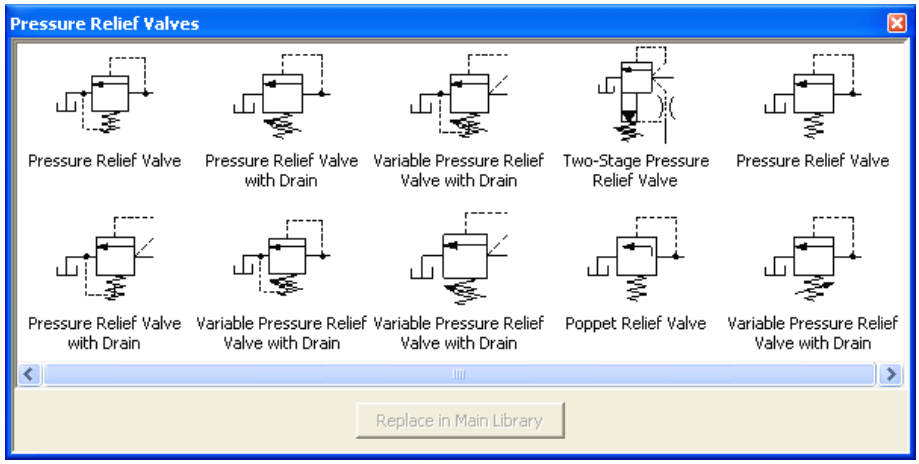


Figure 4-6: Example of Similar Symbols

When a family branch is extended, the 1st branches correspond to similar components if any, and other components with additional options. (Ex. Fixed displacement pump with 1 or 2 shafts, with or without drain, …), or performing a similar function (driveshaft, clutches, and box speeds). All components in a family root without option can be found in the "Other" family.

Personalised Libraries

The personalised libraries that are included in the "Libraries" directory in the installation directory of Automation Studio™ contain many components already assembled.

4.3.1 Manucaturer Pumps

These manufacturer's pumps are only available in the professional version. They are available from the hydraulic main library under the manufacturer's name. When a pump is inserted, the builder automatically opens to offer all models for this specific type of pump. The pump's behaviour in simulation is reproduced based on the manufacturers' specifications as found on their WEB site. Refer to section *5.4 Pump Builder* for builder details.

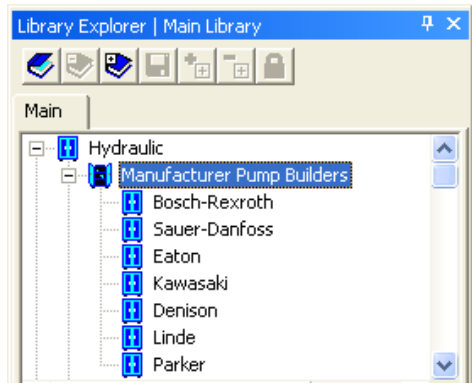


Figure 4-7: Manufacturer Pumps Branch

4.3.2 Pumps and Power Units

This library branch regroups all the power elements such as pumps and pressure sources, and also transmission components such as motor, shaft, clutch and gearbox.

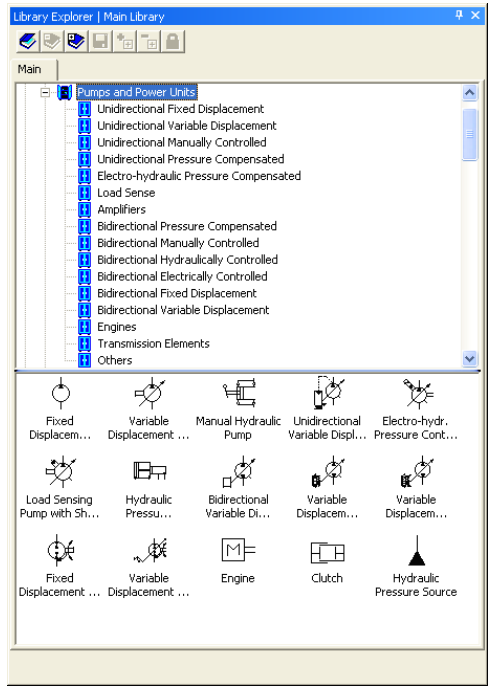


Figure 4-8: Pumps and Power Units Branch

4.3.2.1 Pumps

Pumps are defined with technical data such as displacement or nominal speed. Depending on the selected model other data may be required, such as the pressure setting. When a drain is present, the "Drain Leak" is defined as a flow per pressure unit.

Pump Types

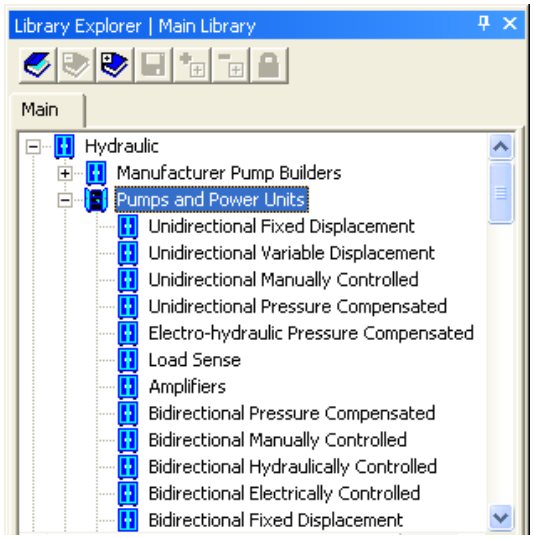


Figure 4-9: Pumps types

Amongst these pumps, it is possible to change the displacement during simulation, just by clicking on the pump. For pumps without transmission shaft, the flow provided is calculated by the technical data and its volumetric efficiency by the characteristic curves of the pump.

For most pumps in Automation Studio™, including the compressors, especially the unregulated ones, the flow supplied by a pump is given as a function of the differential pressure measured between its input and output connection ports. A package of four curves is supplied by default. The possibility to adjust the pumps' displacement with a simple click during simulation justifies the use of several curves, since the simulator calculates the theoretical curve of the new displacement by interpolation of the two envelop curves.

☞ If you increase pump displacement beyond the higher flow curve, you have to adjust it with a flow parameter higher than the calculated flow in the technical data. Without that, interpolation is incorrect.

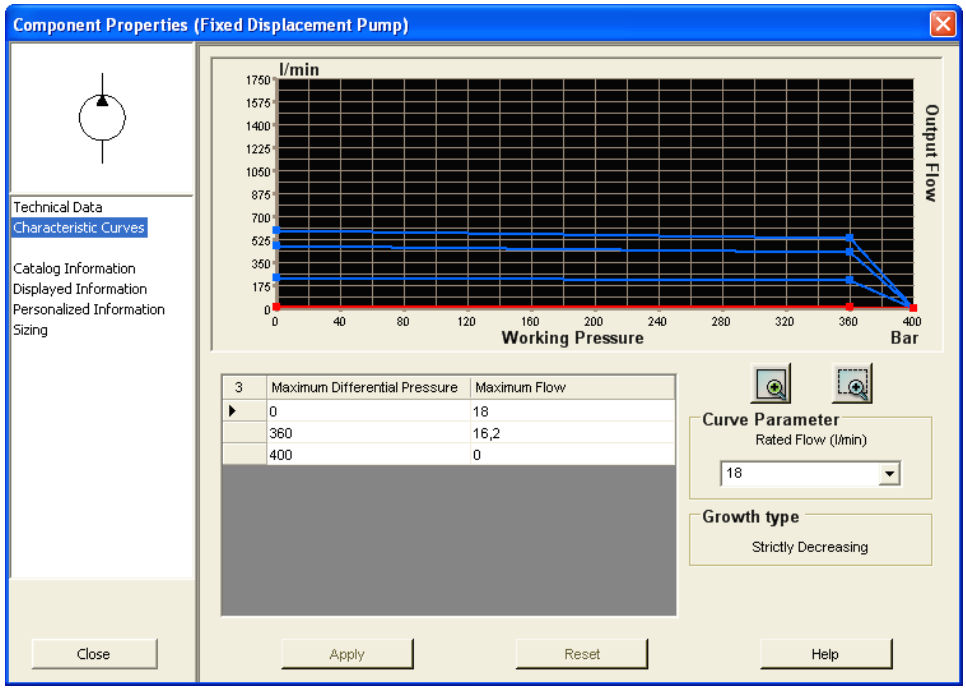


Figure 4–10: Unregulated Pumps Behaviour

The curves can be adjusted to reflect a specific behaviour (power compensation, etc.) or an incorrect behaviour

The pumps with a transmission shaft, or those compensated, do not have characteristic curves. It is the driving motor or the regulation type that modifies the speed as a torque function and, it is the motor power drive that will govern the calculation of delivered flow by the pump.

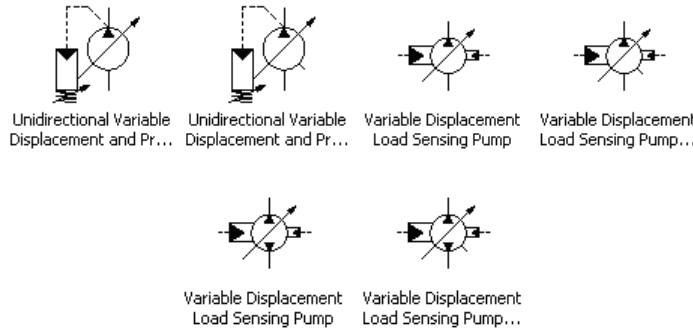


Figure 4-11: Compensated Pumps

A parameter representing the compensated pressure can be added for these pumps.

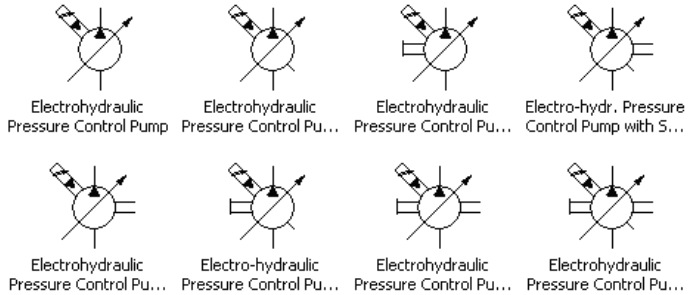


Figure 4-12: Electrically Compensated Pumps

For these pumps, the compensated pressure value depends on the signal.

Scaling up is done by giving the minimum and maximum signals measured in the circuit (example: $Signal_{min}$ = 0.4 A and $Signal_{max}$ = 1.2 A) as well as minimum and maximum compensated pressure (example: P_{min} = 20 bar and P_{max} = +400 bar). On the other hand, the minimum compensated pressure corresponds to the maximal signal.

If the signal is outside the extreme values, then the compensated pressure will acquire the extreme value (example: if the measure is 0.1 A, then the compensated pressure value will be 400 bar).

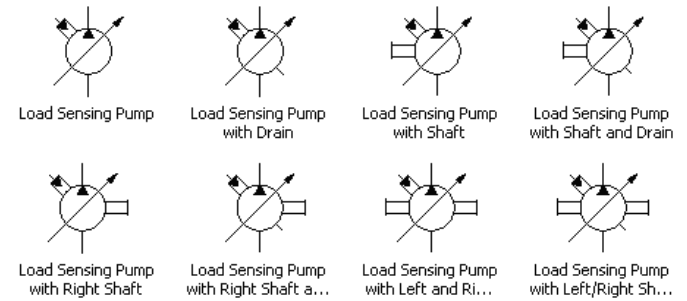


Figure 4-13: Load Sensing Pumps

For these pumps, the pressure setting (compensated pressure) and the differential pressure setting (between pilot and output ports) can be edited in the technical data.

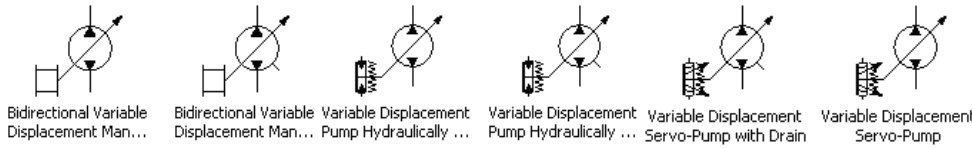


Figure 4-14: Variable Displacement Pumps (Manual, Hydraulic, Pneumatic)

These bidirectional pumps are On/Off pumps. The flow is zero or maximum (in a direction depending on the command).

Pump Type	Command
Manual	Mouse click
Hydraulic	Pressure measure
Electric	Boolean variable

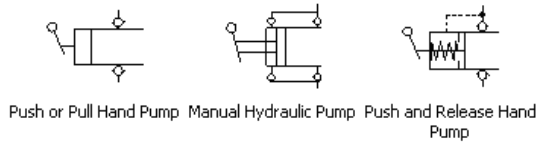


Figure 4-15: Manual Pumps and Similar Symbols

These pumps can be activated by a mouse click, and modelization is the same as for unidirectional pumps without shaft.

For the other regulated pumps refer to the section *5.4 Pump Builder*.

4.3.2.2 Transmission Elements

In order to profit from power flow mechanical calculations, developed in the version 5.5, the transmission shafts, thermal and electrical motors, as well as measure instruments such as tachometers and torque sensors must be replaced in previous folders created under versions prior to version 5.5.

By default, pumps are always driven in rotation, but it is also possible to mechanically link them to electrical or thermal motors. When doing so, the pump driving speed takes the value of the motor speed.

Engines

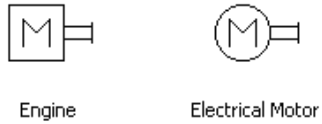


Figure 4-16: Engines

The thermal and electrical motors are standalone components. However for each of the thermal and electrical motors, it is possible to assign two real variables in order to control the speed (receptor) and to write a torque variable (transmitter).

It is also possible, in the hydraulic and pneumatic IEC and NEMA standards, to assign, for each motor, a real variable to control the torque (receptor).

It is equally feasible to modify the speed manually, if the adjustment mode is chosen by the user.

Note that motors of electrotechnic workshop can receive external torque from a fluid simulator. This allows you to calculate the consumed current.

Mechanical power transmission is determined by the transmission component such as gearboxes, shafts, clutches. The simulator computes the torque, and the speed is derived from the conservation of energy laws.

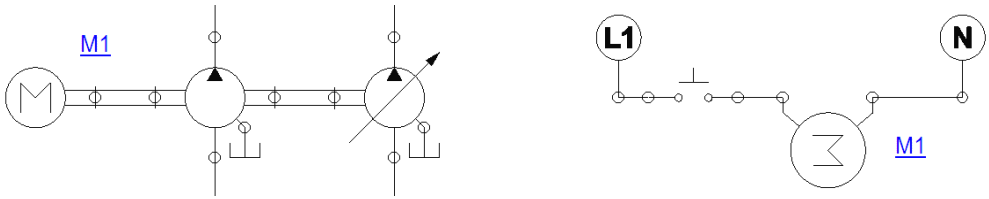


Figure 4-17: Hydromechanical Coupling Example

You cannot connect two motors on a same transmission shaft. For instance, on the next figure, only the left motor is considered (M2 is not considered).

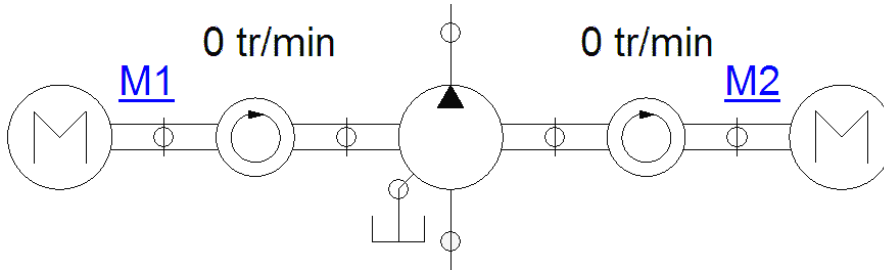


Figure 4–18: Two Motors on a Same Transmission Shaft

A model for a thermal or electric motor is based on a curve that relates maximal speed to power.

For a speed ω, the maximum corresponding power is $(\omega \cdot P_{max})/ \omega_{max.}$.

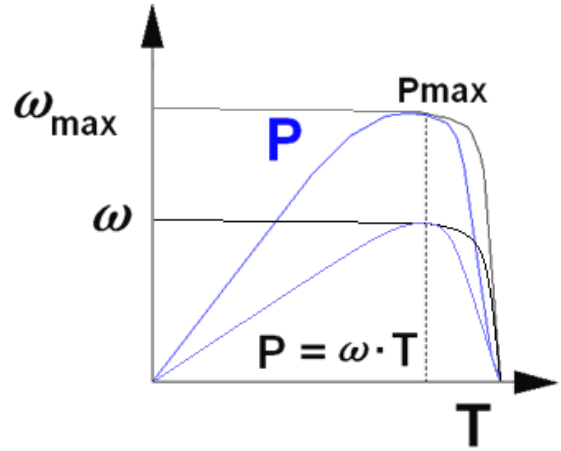


Figure 4-19: Modelisation of a thermal/electric motor

Transmission Components

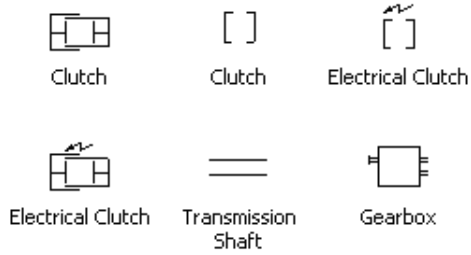


Figure 4-20: Transmission Components

The simulator calculates mechanical power transfer for transmission components as gearboxes, shafts, and clutches. The torques are calculated and speeds vary depending on energy conservation law.

For each clutch, it is possible to assign a Boolean variable in order to engage or disengage transmission with a control variable.

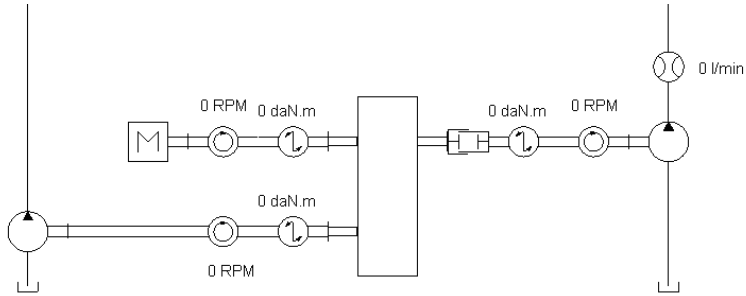


Figure 4-21: Example of Hydromechanical Coupling

If more than two mechanical connections are overlapped, the simulation cannot start. The faulty components will be listed in the message window with their identifiers.

4.3.2.3 Amplifiers and Others

Transformers are devices to amplify or convert power. These components can be found in the power components of hydraulics and pneumatics workshops.

☞ However, for the components that convert pneumatic energy into electrical energy, you have to possess two licences. You can find them in the pneumatics workshop.

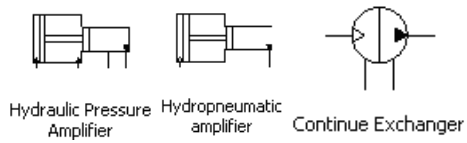
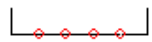


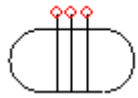
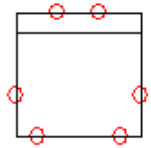
Figure 4–22: Amplifiers and Others

4.3.3 Fluid Storage

For pneumatic systems, air comes from the ambient atmosphere, which has an unlimited supply of air or from an accumulator which fills up from the ambient atmosphere.

For hydraulic systems, oil is held in a tank. Tanks available in Automation Studio™ are defined by their volume, internal pressure and temperature. Following are the five available tanks:

Name	Symbol	Volume	Temperature	Pressure
Atmospheric Reservoir		Infinite/Finite	Constant (ambient)	Atmospheric
Hydrostatic Reservoir		Infinite/Finite	Constant (ambient)	Hydrostatic
Atmospheric Reservoir	 Configurable port quantity	Infinite/Finite	Variable Temperature	Atmospheric
Reservoir		Finite	Variable Temperature	Hydrostatic

Name	Symbol	Volume	Temperature	Pressure
Réservoir sous pression		Fini	Température variable	Pré charge
Pressure-sealed Reservoir *Refer to section 5.5 Reservoir Builder.*		Finite	Variable Temperature	Pre-charged

To simulate a hydrostatic pressure, the reservoir is located under the branch “Flow Lines and Connections”.

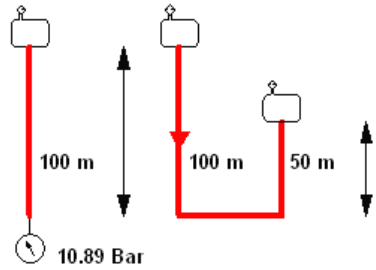


Figure 4–23: Hydrostatic Pressure and Transfer

Using a reservoir is the same as assigning an elevation to a tank. The hydrostatic pressure in a circuit is provided by the reservoir. If multiple reservoirs are used, fluid transfer may happen.

It is possible to assign a finite volume to a reservoir from the properties dialog box.

The thermal fluctuation in a fluid is calculated as follows:

- For finite volume reservoir: the initial volume and then the actual volume is used;
- For infinite volume reservoir: the residual volume is used for atmospheric reservoir, and the initial volume for the pressure-sealed tank;
- The surface considered for calculation of the reservoirs’ thermal dissipation, is equivalent

to a cube, whose volume is the reservoir's "total volume".

The pressure read at the reservoir's connector always corresponds to pressure at a zero altitude.

Refer to section *5.5 Reservoir Builder*, if you want a reservoir with multiple orifices at different heights or use level, flow or pressure sensors.

4.3.4 Flow Lines and Connections

4.3.4.1 Fluid Lines

The fluid lines are the only components that do not have symbol in the main library because they can be drawn directly on a diagram (Refer to Automation Studio™ User's Guide for more details). However, once represented on the diagram, the properties dialog box is accessible the same way than for other components.

Except the visual configuration (pressure lines, pilot lines, outlet pipes or customized lines), every line can be given specific technical properties. This is done through the "Component Properties" dialog box, as shown below.

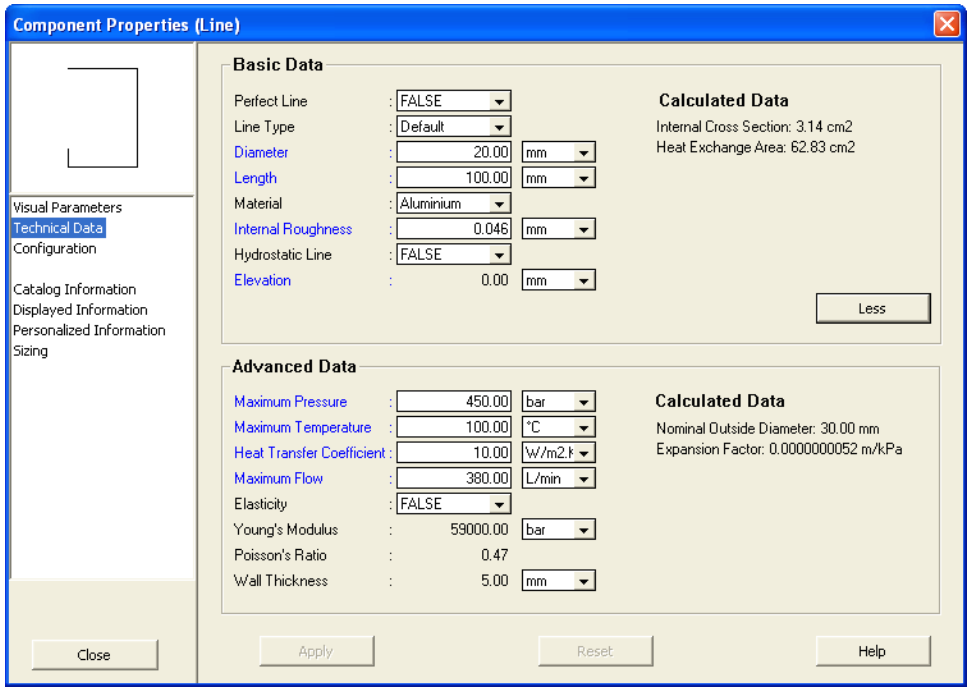


Figure 4-24: "Component Properties" Dialog Box for Lines

For an explanation of most of the fields in the above figure, refer back to section *2.2.3 Line Management*. The explanations will be completed later.

It is recommended to respect the ISO Standard when editing a circuit. (Refer to section *2.1 Configuring Installations and Circuits* to see advantages). However, you can modify the appearance of lines in the properties dialog box.

Lines Simulation

Perfect Fluid Line

Very often, in the product catalogues, the symbol that emulates behaviour is composed of simple components and fluid lines.

In order to easily match technical specifications from a catalogue with the components of Automation Studio™, it is now possible to declare locally one or several lines as being perfect. When a line is perfect, it does not generate any losses, does not influence viscosity

and does not create or evacuate heat.

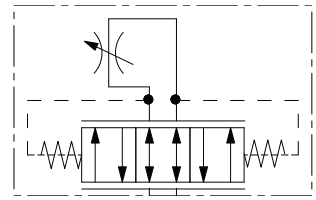


Figure 4-25: Example of Component, with Restriction, Lines, Pilots and Directional Valves

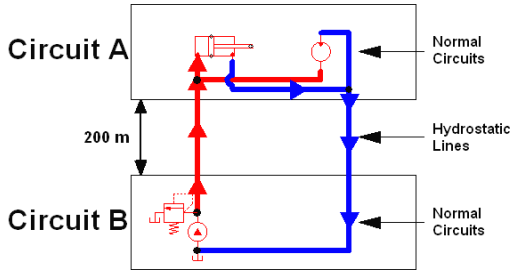
When this option is activated, the technical data is shaded and is not interpreted by the simulator. The line behaves as two jump-to lines looking like the original line.

☞ The “Perfect Lines” option in the “Project properties” remains global. If this option is not active, the pressure drops are evaluated in the project, while a fluid line could be declared perfect locally.

Hydrostatic Line

It is possible to assign an elevation to lines (accessible in the advanced technical data). This is equivalent to placing the circuit in the Earth's gravitational field, which value is adjustable in the simulation general options. Only pressure drops are balanced, and a flow becomes necessary to calculate the gravitational influence. For lines, if the elevation helps the fluid flow, the hydrostatic value is subtracted to the pressure drops, and if the hydrostatic pressure works against the fluid flow, the hydrostatic value is added to the pressure drops.

☞ The simulation of hydrostatic lines is costly in computing time. For this reason, it is strongly recommended to use this option for significant elevation differences only. For example, in the following figure, it is recommended to use normal lines for the A and B circuits, and hydrostatic lines for the 200 m lines between these circuits.



This elevation will be considered only if the hydrostatic line option is chosen.

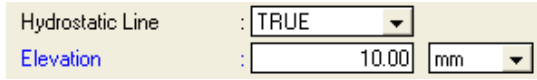


Figure 4-26: "Hydrostatic Line" Option

Zone	Description
Hydrostatic Line	Specify whether to use the "Elevation" data.
Elevation	Specify the height difference in the line.

This option is only available for hydraulic lines.

When drawing a hydraulic (pneumatic) line, the first connector is identified A(1) and the second B(2); if the elevation is positive, the line is shown as rising from A to B (1 to 2). If the elevation is negative, the line is shown to lower from A to B (1 to 2).

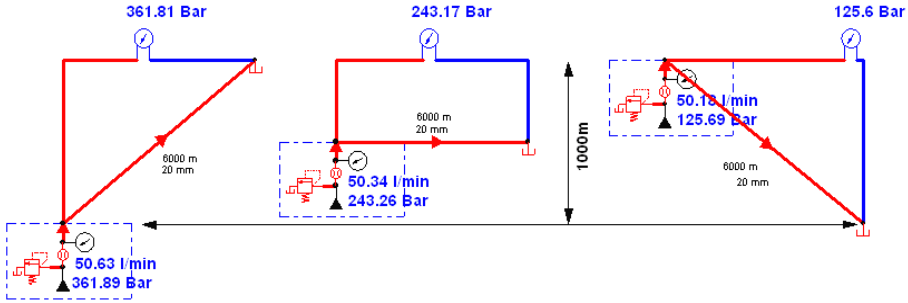


Figure 4–27: Elevation Effects on Lines

Elasticity

An option "Elasticity" is available in the "Component Properties" dialog box (hydraulics and pneumatics). It is possible to specify the dilatation of the line as a function of pressure. This information is only used when the "Elasticity" option is checked.

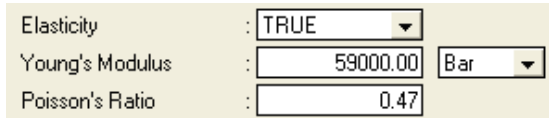


Figure 4-28: "Elasticity" Option

Zone	Description
Elasticity	Specify whether to use the elasticity information.
Young's Modulus	Specify the Young's Modulus for the line material in order to characterize line extension in longitudinal direction as a function of the effort applied on the pipe by the pressure
Poisson's Ratio	Specify the Poisson's Ration for the line material in order to characterize line contraction in perpendicular direction as a function of the effort applied on the pipe by the pressure.

Other Properties

Pressure drop in a line is influenced by properties such as diameter, length, material and roughness.

Validity of Lines

- Diameter must be greater than 1 mm;
- Length must be greater than 10 mm;
- Length must be greater than diameter (ideally at least 5 times greater).

It is possible to set the *Kv* or *Cv* flow index to model pressure drops associated with connections. The default value is very large to minimize the drops. This coefficient corresponds to the equivalent flow coefficient of the line's ends. It doesn't depend on the fluid used.

There will always be fluid in lines even if there is suction and a resulting depressurization is calculated. You must use the component displayed functions indicating the lines status.

Lines with a flow lower than 6 l/min (1.59 GPM) does not display the flow direction.

Assigning a Line Function

When you access to a hydraulic (or pneumatic) line shortcut menu, you can choose between 6 (or 5) distinct line types. When you change the line type, the properties, diameter and length in particular are re-initialized to the default values of the project.

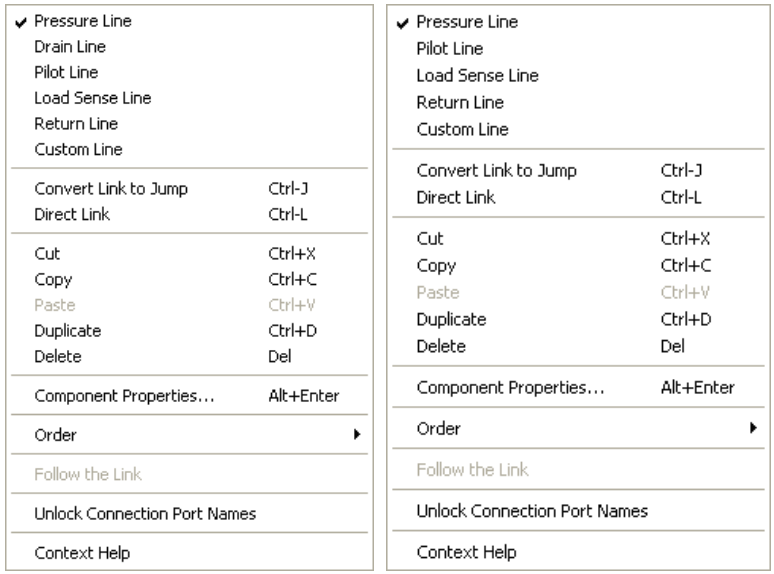


Figure 4–29: Hydraulic and Pneumatic Lines Shortcut Menu "Lines Function"

This control is found in the Diagram Editor for hydraulics and pneumatics.

The visual properties of the lines can be modified through "Project Properties" → "Options" → "Fluid Edition".

The default representation for the lines can be modified for each workshop.

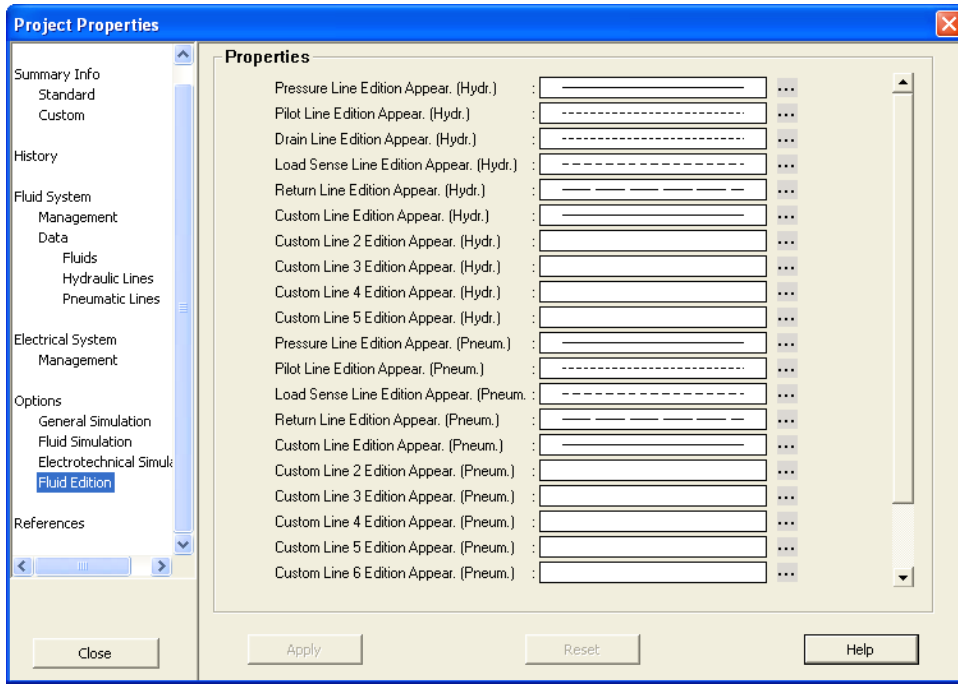


Figure 4–30: Visual Properties for the Lines

It is now possible to add specific forms to line ends (arrows, dots, etc.). The visual parameters are available in the line properties dialog box.

If a line is cut in two, both sections will have its own properties. It is possible to change the representation of several lines at a time by selecting all lines and accessing the lines' dialog box.

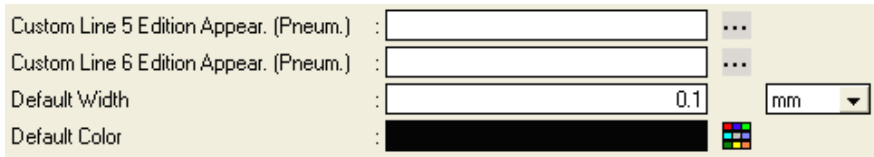


Figure 4-31: Symbol Appearance Configuration

In order to standardize components' layout, as it can be done for the fluid lines, you can specify line thickness and color of symbols inserted from the library to the diagram. This option is global to the project.

Converting Links to Jumps

This option is available from the line shortcut menu (or CTRL+J) and allows to visually simplifying the diagrams by converting a line into two jumps within the same diagram.

Turning a line into a link results in a line without pressure drop.

To create two jumps between two diagrams, the links provided in the library under “Flow Lines and Connections” must be used. The same tag name must be assigned in order to connect two links.

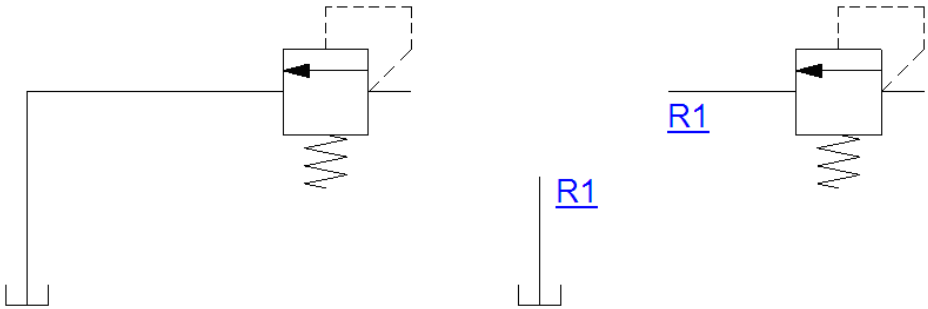


Figure 4–32: Jump Format with Hyperlinks

Automatically Processing Intersections

When a 3-way (or more) junction is created, a black connector is automatically created as required by ISO standards.

4.3.4.2 Connections

In the “Flow Lines and Connections” branch, there are components allowing to connect or to block fluid lines.

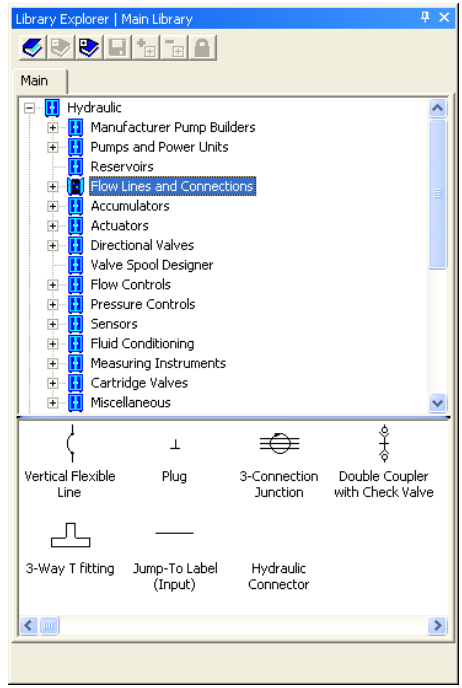


Figure 4-33: Connection Components

Other Connections

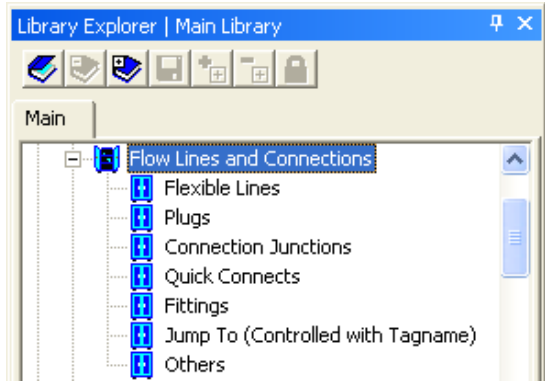


Figure 4-34: Other Connection Types

4.3.5 Accumulators

This library branch regroups the 5 accumulator types that exist in Automation Studio™.

Accumulator technical data vary according to the type of accumulator.

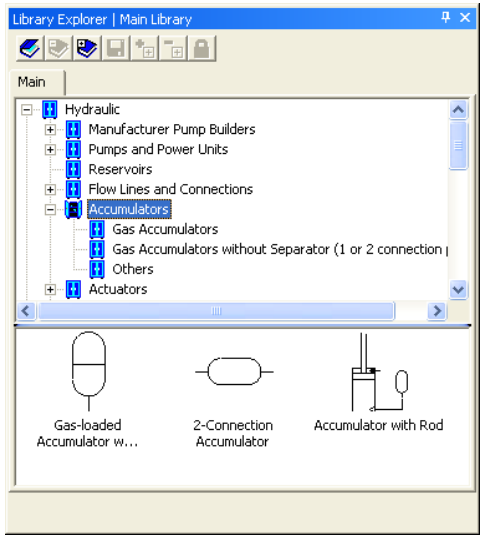


Figure 4–35: Accumulators

4.3.5.1 Gas-loaded Accumulator

The preload pressure for the gas-loaded accumulators is the minimum pressure that must prevail in the circuit for the accumulator to begin charging.

The accumulator is automatically loaded with oil up to 5% of its total volume. This oil quantity ensures stability at the beginning of the simulation.

The gas transformation is considered as isothermal and the gas is considered as perfect. While the accumulator is loading and unloading, the simulator considers the flow coefficient of the connection ports.

4.3.5.2 Other Accumulator Types

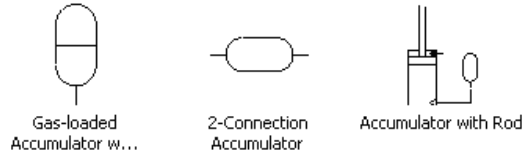


Figure 4-36: Accumulator Types

Weight-Loaded Accumulator

During its loading, the pressure will stay constant and equal to the "Mass" weight divided by the accumulator's section area as a diameter function.

Spring-Loaded Accumulator

During its loading, the pressure will increase proportionally to the "Spring Stiffness" coefficient.

Accumulator with Rod

This accumulator has the same behaviour as the gas-loaded accumulator, except for the sensors that can be declared cylinder sensors.

4.3.6 Actuators

Linear or rotary actuators can all be found in the actuator library branch. Linear actuators are grouped by type. To change the cylinders' properties, refer to section *5.1 Cylinder Builder*.

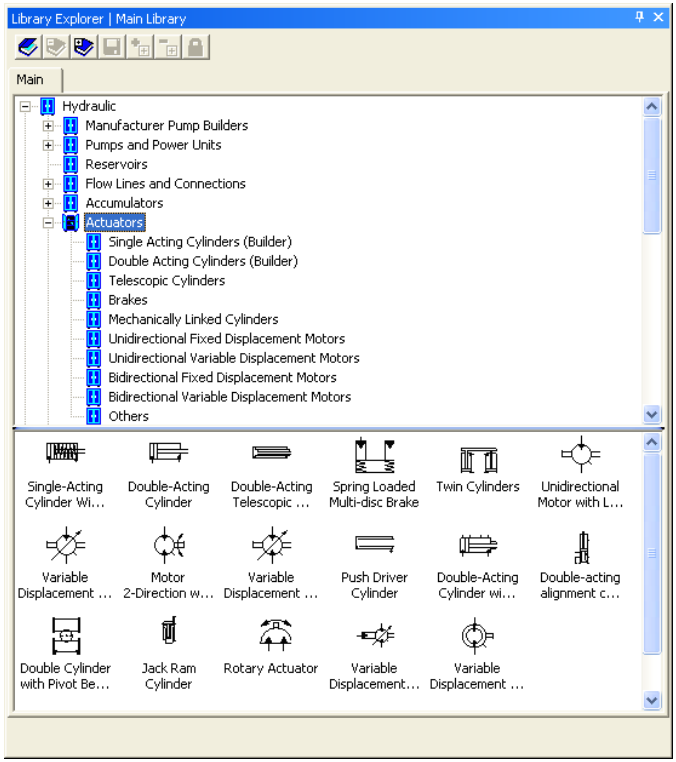


Figure 4-37: Actuators Components

Actuator Modeling

The actuator's behaviour complies with the equation of rotation or translation mechanics of solids. Mass and inertia can be adjusted from the basic or advanced technical properties.

Linear and rotary actuators have a default inertial mass. A minimal displacement must be provided. Depending on the technology used, Automation Studio™ might compute the displacement, as follows, for stability considerations:

Minimal inertia for pneumatic motor is 0.01 $Kg.m^2$;

Minimal inertia for hydraulic motors depends on the displacement:

- Displacement>=10000 cm^3: minimal inertia = Displacement (cm^3)/300.0 $Kg.m^2$;

- Displacement >= 200 cm^3: minimal inertia = Displacement (cm^3) /600.0 $Kg.m^2$;
- In other cases, minimal inertia = 0.01 $Kg.m^2$.

On the other hand, the minimum cylinder diameter is 5mm. The simulator is not calibrated for very small volumes. Beyond this limit, the extension and retraction velocity, the flowmeter and the pressure gauge measures can deviate from reality.

For all types of actuators, two external forces are present, driving forces helping movement, and resistive forces opposing movement.

Each force can be steady, periodic, or changing.

Periodic Forces

Resistive or driving, periodic forces can be modeled using a curve providing the force versus the linear or angular position. For cylinders, the period is an extension from 0 to 100% and for the retraction from -100 to 0%. For motors, the period is defined.

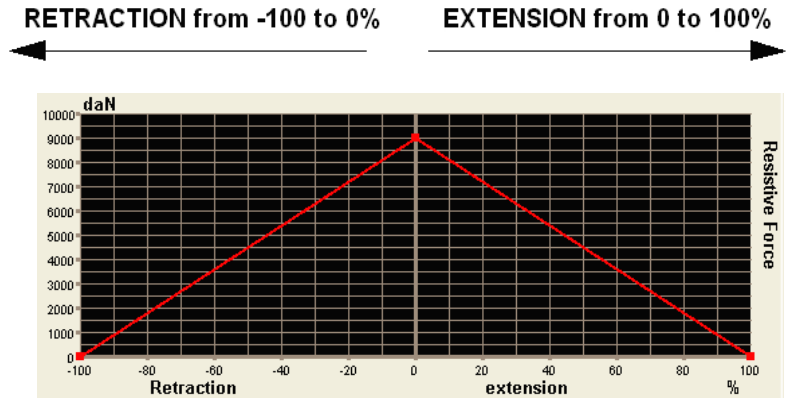


Figure 4–38: Cylinder Resistive Force

In the figure above, the force is resistive (>0); it is decreasing during the extension (from 0 to 100%) and increasing during retraction (100% to 0).

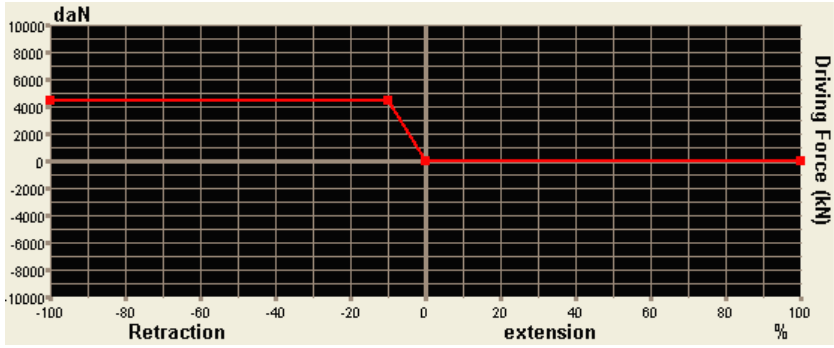


Figure 4–39: Cylinder Driving Force for One Direction

In the example above, the driving force is applied in one direction only. The driving force is applied only during the retraction period. During extension, the driving force is null.

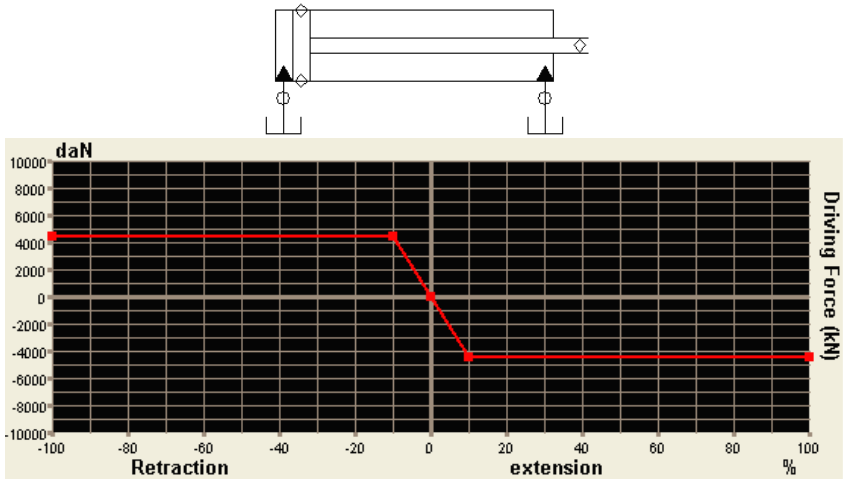


Figure 4–40: Cylinder Driving Forces for Two Directions

In the above case, the driving force is applied during both phases of movement and the cylinders will self-propel due to the dynamic forces.

Motors have the same curves. The only difference is that the movement period must be defined in the motor advanced data (Angular measurement corresponding to number of turns).

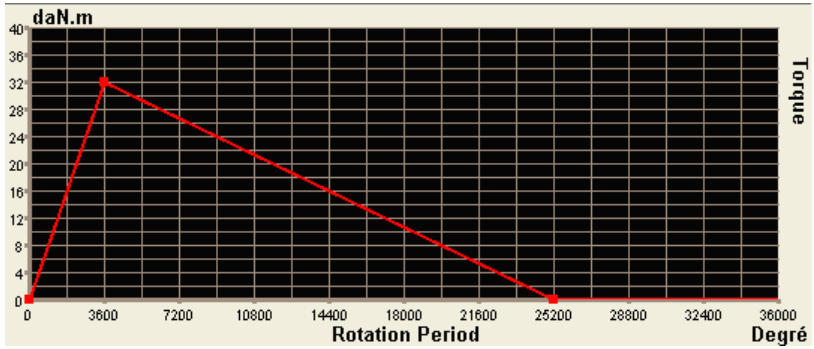


Figure 4–41: Resistive Torque over 1000 Turns

The above curve applied to a bidirectional motor is read for 0 to 1000 turns when the motor is turning clockwise, and read from 1000 to 0 when the motor is turning anti-clockwise. The following figure illustrates the symmetry between the pressures at the motor's connectors, based on the direction of rotation.

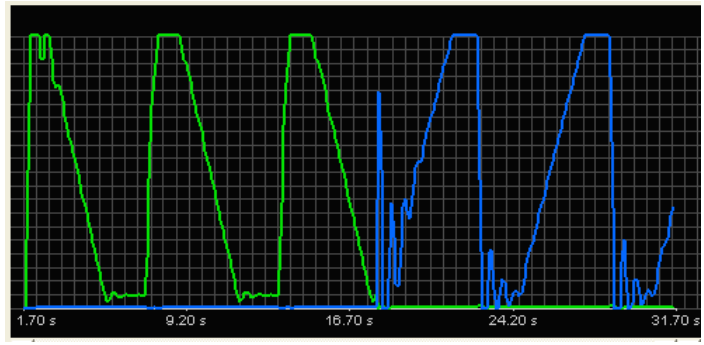


Figure 4–42: Changes in the Direction of Rotation at t ≈ 17s

Variables Forces

They are supplied through a variable which is provided from a control device (wheel, joystick), measuring device (transducer), an external variable defined the SFC or Ladder workshop or from an OPC Server. The proportional hydraulic generator, generic sensors, association of controller can be used for this purpose.

- Force: units are in kN (Kilo Newton);
- Torque: units are in kN·m (Kilo Newton·meter).

Fixed Forces

They can be tuned from the technical data. The push and pull forces are resistive and do not have an equivalent for motors. For the driving force, a mass and the angular position can be specified for the cylinders, while for motors, you can find an external torque.

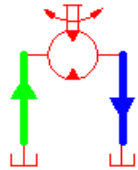


Figure 4–43: Effect of a Fixed External Torque.

Actuator Connectors

The bigger the cylinder input flow is, the larger is the pressure drop at the connector port. For cylinders with different diameters, connected in parallel, the applied forces must be larger than the amount of lost forces due to pressure drop at the ports for the cylinder with the largest diameter to extend first.

This condition is even more apparent in pneumatics, because larger flow is required due to the compressibility. When the diameter is very large, the force required to noticeably raise the pressure is proportional to the surface.

Cylinder Diameter (cm)	Pressure Drop (bar/l/min)
1	0.0034
10	0.00084
20	0.0002
30	9.25 E-5
40	5.2 E-5
50	3.4 E-5

Figure 4-44: Pressure Drops in Cylinder Ports

Below is the equivalent table for fluid motors.

Motor Diameter (cm)	Pressure Drop (kPa/l/min)
1	0.34
10	0.084
20	0.02
30	0.00925
40	0.0052
50	0.0034

Figure 4-45: Pressure Drops in Fluid Motor Ports

Cylinder Positioning

The resistive force curve is read every 0.01 second. If the piston velocity is high, it is highly possible that the piston position will pass the position where the first resistive force is applied before an effect can be seen. Also, the rod inertia (artificially larger than normal on small cylinders) must be null by the resistive force before the speed goes down to zero. Furthermore, the cylinder may not always stop at the same position if it is in a dense resistive force zone just before or after the reading. Hence some visual imperfections in the position.

4.3.7 Directional Valves

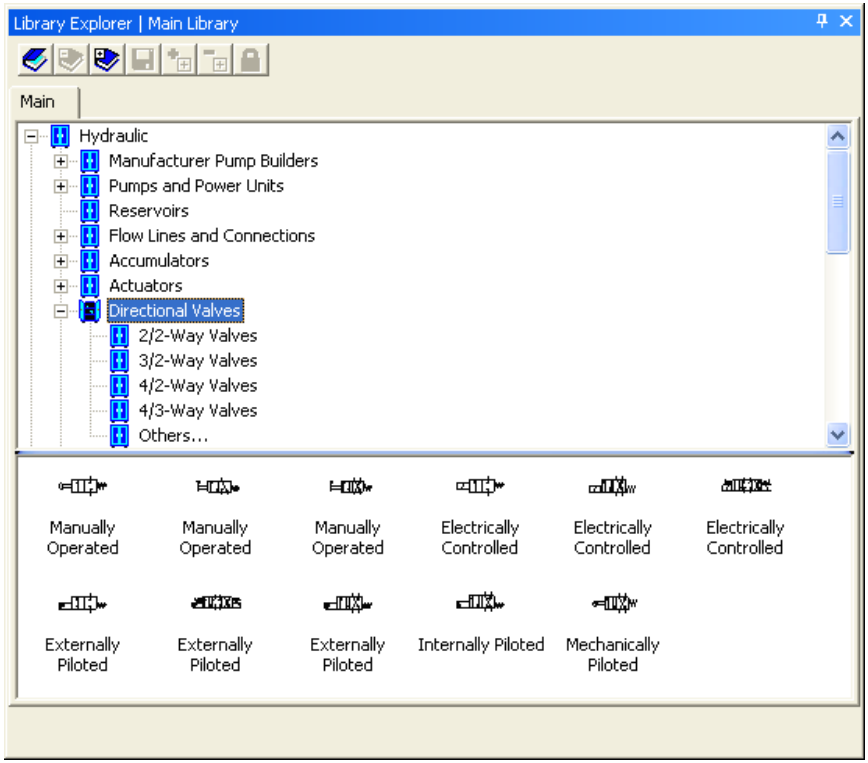


Figure 4-46: Directional Valves

4.3.7.1 Controls Configuration

Control Types and Properties

To access the full directional valves specifications, refer to the "Component Properties" dialog box. This section contains the control properties used by the simulator.

- Mechanical: Force *F*;
- Manual: Force *F*;
- Hydraulic: Surface *S* and simulator pressure variable *P*;
- Pneumatic: Surface *S* and simulator pressure variable *P*;
- Electric: Force *F* + Boolean Variable;
- Spring (PB with spring): *F* variable + (Boolean Variable);

- Position Sensor: Passive Control;
- Timer: Force + Delay;
- Thermal: Force *F*.

Introduction

Once a directional valve is included in a circuit, the software must know the valve's behaviour during simulation.

The valve's role is to direct fluid towards a chosen direction. To do so, the simulator requires knowing the valve spool position at every simulation cycle.

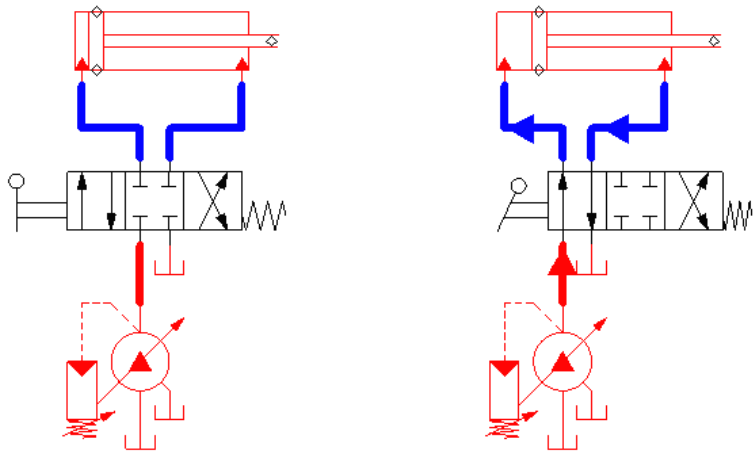


Figure 4–47: Directional Valve Implementation Example

Generally speaking, a directional valve receives two opposing control forces on each side of the spool. When one force is larger than the other, the spool moves.

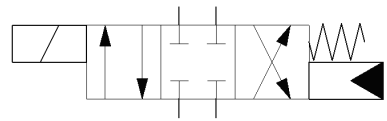


Figure 4–48: Directional Valve Configuration Example

Electrical force on the left – Hydraulic and/or mechanical force on the right

These forces are controlled through different types of signal. From the examples:

- Control on the left side: if the electromagnet is activated by a SFC, a Ladder diagram or an electric circuit, the force applied will vary according to the simulation output of these circuits (injected force) and could move the spool to the right.
- Control on the right side: if the hydraulic pilot is activated with a pressure, then a force will be applied on the spool based on the simulation parameters provided (pressure, surface). The spring, a mechanical control, provides a variable force whenever it is under caged or not.

Behaviour Calculation Rules

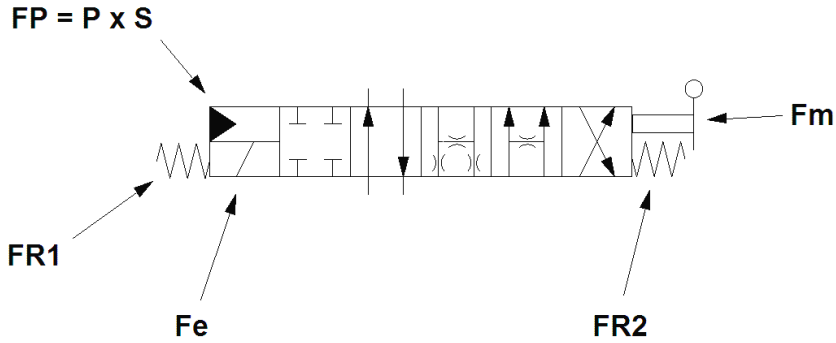


Figure 4–49: Forces Applied on a Directional Valve

A directional valve is exposed to lateral forces (generally opposed but not necessarily), *Fl*, the left force and *Fr* the right force. When they are different ($Fl\text{-}Fr \neq 0$), then the spool moves in the direction of the larger force.

Combining Commands in Parallel

The lateral forces are the sum of the parallel forces; each parallel force is the total of controls placed in series. Serial controls are a set of controls automatically placed one after the other by the valve builder. If a control is moved from its original position, the simulator will not take this new control position in its forces calculation. This means that if a control, initially in parallel to another, is moved to be in series, the simulator will still consider the valve as being in parallel. In the following figure, *F1*, *F2* and *F3* each generate a force: added together, they give *Fg*.

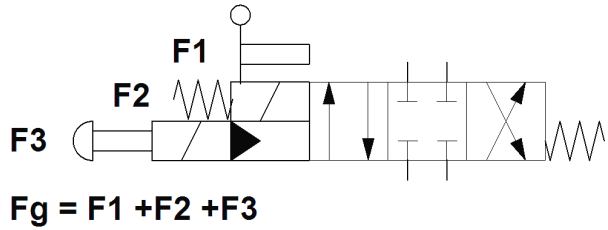


Figure 4–50: Example of Parallel Commands

Combining Commands in Series

If N controls are in series, only one is shown as being in contact with the valve, and according to the ISO standard, the control in contact with the valves applies the force while the other ones only activate the command. For the control force to be activated, all control commands must be active or true.

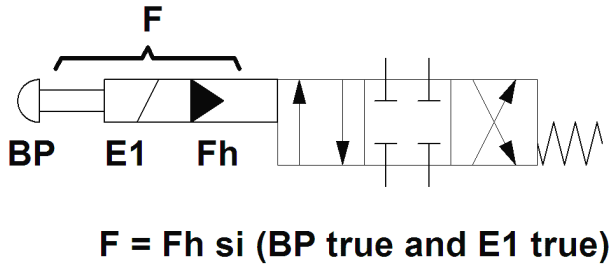


Figure 4–51: Example of Serial Commands

Resulting Behaviour of a Directional Valve

We can now find the total force *Fl* or *Fr* from the various applied forces. For configurations must be detailed:

- No Spring or Detent;

Nothing creates an opposition to a movement if $Fl\text{-}Fr \neq 0$, therefore the spool moves from one extreme end to the other.

Figure 4–52: Click on the Lever with Fm > FP

- Detent without a Spring;

In general, a detent mechanism is associated with the presence of a manual command. In this case, when the manual command is activated, the sum of *Fr* and *Fl* force will move the spool to the first detent position. Each new manual command activation will move the spool to the next detent position. The outcome of a click on the lever with *Fm > FP* followed by a second click is displayed in the following figure.

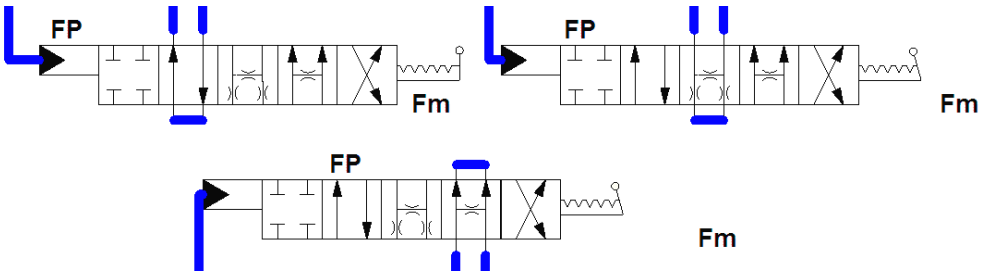


Figure 4–53: Resulting force without a spring with a detent device

- Spring without a Detent Device

The springs force changes with each new position. Therefore, the position is calculated in two steps:

a) Phase 1: The sum of forces is calculated (the current spring force is used) to find the displacement and direction;

b) Phase 2: The sum of forces is calculated at the new position (displacement size) and its stability.

Spring Forces Calculation:

The total force provided by springs is the sum of the force of each spring.

A spring always works when caged but it can also be configured to work under tension. This

is the default configuration which can be changed in the control configuration. When a spool can move, it meets with a resistance caused by all springs included in the directional valves.

Spring force: *FR* is the force set to a spring. It is the force required to be added to the present force balance to move the spool from position *N* to *N+1*.

A spring can be set to work in compression only or under compression/tension from the properties dialog box. An example of a directional valve that uses springs is shown in the next figure. If *F > FR/3* the spool gets to position *P1*; if *F > 2FR/3* the spool gets to position *P2*; if *F > FR* the spool gets to position *P3*.

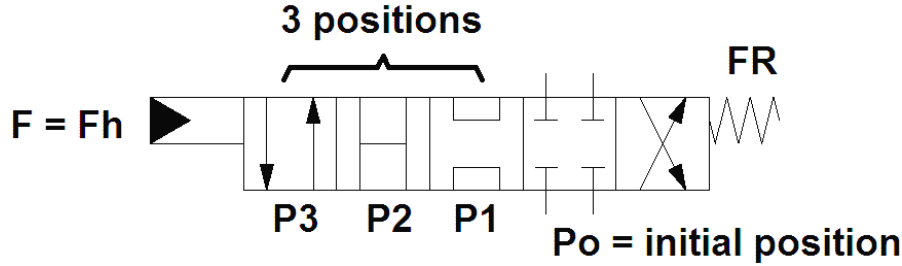


Figure 4–54: Resulting Force with a Spring Without a Detent Device

- Spring and Detent Device

The force equilibrium is done like the spring example, but the displacement is limited to the next detent position.

4.3.8 Spool Position Symbols

The "Spool Position Symbols" branch gathers symbols used to develop customized spool positions. Refer to section *5.8 Position Editor* for further details.

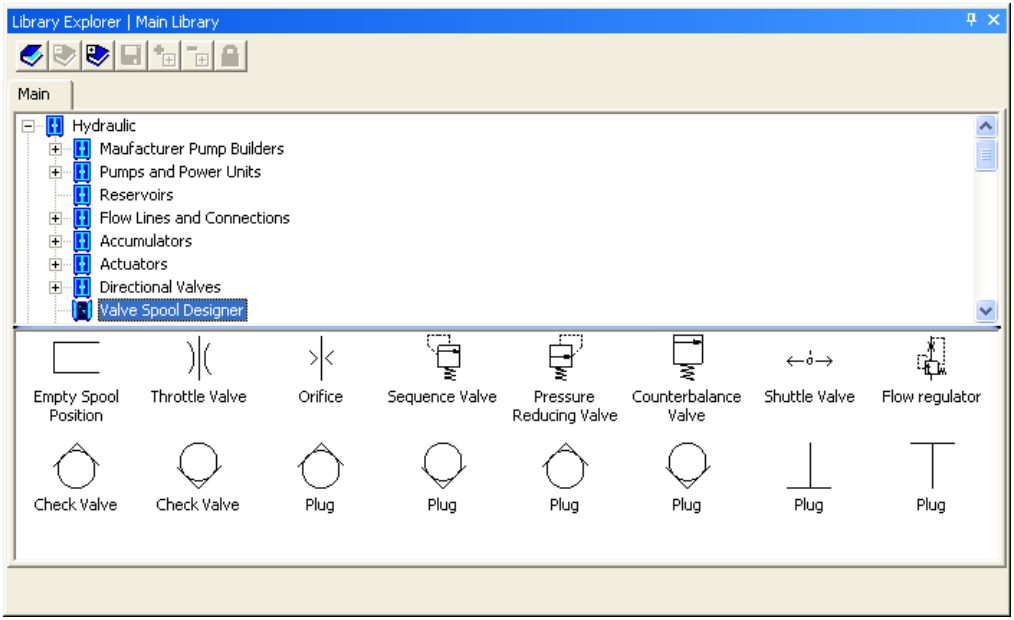


Figure 4-55: Spool Position Symbols

4.3.9 Flow Controls

Various flow control valves are available and most are common to both hydraulic and pneumatic environment.

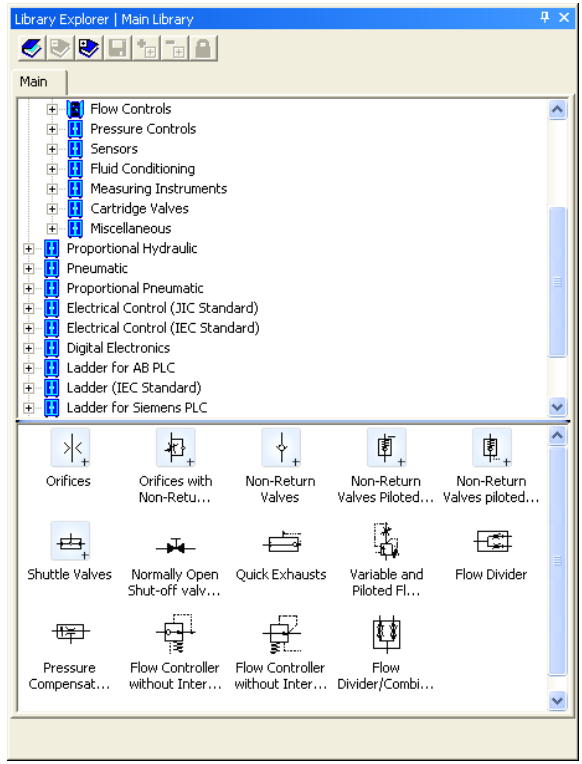


Figure 4-56: Flow Controls Components

4.3.9.1 Flow Restrictors and Orifices

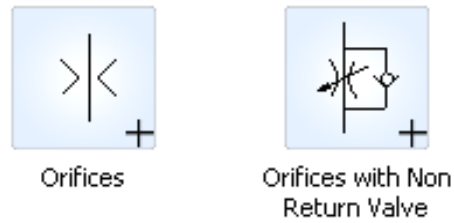


Figure 4-57: Flow Restrictors and Orifices Components

The flow restrictors do not control flow; it is only by making the connections differential pressure constant that the flow also becomes constant. The opening diameter can be adjusted.

The Bernoulli equation is used to model the restrictors. The thermal variations are immaterial

on diaphragm restrictors, the pressure drop will not change with the temperature.

The computation sheet for orifices and restrictors is the same. For an orifice, however, if the viscosity is changed, it makes sense to change the discharge coefficient which is accessible within the component properties.

4.3.9.2 Check Valves

Independently of the operational logic, the opening pressure on a check valve can always be modified. Furthermore, opening or closing the piloted check valves requires a control ratio between the piloted pressure and cracking pressure.

The pressure drops are calculated from the pressure versus flow curves.

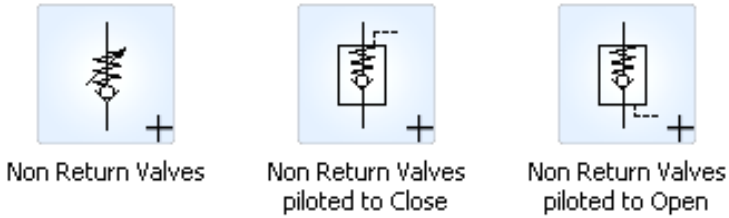


Figure 4-58: Check Valves Components

4.3.9.3 Shuttle Valves

The shuttle valve is configured just like the check valves.

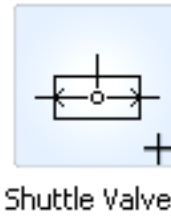


Figure 4-59: Shuttle Valves Components

4.3.9.4 Shut-Off Valves

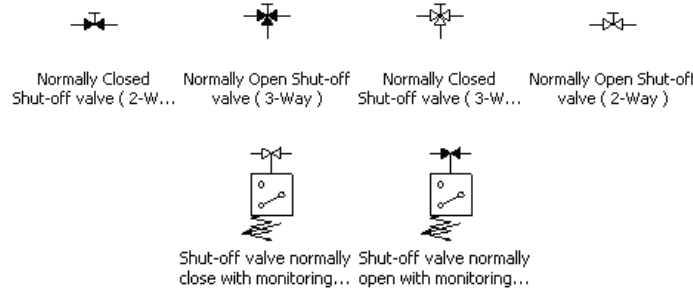


Figure 4-60: Shut-Off Valves Components

During simulation, the valve state changes with a mouse click.

The pressure drop is set with the flow coefficient. Neither unit system is favoured; the flow coefficient is *Cv or Kv.* This value is measured for an oil viscosity of 32 cSt. Refer to section *4.3.10 Pressure Controls → Pressure Drops* for model details.

For components controlled by pressure sensor, it is possible to read the state of the associated Boolean variable in order to use it, for example, in a control circuit.

British Standard 2917-1

These symbols respect (comply with) the British standard, which is, in this case, the opposite of the ISO standards.

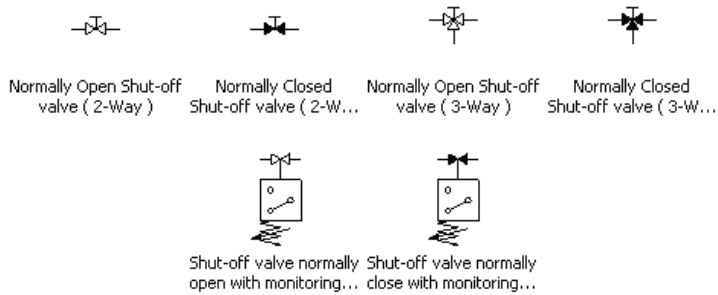


Figure 4-61: Shut-Off Valves Components, British Standard

4.3.9.5 Quick Exhaust

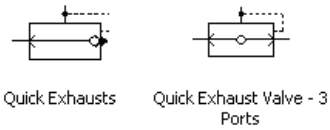


Figure 4-62: Quick Exhaust Components

These components can return the fluid to the reservoir when the pilot pressure is sufficient.

4.3.9.6 Flow Controllers by Differential Pressure

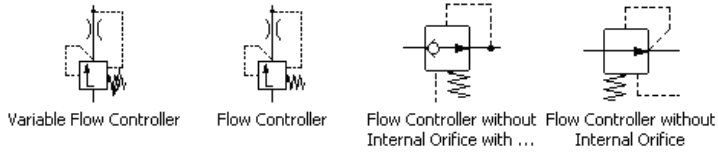


Figure 4-63: Flow Controllers Components Regulated by Differential Pressure

The flow control is carried out by providing a measured flow or the required differential pressure. The considered differential pressure is the pressure difference between the output and the pilot ports.

The pressure drops are calculated from the pressure versus flow curves, but curves are only considered when the flow through the component is smaller than the requested flow or differential pressure.

4.3.9.7 Flow Dividers

Mechanical Flow Rate Controllers

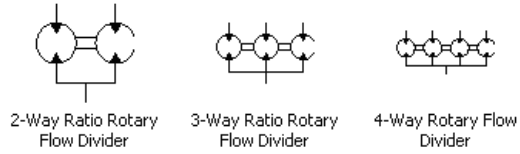


Figure 4-64: Mechanical Flow Rate Controllers Components

The flow rate is controlled by specifying the displacement for the different chambers.

Other Flow Rate Controller

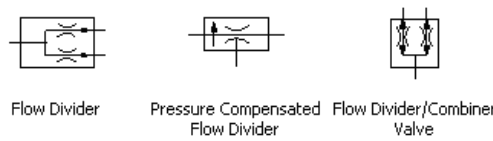


Figure 4–65: Other Flow Rate Controller

These flow controllers have a simple simulation model and are operated like the mechanism they represent but without a compensated pressure. However, all controllers are compensated to the temperature.

4.3.10 Pressure Controls

4.3.10.1 Introduction

All pressure valves can be found in the root of both hydraulic and pneumatic branches. However, for those that are not familiar with these components, it is possible to extend these branches to find classified symbols. For each type of valve, you have a cracking, reseating or closing pressure.

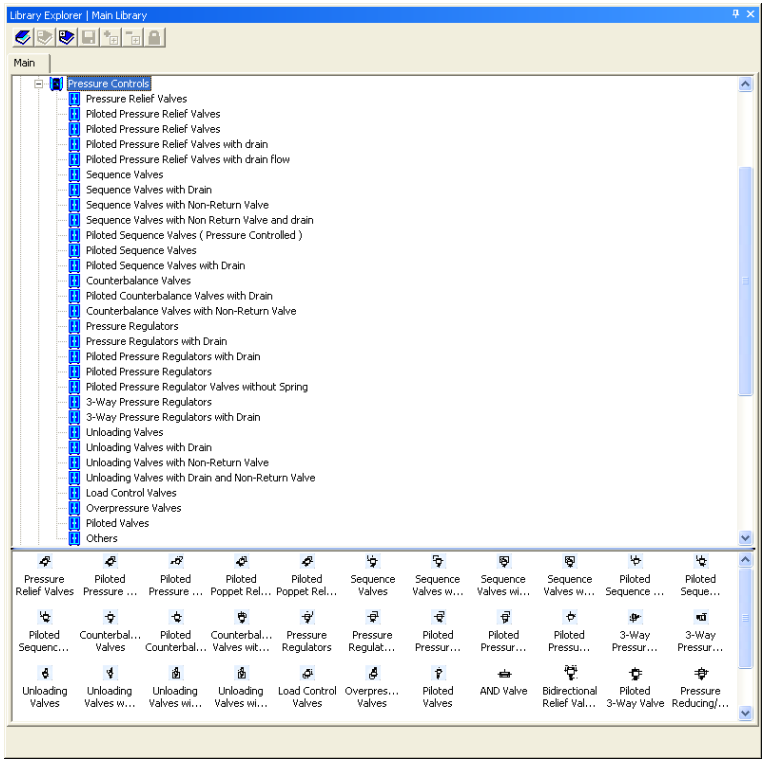


Figure 4-66: Pressure Controls Components

Pressure Drops

In the pressure controlling valves, they are modeled with an overpressure proportional to the flow through the valve. By default, the pressure controlling valves are almost perfect. This status allows you to modify the circuit component behaviour and is very useful for maintenance. It is also possible to characterize a curve in terms of the flow coefficient.

Independently of the valve behaviour, when there is a flow, the simulator calculates pressure drops based on a pressure drop curve. This curve defines, as a flow function, the overpressure created when the fluid passes through the valve. The curve defines the fluid pressure drops when the fluid passes through the valve for a viscosity of 32 cSt at atmospheric pressure.

For each valve whose pressure drops are modeled by a curve, the simulator finds in which circuit it is inserted in order to determine the corresponding temperature of this fluid for a 32 cSt viscosity. Once this temperature is known, the interpolation can be performed in order to

find current viscosity, density and pressure drops. You should know that the limits of research on temperature range from -100 °C to +200 °C. In other words, the viscosities and densities used will be enclosed by their values at -100 °C and 200 °C.

If you want constant viscosity for all circuits in your project, you have to uncheck "Thermal Simulation" in the "Project Properties" → "Fluid system" → "Options".

If you want constant viscosity for some circuits of your project, you have to specify, in the fluid properties of the used fluids in the desired circuits, the same kinematic viscosity and density for the two temperatures, because these properties are closely related.

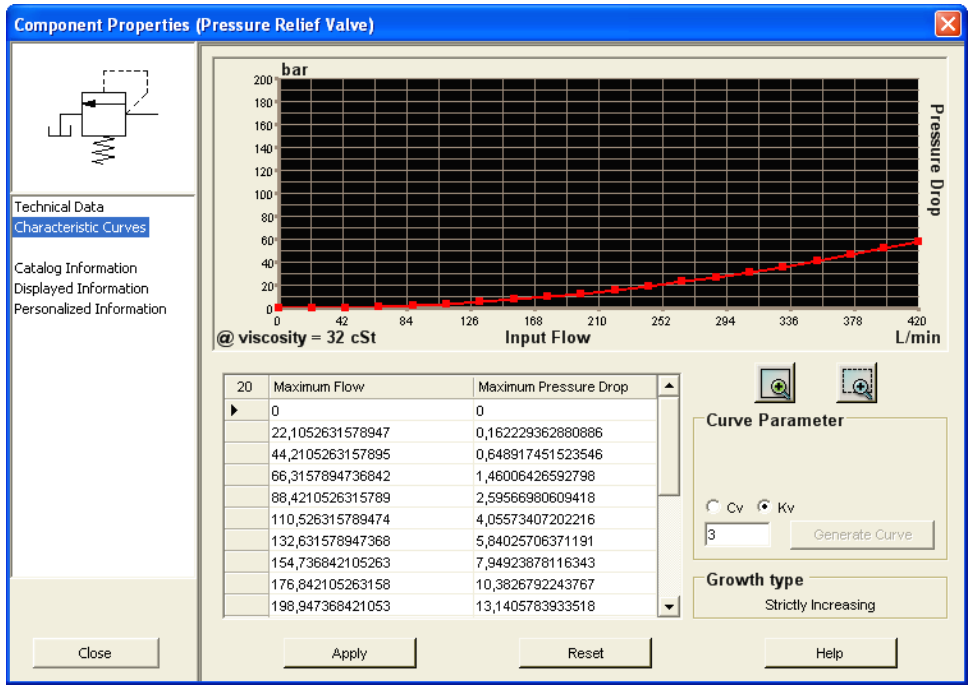


Figure 4-67: Curve of Hydraulic Pressure Drops

You have to consider the fact that the viscosity increases with the pressure. The viscosity doubles when the pressure is 400 times higher than the normal atmospheric pressure (approximately 400 bar).

It is possible to specify a flow coefficient instead of editing the curve. Directly type a *Kv* or *Cv* and click on "Generate Curve" button.

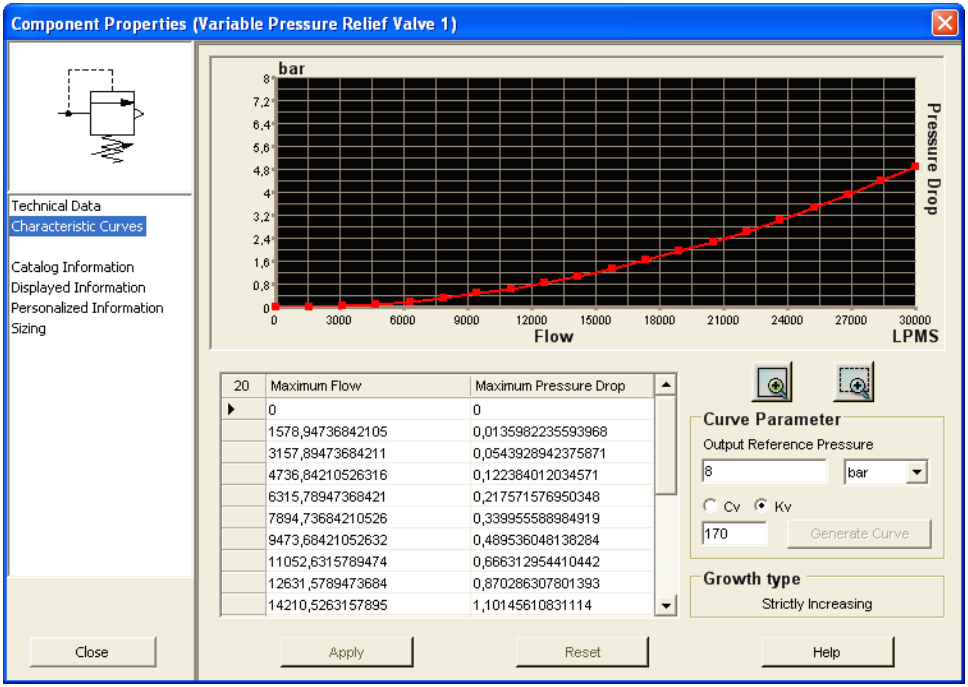


Figure 4-68: Curve of Pneumatic Pressure Drops

When a pressure valve contains a check valve, the pressure drop curve is also used for the check valve.

In pneumatic, the pressure drops from the curve are compensated relatively to the output pressure. This means that the pressure drops are inversely proportional to the absolute pressure, i.e.:

Pressure drop = Curve Pressure Drop * Atmospheric Pressure / (Output pressure + Atmospheric Pressure)

Hydraulic Flow Coefficient

For hydraulics, the flow coefficient is calculated from the following formula:

$$Flow\ Coefficient = Q\ .\ (\rho / \Delta P)^{1/2}$$

According to the unit standard used, you can use *Cv* or *Kv*.

For *Kv*: *Q* flow is in m^3/h, ΔP differential pressure is in bar and ρ specific weight or density is in kg/dm^3.

For *Cv*: *Q* is in Gallon US/min, ΔP is in psi.

The conversion from and to *Cv* and *Kv* is:

$$Kv \approx 0.865\ Cv\ or\ Cv \approx 1.15\ Kv$$

If the thermal option is not activated, then the fluid is considered oil as set in the project options with characteristic set for 40 ºC or 104 ºF.

Pneumatic Flow Coefficient

In pneumatics the flow coefficient *Kv* or *Cv* depending on the unit convention, are given by the following formulas:

$$Flow\ Coefficient\ Kv = Q/(28.16(\Delta P.P_s)^{1/2})$$

P_s = Absolute exit pressure of the component (in Bar);

ΔP = Pressure drop (in Bar);

Q = Pneumatic flow through the component (at standard atmospheric reference pressure);

$$Flow\ Coefficient\ Cv = Q/(0.978.(\Delta P.P_s)^{1/2})$$

P_s = Absolute exit pressure (psia);

ΔP = Pressure drop (psi) between the entry pressure P_e and the exit pressure P_s;

Q = Pneumatic flow through the component at standard atmospheric reference pressure (SCFM);

Cv: Imperial pneumatic flow coefficient.

The existing relation between *Cv* and *Kv* in pneumatics is:

$$Kv \approx 14.3\ Cv \text{ or } Cv \approx 0.07\ Kv$$

In pneumatics, the horizontal axe of the load drop curve is given in standard air flow, namely at the standard atmospheric reference pressure.

In either hydraulics or pneumatics, it is possible to characterize the load loss in terms of the flow coefficient.

In hydraulics, it is sufficient to specify Cv or Kv, whether in pneumatics, the component exit pressure needs to be specified. To create or display a simulation curve, click on the button "Generate Curve".

Rules to know:

If n components are in series, then the equivalent flow coefficient to the n components is verified by the following rule:

$$1/Cv_{équivalent} = \Sigma\ 1/Cv_i$$

If n components are in parallel, then the equivalent flow coefficient to the n components is verified by the following rule:

$$Cv_{équivalent} = \Sigma\ Cv_i$$

Operating Behaviour

In the following sections, general operating behaviour of components is explained. In the component shortcut help (F1), the component behaviour is more detailed.

4.3.10.2 Pressure Relief

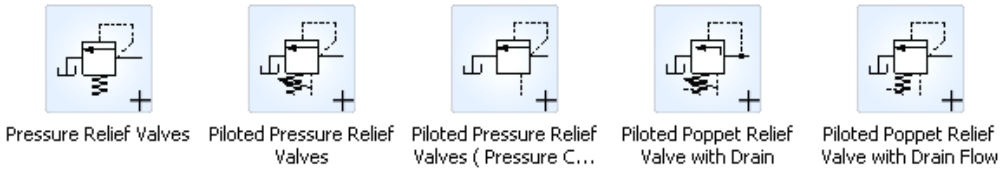


Figure 4-69: Pressure Relief Valves Components

1. The pressure relief valves open at the "Cracking Pressure";

2. The piloted pressure relief valves open at the minimal value between “Cracking Pressure” and existing pilot pressure;

3. The pressure limiters with drain open at the minimum value chosen between “Cracking Pressure” increased by the drain pressure, and the existing pressure in the pilot.

4.3.10.3 Sequence Valves

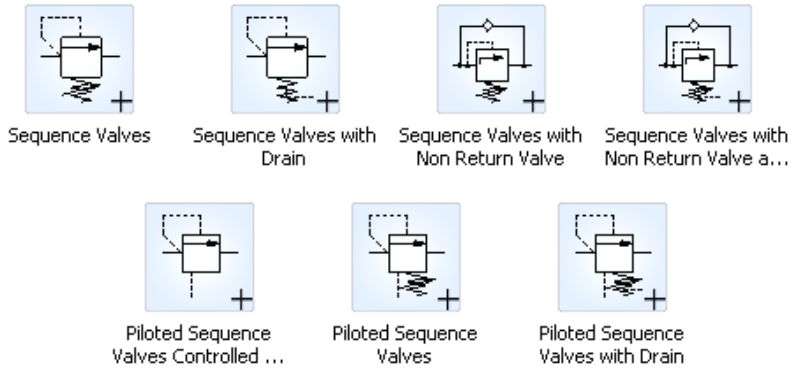


Figure 4-70: Sequence Valves Components

1. The sequence valves open at a pressure higher than or equal to the “Cracking Pressure”;

2. The sequence valves with drain open at a pressure higher than or equal to the “Cracking Pressure” increased by the drain pressure;

3. A check valve allows a flow in the opposite direction;

4. The piloted sequence valves without spring open at a higher pressure than the pressure prevailing in the pilot;

5. The piloted sequence valves open at the minimal value between “Cracking Pressure” and pressure prevailing in the pilot;

6. Two-Stage sequence valve opens at the “Cracking Pressure” when the pilot pressure is less than the “Stage Pilot Signal Pressure”. Otherwise this valve will open at the “Active Stage Cracking Pressure”;

7. The piloted sequence valves with drain open at the minimal value between “Cracking

Pressure" and pressure prevailing in the pilot increased by the drain pressure;

8. A check valve allows a flow in the opposite direction.

4.3.10.4 Counterbalance Valves

For most of these valves only the "Cracking Pressure" is available. However, some of them allow to be closed with a pressure less than the pilot pressure needed to overcome the force developed by the spring. For this secondary pilot, it is possible to characterize the required pressure.

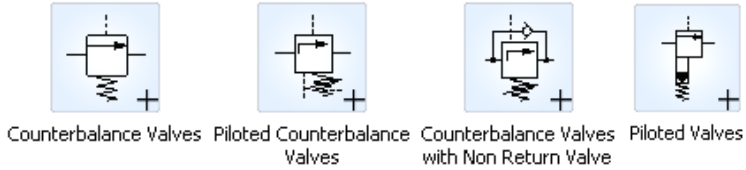


Figure 4-71: Counterbalance Valves Components

1. The counterbalance valves open at a pressure higher than or equal to the "Cracking Pressure";

2. The piloted counterbalance valves with drain open at a pressure in main pilot higher than or equal to the minimum value between secondary pilot pressure and the "Cracking Pressure" increased by the drain pressure;

3. A check valve allows a flow in the opposite direction;

4. The piloted valve opens when the input pressure is higher than or equal to the sum of the "Cracking Pressure" and the pilot pressure.

4.3.10.5 Pressure Reducing Valves

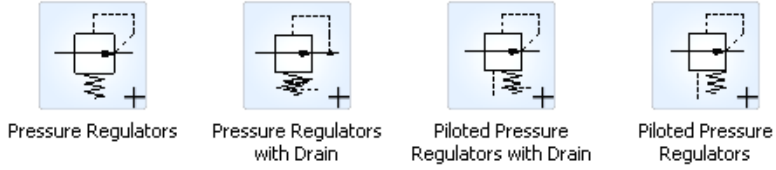


Figure 4-72: Pressure Reducing Valves Components

1. The pressure reducing valves limit the pressure downstream of the valve to the "Setting Pressure". Without drain, the overpressure cannot be evacuated;

2. The pressure reducing valves with drain limit the pressure downstream of the valve to the "Setting Pressure" increased by the drain pressure. A drain allows to evacuate overpressure downstream of the valve up to a value equal to the drain pressure;

3. The piloted pressure reducing valves limit the pressure downstream of the valve to a regulated pressure equal to the minimum value between the "Setting Pressure" and the pilot pressure;

4. The piloted pressure reducing valves with drain limit the pressure downstream of the valve to a regulated pressure equal to the minimum value between the "Setting Pressure" increased by the drain pressure and the pilot pressure;

5. The 3-Way pressure reducing valves limit the pressure downstream of the valve to the "Setting Pressure", while excess fluid is evacuated by the 3rd port;

6. The 3-Way piloted pressure reducing valves with drain limit the pressure downstream of the valve to the "Setting Pressure" increased by the drain pressure, while excess fluid is evacuated by the 3rd port;

7. The 3-Way piloted pressure reducing valves with drain limit the pressure downstream of the valve to a regulated pressure equal to the minimal value between the "Setting Pressure" increased by the drain pressure and the pilot pressure, while excess fluid is evacuated by the 3rd port;

8. The pressure reducing/relieving valve remains open as long as the pilot pressure is less than the "Settings Pressure" increased by the drain pressure. If it is higher, then the flow is diverted to the other port.

4.3.10.6 Other Valves

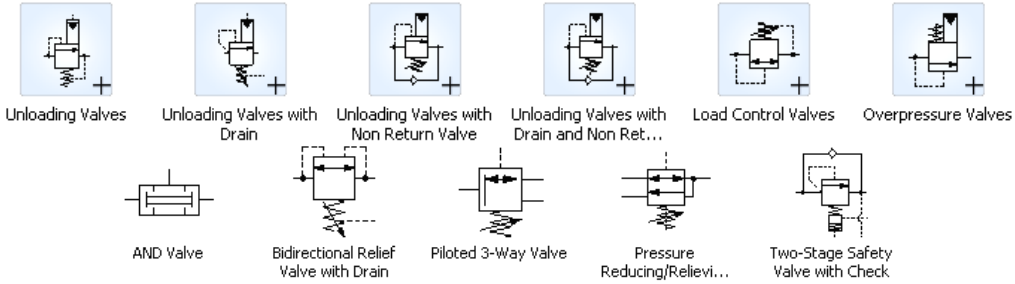


Figure 4-73: Pressure Reducing Valves Components

1. The unloading valves open at a pressure higher than or equal to the "Cracking Pressure";

2. The piloted counterbalance valves with drain open at a pressure at the main pilot higher than or equal to the minimum value between the secondary pilot pressure and the "Cracking Pressure" increased by the drain pressure;

3. A check valve allows a flow in the opposite direction;

4. The load control valves opens at a pressure higher than or equal to the "Cracking Pressure" minus the pressure downstream of the valve;

5. Two-Stage safety valve opens at the "Cracking Pressure", if the pilot pressure is less than the "Stage Pilot Signal Pressure". Otherwise this valve will be opened at the "Active Stage Cracking Pressure";

6. The replenishing (overpressure) valves open at a pressure higher than or equal to the "Cracking Pressure" increased by the product of the pilot pressure and the "Pilot Ratio (1:n)" increased by the pressure drain;

7. Bidirectional relief valve with drain opens at the "Cracking Pressure" increased by the drain pressure. The flow occurs from high pressure to low pressure.

4.3.11 Sensors

4.3.11.1 Digital Sensors

This branch includes all ON/OFF sensors available. The thresholds of measures such as

pressure, differential pressure, flow, temperature, proximity and mechanical position can be specified.

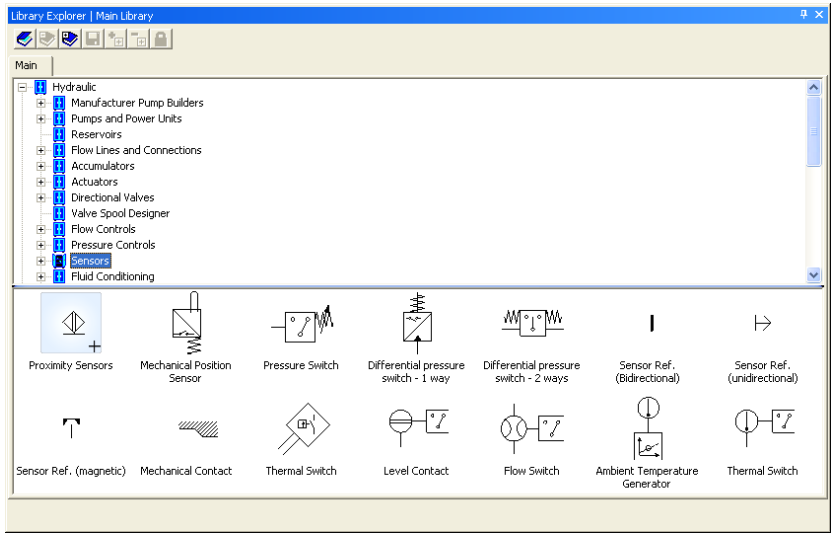


Figure 4-74: Sensors Components

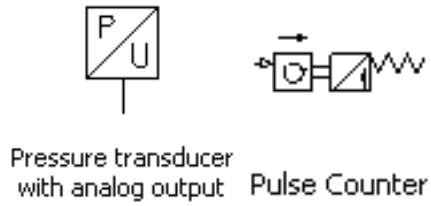


Figure 4-75: Pneumatic Sensor Only

Digital sensors allow to output or change the state of a variable or of a component in comparison with the measured value and a predefined threshold. The variable or component returns to its original state when the measured value is not located within the threshold scope.

For more details on ambient temperature generator see section *3.2.1 Ambient Environment*.

Mechanical Contact Sensors

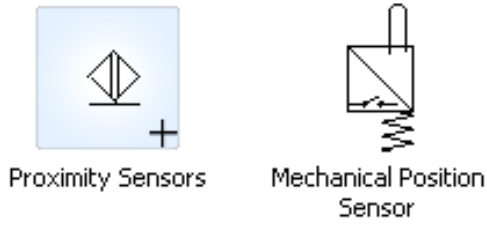


Figure 4–76: Mechanical Contact Sensors

These sensors are activated when their contact point meets with the contact point of a mobile device (cylinder shaft, directional valves roller, etc.).

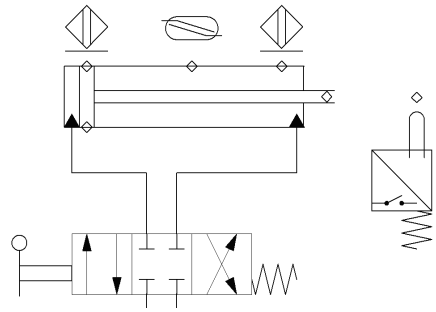


Figure 4–77: Examples

The proportional valves with rollers as controls can be used as mechanical sensors without associated variables.

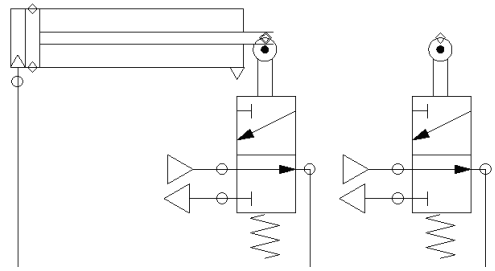


Figure 4–78: Proportional Valve Roller Example

If space is limited on the diagram to place the sensors, it may be useful to use the sensors' reference symbols with a cam which would be located somewhere else on the diagram.

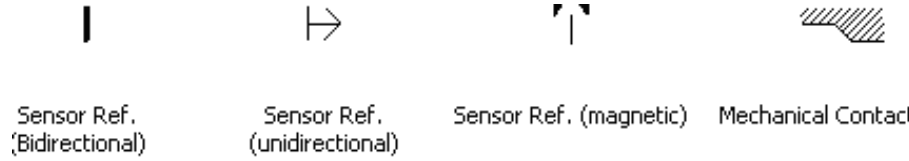


Figure 4–79: Sensor Mechanical Contact References

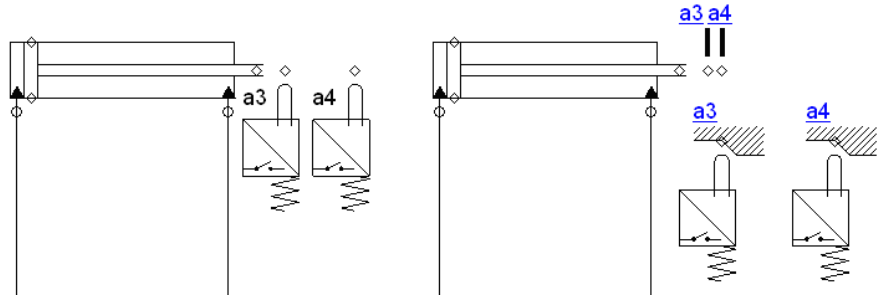


Figure 4–80: Practical Example for Referring Contact Points.

Fluid Sensors

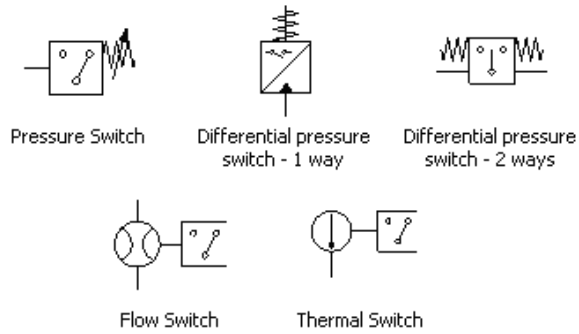


Figure 4–81: Fluid Sensors

These sensors are activated based on the comparison of the measured value and the threshold value of the sensor.

Manually Activated Sensors

If information cannot be measured (level) or if the activation would take too much time, the sensor contact status can be changed by clicking on the sensor.

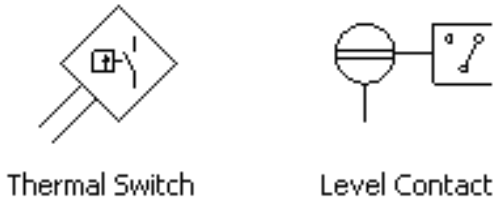


Figure 4–82: Fluid Sensor

All these sensors have a matching or compatible electrical contact. These contacts can be found in the Electrical Control Workshop.

4.3.12 Fluid Conditioning

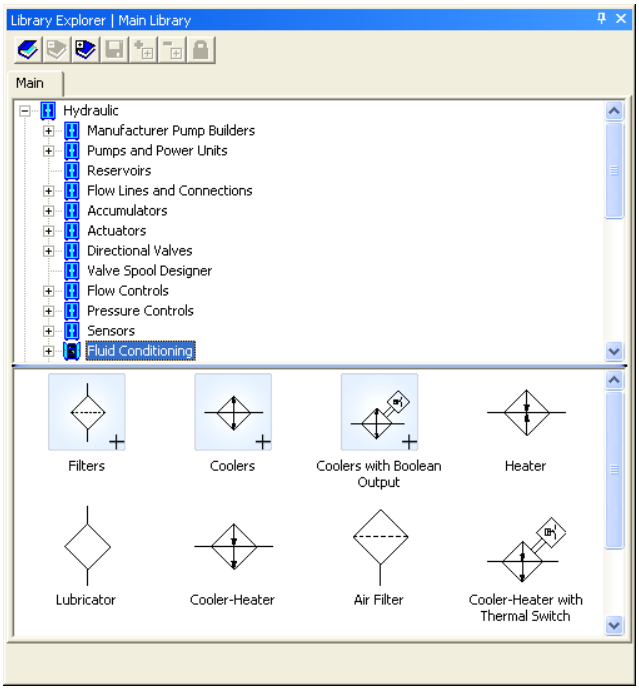


Figure 4-83: Fluid Conditioning Components

The components whose symbol illustrates a function other than thermal evolution of fluid are passive and offer only flow resistance, defined by the pressure drop curve.

Heater

It is possible to specify a "Heat Power" that is triggered from the "Switching Temperature" and stops when the "Deactivation Temperature" is reached.

Cooler-Heater

They combine the functions of every previous component.

For components with a thermal contact, this variable state can be read and used for command circuit logic.

Fan

This component does not exchange thermal data but it opposes a resistive torque when activated.

4.3.12.1 Filters

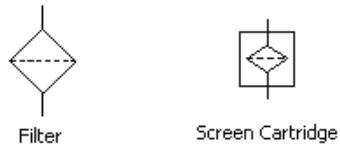


Figure 4-84: Filters

4.3.12.2 Cooler

It is possible to specify, with the dissipation curve, the power dissipation as a function of differential temperature between oil and environment.

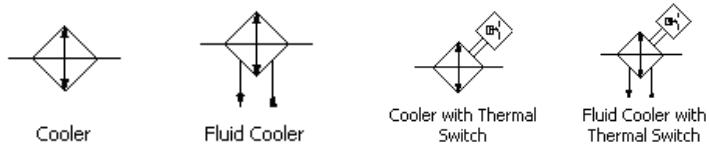


Figure 4-85: Coolers

4.3.13 Measuring Instruments

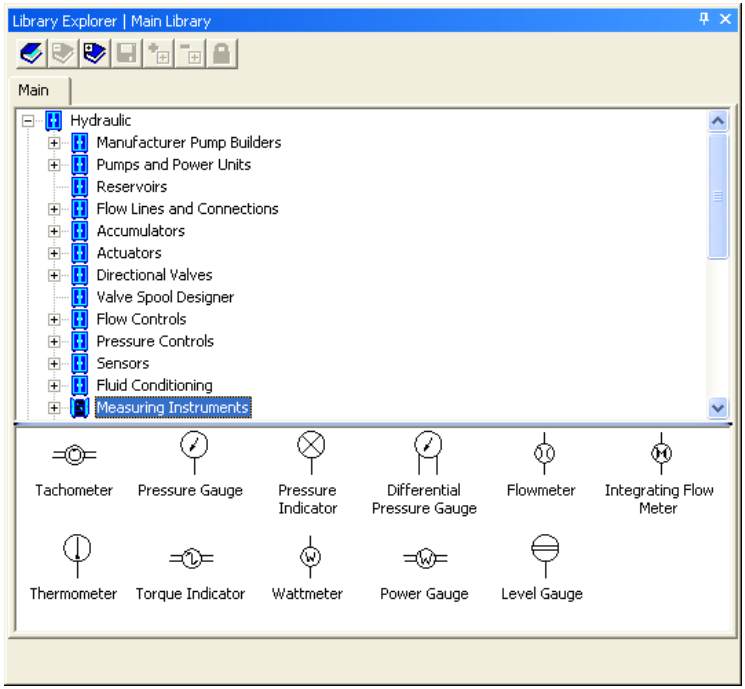


Figure 4-86: Measuring Instruments

It is possible to access some of the circuit values, calculated by the fluid simulator, through measuring instruments. The values can be displayed using a variety of unit scales. Following, is a list of the values available for display:

- Relative pressure (to atmospheric pressure);
- Differential relative pressure (to atmospheric pressure);
- Flow;
- Flow normal air: for pneumatics only (air flow measured at atmospheric pressure) – SCFM (LPMS) = CFM (LPM) * (compression ratio);
- Oil temperature: for hydraulics only; in order to observe a temperature evolution, thermal simulation should be activated in the "Fluid Dashboard" or in the project options;
- Integral flow meter: for hydraulics only;
- Fluid power;
- Tachometer: this component has a similar component with only 1 connector;

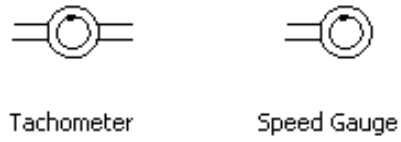


Figure 4-87: Tachometers

- Torque gauge: This measuring instrument is inserted between a transmission component and an actuator. It allows measuring the developed torque upstream of the actuator. In other words, a torque indicator with one free mechanical connector will measure a torque equal to zero.

The following measuring instruments are not available in the library but are restricted to the symbol.

- Level meter.

When this last value is not directly available, it is still possible to get it from the configurable reservoir if the right sensors are used.

4.3.14 Cartridge Valves

As opposed to regular, on-off valves, cartridge valves have a progressive opening, which is influenced by the maximum flow property: the lowest this property, the faster the valve opens.

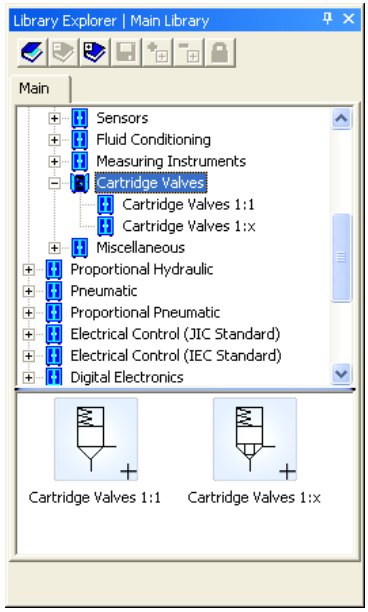


Figure 4-88: Cartridge Valves Components

4.3.15 Miscellaneous

There is a "Miscellaneous" branch which contains, among others, the "Functional Box" component and the sub-branches such as shown in the following figure.

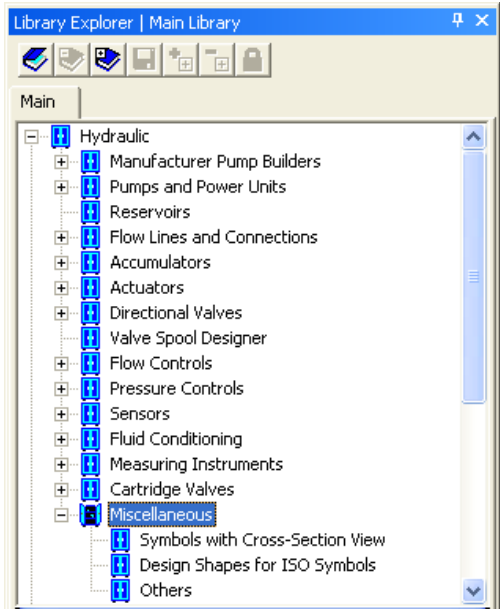


Figure 4-89: "Miscellaneous" Branch and Functional Box Component

The family "Symbols with Cross-Section View" gathers all symbols offering a cross-section view during simulation. These components can also be found in other library branches based on the specialty they are related to.

The family "Design Shapes for ISO Symbols", as its name specifies, includes all basic symbols recommended by the ISO standard used to create additional components. These additional components are passive; they do not have behaviour. Please refer to section *4.2 Standards and Specifications* for details.

The "Others" branch contains the "Fuctional Box" component which allows you to create your own fucntions and others specific and complex components in Automation Studio™. Refer to section *5.9 Functional Box* for détails.

4.4 Specific Pneumatic Components

The behaviour of pneumatic components is similar to their hydraulic counterparts. The main differences between the simulation of a pneumatic circuit and a hydraulic system are that there are no hydrostatic pressure, no heat and the air compressibility is incomparably higher than that of hydraulic fluids.

4.4.1 Compressors and Motors

The compressors are modeled as hydraulic pumps.

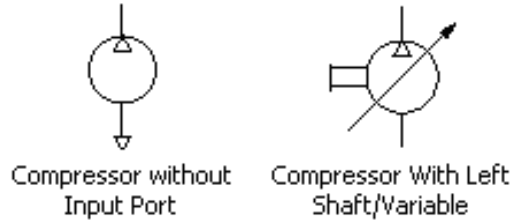


Figure 4-90: Compressors Components

4.4.2 Sensors

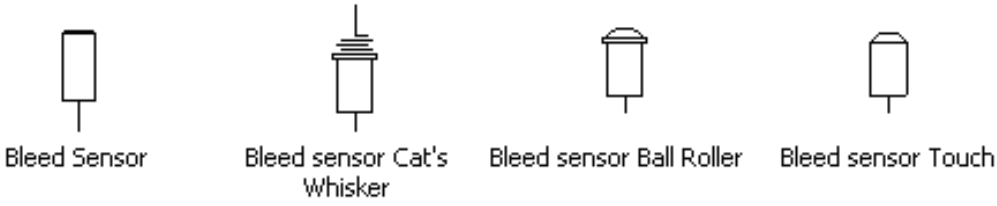


Figure 4-91: Sensors Components

4.4.3 Sequencers

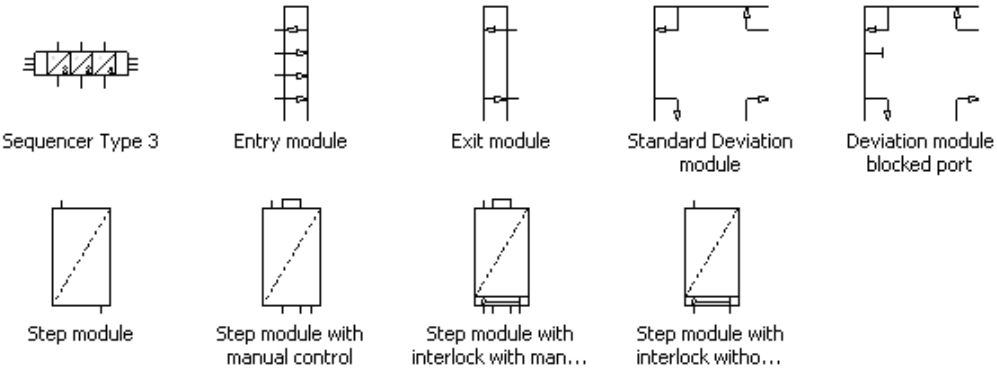


Figure 4-92: Sequencer Component

4.4.4 Relays

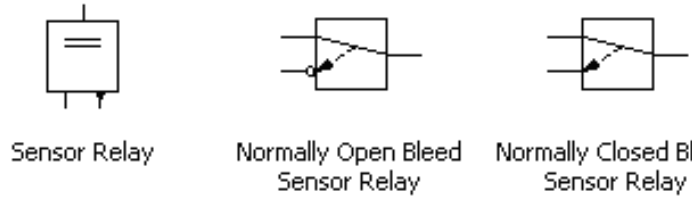


Figure 4-93: Relays Components

4.4.5 Modular Logics

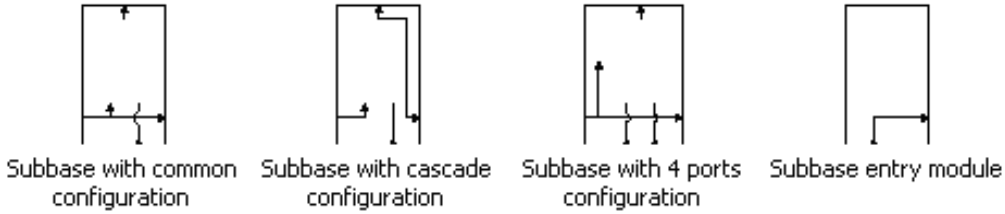


Figure 4-94: Modular Logics Components

4.4.6 Logic Units

Each of these valves has the logic behaviour of its function. The logic behaviour is described in the "Component Properties" dialog box.

4.4.6.1 Logic Valves

The logic valves are pneumatic components.

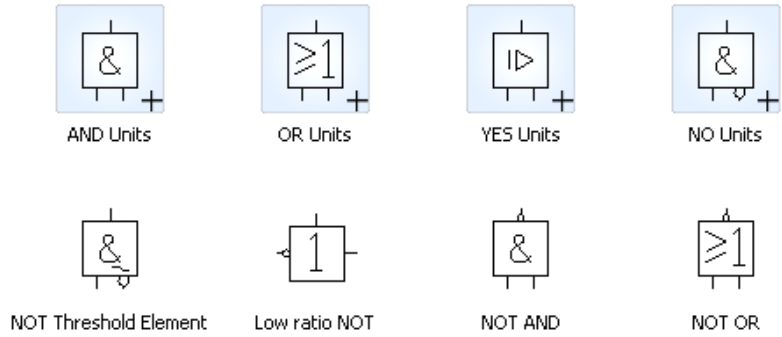


Figure 4-95: Logic Valves Components

In the properties dialog box of each of these components, there is a truth table which explains the component behaviour.

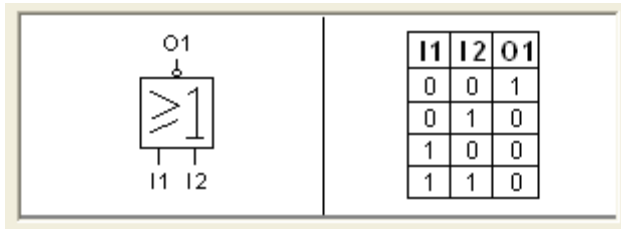


Figure 4-96: Example of NOT OR truth table

4.4.6.2 Memory Unit

Figure 4-97: Memory Unit Components

4.4.7 Timers

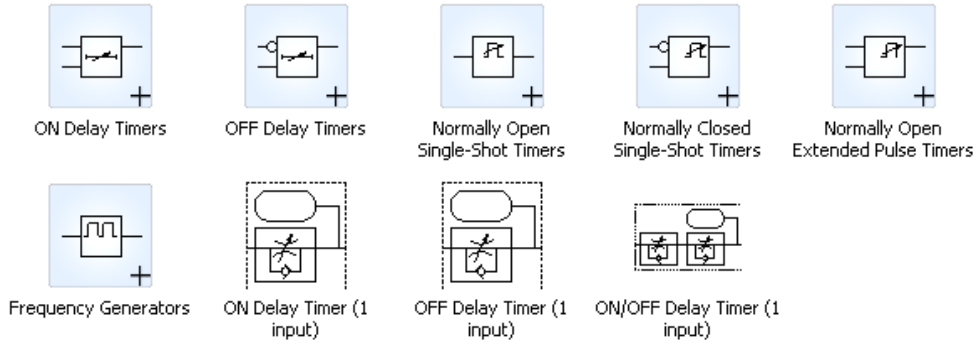


Figure 4-98: Timer Components

4.4.8 Counters

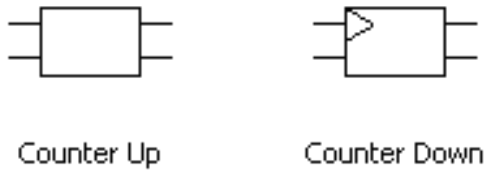


Figure 4-99: Couters Components

4.5 Proportional Workshops Components

The hydraulic and pneumatic proportional components receive a signal to move or modify the variable setting (directional valves, flow valves, pressure valves, pumps). Displacement linearity can be adjusted with the "Displacement curve" of each component.

4.5.1 Pumps

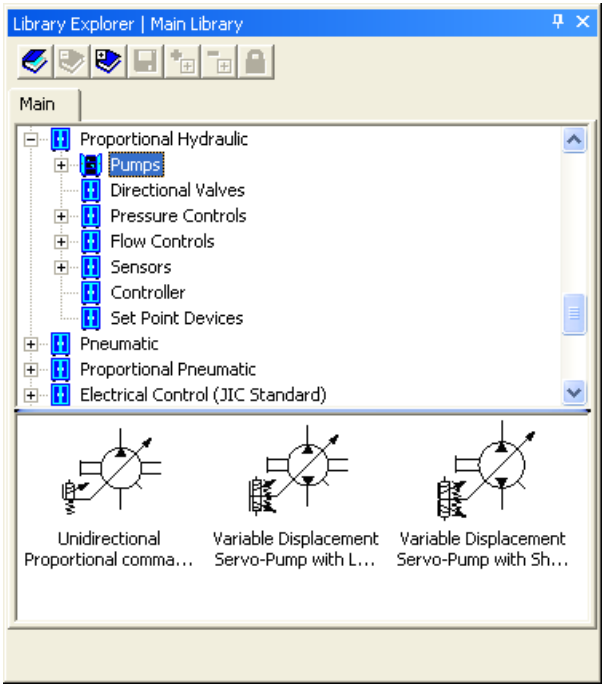


Figure 4-100: Proportional Pumps Components

4.5.2 Proportional Directional Valves

4.5.2.1 Control Configuration

Introduction

A new modelization for the directional valves is available since Automation Studio™ version 5.4. The improvements are:

- Control's configuration and modelization; it is now possible to use all control technology (hydraulic, electric, manual, etc) on the same valves;

- Spool position configuration; you can now create its own spool positions;

- Spool simulation; each internal channel is set with its own flow specifications.

These changes required a new step to configure directional valves. In previous versions, the initial position in edition or in simulation mode was the same. Now, if springs are used, you must properly calibrate them to achieve the same initial position during simulation as in the edition mode.

Spool Displacement

For on/off directional valves, the spool position is one of the available positions. For proportional directional valves, the stroke must be known since the displacement is continuous. This technical data is set in the valve component properties dialog box.

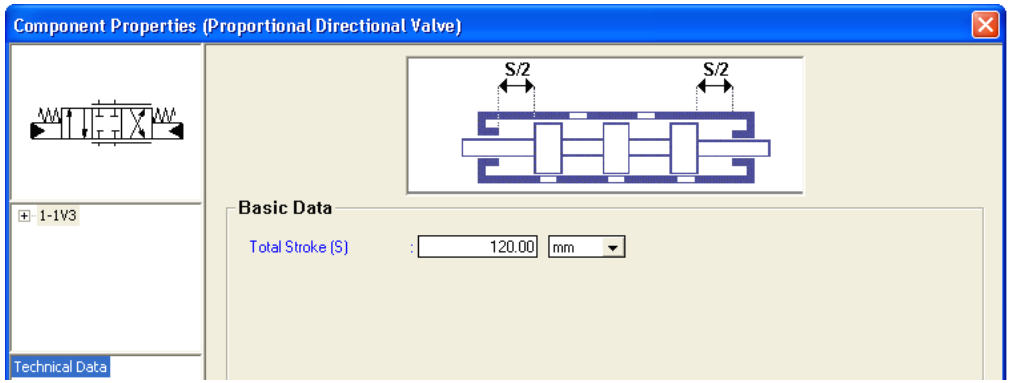


Figure 4–101: Proportional Directional Valves "Component Properties" Dialog Box

The following explanations are developed using the properties default values. *S* is the spool maximum stroke, *Lo* is the spring free length, *Lw* is the installed length, *Fw* is the installed force, *k* the spring stiffness, *Fsa* spring stiffness coefficient, and *Fmax* the maximum compression force.

In the previous figure, the spool has a stroke of 120mm (default value). For all types of directional valves, the half-stroke is always present on the left and on the right. In this case, 60mm on each side.

If there is no spring present and no forces applied upon simulation start, the spool initial position will be the same as in the editing mode.

If springs are part of the valves and/or the force balance on the spool is not zero, when starting the simulation mode the sum of all forces is calculated, and the spool is moved to the

balanced position. In the following figure, a pressure is applied to the valve control, therefore the spool will move to the appropriate position.

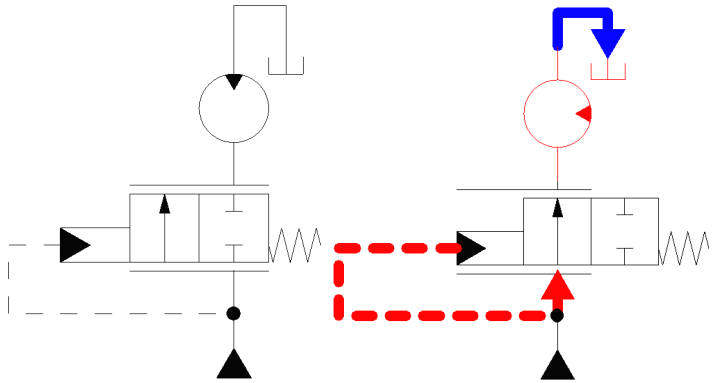


Figure 4–102: Spool Position in Edition Mode and Simulation Mode

Spring Configuration

It is important to understand the spring's configuration because once in simulation, the spool will take its true balanced position. Therefore, a bad configuration will change the spool position shown in the Edition mode, and create unwanted flow distribution.

Unidirectional Displacement

If the spring has been added on the proper valve side, then it is, by default, properly positioned.

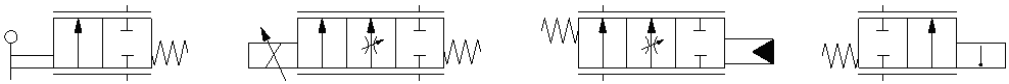


Figure 4–103: Unidirectional Valves

Bidirectional Symmetrical Displacement

If the valves are symmetric (initial position in the middle) and that an identical spring is added on both sides then, the spool is, by default, centered.

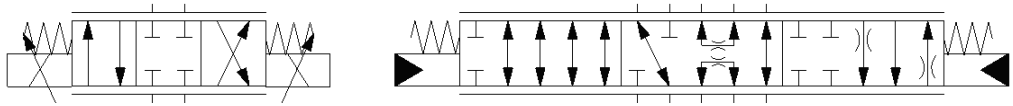


Figure 4–104: Bidirectional Symmetrical Valves

Bidirectional Asymmetrical Displacement

If the directional valve is asymmetrical (initial position not centered), the spring's free length has to be adjusted.

The rule to apply is the following: for a valve with *N* positions, *Nl* being the number of positions to the left of the initial position, and *Nr* the number of positions to the right of the initial position: $N = Nl+Nr+1$.

The left spring must be able to take a displacement of *Nr* positions and the right spring, the displacement of *Nl* positions. Therefore, for the edition mode initial position to remain the same in simulation mode, the spring's free length must be:

The left spring free length, $Lo\ left = (Nr/(N-1))\cdot S$

The right spring free length, $Lo\ right = (Nl/(N-1))\cdot S$

With these values the springs installed force is null.

The following configurations set the spool position in simulation mode at the same position as in the edition mode.

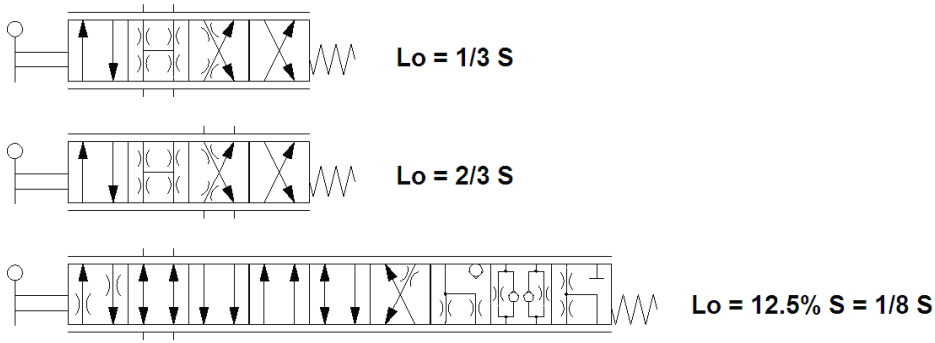


Figure 4–105: Centered Spool Configuration

Spring Installed Force

To create an installed force, the spring's length must be adjusted. If the directional valves are unidirectional or bidirectional symmetrical, the calculated data facilitates understanding.

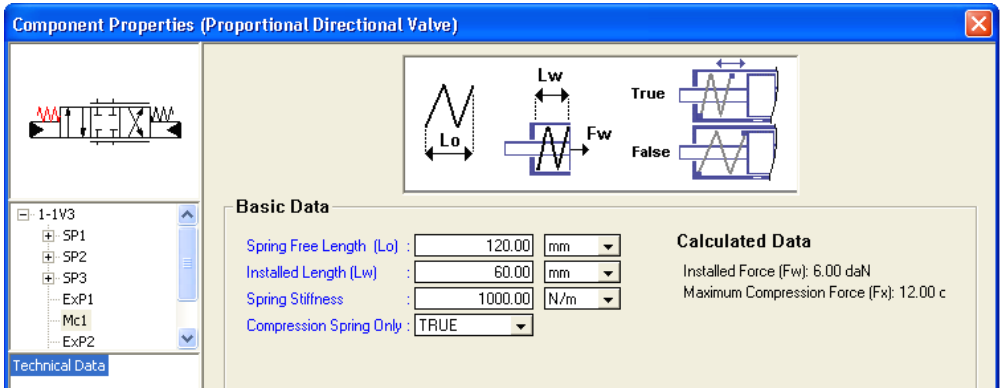


Figure 4–106: Spring Technical Data

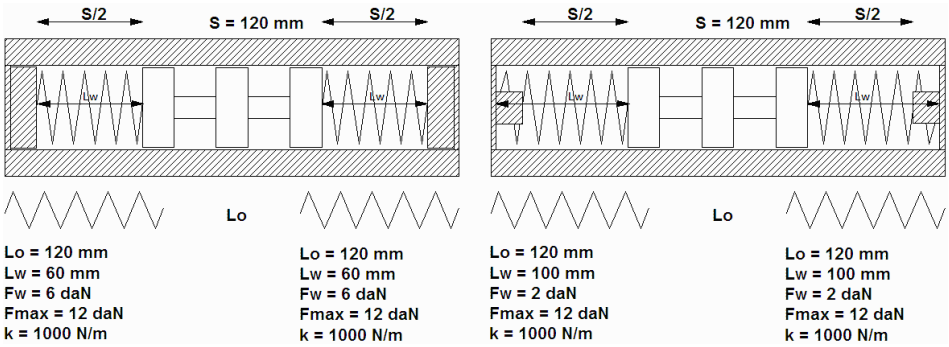


Figure 4–107: Different Installed Force on a Bidirectional Symmetrical Valve

For bidirectional asymmetrical valves, the calculated installed force can only be applied to the spring itself. All springs' installed forces are only taken into account when switching to the simulation mode. The sum of all forces re-positions the spool and re-calculates the installed force.

Control Generated Forces

This table summarizes the forces applied by the controls to the spool:

Name	Technical Data	Additional External Parameters	Applied force (F) calculation
Internal Pilot	S: Surface Pilot Connection Minimum Pressure Threshold Maximum Pressure Threshold	P: Pressure at the pilot input	F = 0 if P ≤ Pmin F = (P- Pmin).S if Pmin ≤ P ≤ Pmax F = Pmax.S if P ≤ Pmax
External Pilot	S: Surface Minimum Pressure Threshold Maximum Pressure Threshold	P: Pressure	F = 0 if P ≤ Pmin F = (P- Pmin).S if Pmin ≤ P ≤ Pmax F = Pmax.S if P ≤ Pmax
Electromagnet Thermal	Minimum and maximum force Minimum and maximum signal	Minimum and maximum signal	Linear scaling between signal and force signals.
Spring	Stiffness constant, free length Chamber length	Potential spool stroke Other spring present	Refer to *Figure 4–106: Spring Technical Data*.
Manual or mechanical control	Minimal and maximal force	Click during simulation between minimum (-10) and maximum (+10) signal	Linear scaling between signal and force signals.

Valve Control Configuration Examples

For the following directional valve, though it has two solenoids, the way it is configured

allows it to be controlled by a single signal applied to only one solenoid, or by a single signal applied to the both solenoids. Be careful to set correctly minimum and maximum forces to obtain the desired behaviour. The following figure shows its possible settings.

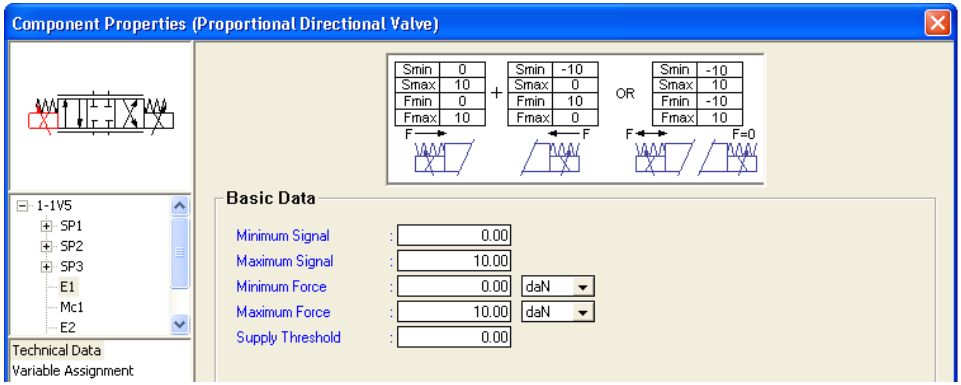


Figure 4-108: Signals of Solenoids

For the following directional valves, the thermal control reads the pressure at connector T; at 25 C or less no force is being applied; at 75 C, it applies a force of 12 daN.

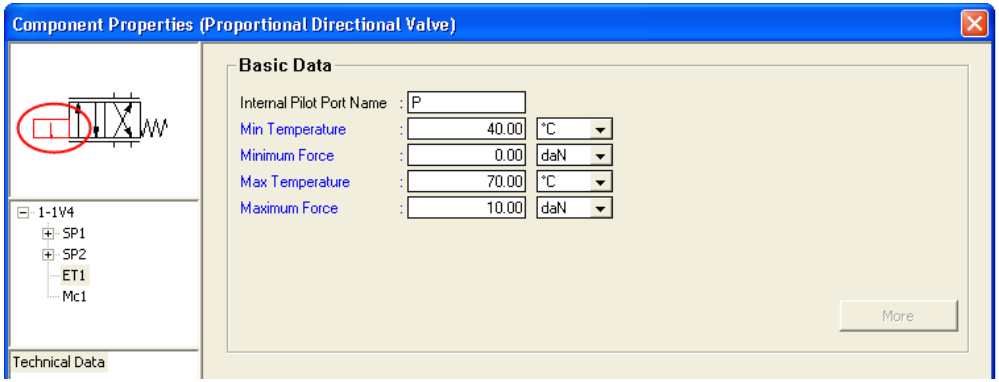


Figure4- 109: One Signal on a Thermal Control

4.5.2.2 Displacement Configurations

For a proportional valve, the relation between the pilot signal and the spool real position is modeled using displacement curves. By default, the relation is proportional.

This property is the same for the flow and pressure proportional valves.

4.5.2.3 Internal Flow Configurations

Flow Curves

For proportional valves, each spool section has curves defining the section connector opening as a function of the stroke (in %). The flow curve for each section is matched to a preset spool position. By default the curves do not overlap. By changing these curves, the component's behaviour can match the manufacturer's specifications.

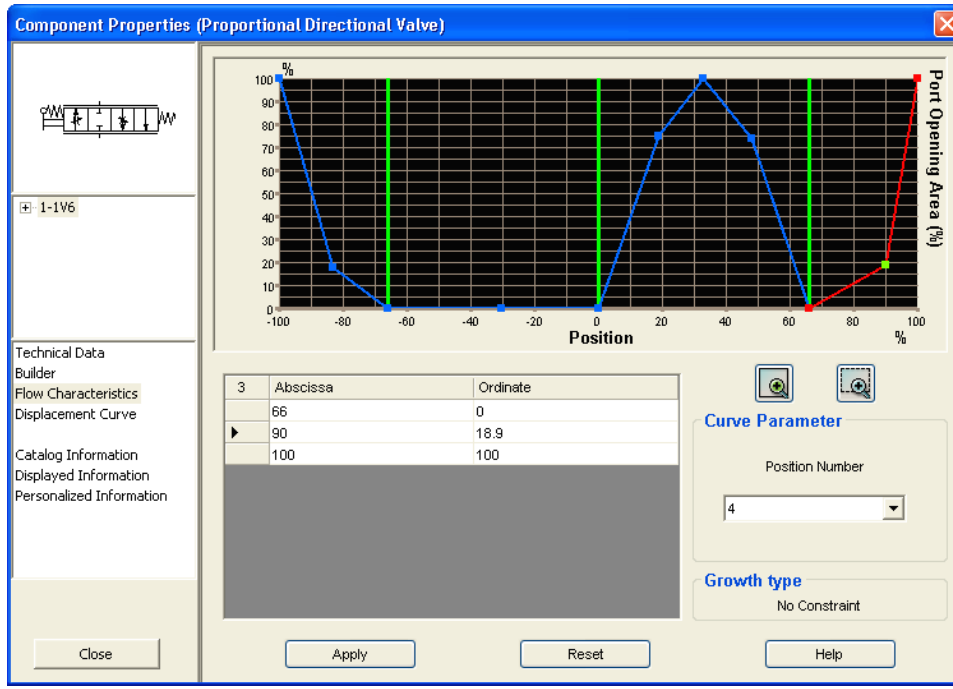


Figure 4- 110: Opening vs. Spool Position

By adjusting these curves, it is possible to overlap, so that flow is authorized simultaneously for two positions or more.

In the following figure, if the spool position is between ¼ and ¾ of the total stroke, position 1 and 2 will be active. The flow from P to A and from P to B are both active.

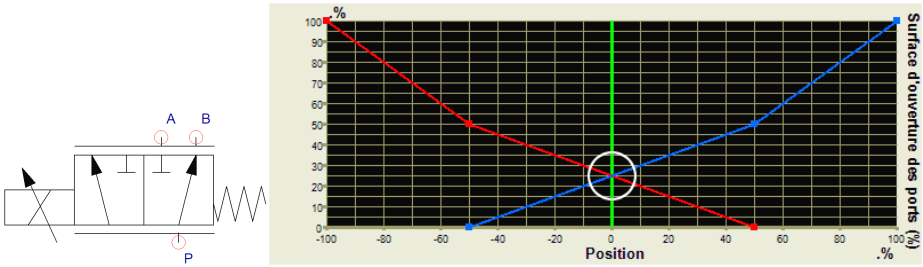


Figure 4- 111: Overlapping Between Two Positions.

Channel Properties

Each internal channel for a position is made of sub-components. Each of them has technical data just like any Automation Studio™ component.

When a channel has only one line, it is possible to set its flow rate *Cv* or *Kv* (depending on the standard used) and to specify the differential generated pressure as a function of the flow. These values correspond to fully open orifices. The pressure drop is then dynamically recalculated according to the flow curves (connector % of opening).

The following example will be used to demonstrate the calculation made by the simulator. For the following internal channel, the differential pressure is 5 bars to the nominal flow of 60 l/min when the channel is fully opened.

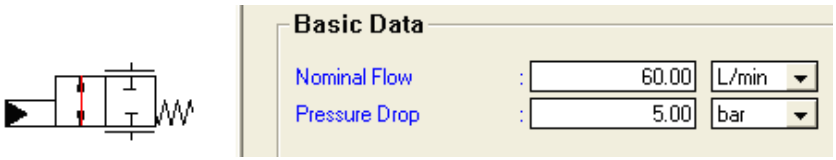


Figure 4- 112:Channel Configuration

These properties are defined for a simulation with oil of 32 cSt viscosity. This value is measured for an oil viscosity of 32 cSt. Please refer to section *4.3.10 Pressure Controls → Pressure Drops* to see what this model implies.

Therefore, if section 1 is at mid-travel or ¼ of the total spool stroke (circled area) then, for a flow of 15 l/min, the differential pressure will keep a value of 5 bar. For a flow of 60 l/min, the differential pressure will be 80 bars.

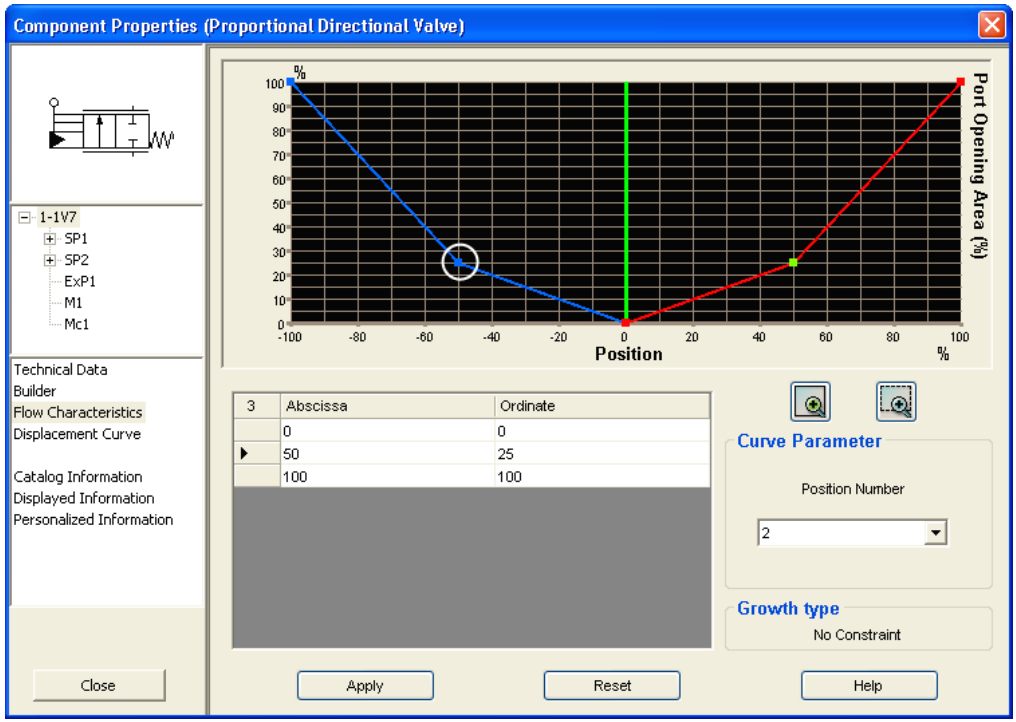


Figure 4- 113 Flows for Each Section is at 50% of the Full Spool Travel (No Overlapping)

Each connection port also has a flow rate which generates a differential pressure of 1 kPa for 60 l/min (➔ 2 kPa for 60 l/min for the channel) which cannot be modified.

4.5.2.4 Spool Position Properties Modification

It is possible to configure each line in a section by clicking on the sub-component (click on the zone and use the mouse scroll wheel to zoom) or by opening the sub-section on the section tree with the same number (in the following picture, the spool section is the forth from the left, therefore click on SP4).

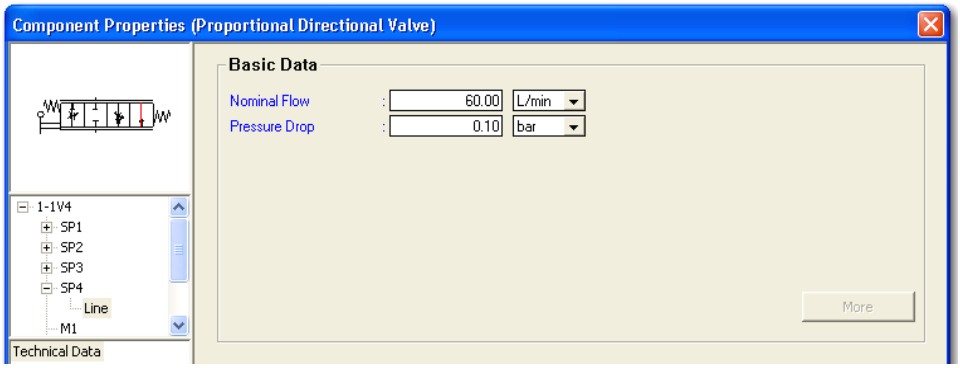


Figure 4–114: Channel Properties in Spool Section 4

For all internal channel sub-components, it is possible to set the technical data very precisely.

4.5.3 Proportional Regulators

4.5.3.1 Pressure Controls

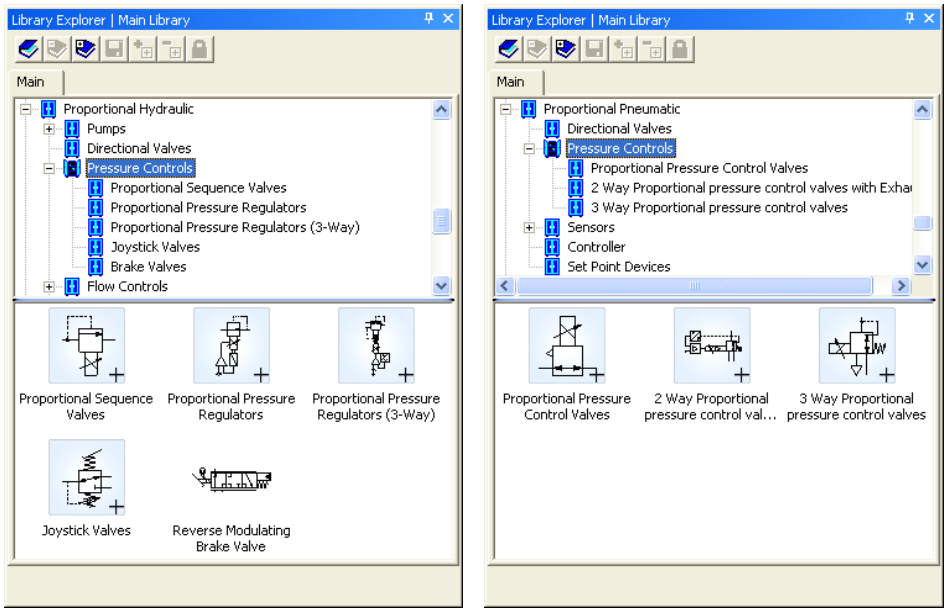


Figure 4-115: Proportional Pressure Controls Components

Pressure drops are modeled as the On/Off valves.

4.5.3.2 Flow Controls

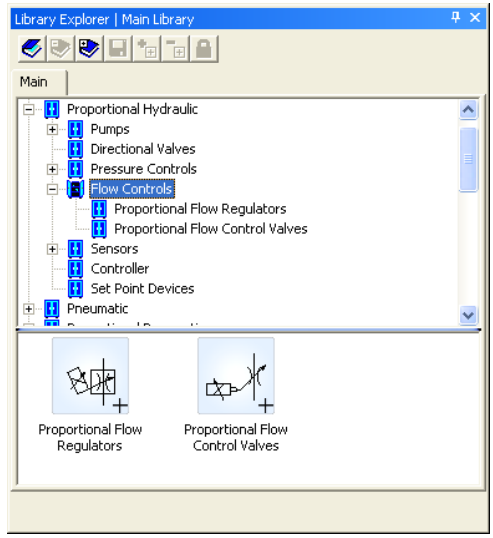


Figure 4-116: Proportional Flow Controls Components

4.5.4 Analog Sensors

These sensors are available in the sensor branch of the hydraulic and pneumatic workshops.

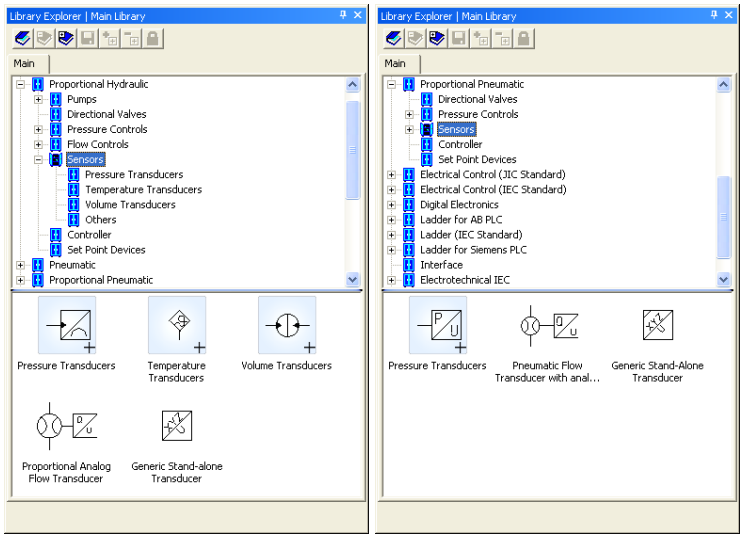


Figure 4-117: Analog Sensors Components

The analog sensor allows outputting a measured value in a system variable; this variable must be properly scaled using the "Component properties", "Configuration" option. These sensors are generally associated with the proportional technology.

4.5.4.1 Fluid Sensors

They are found in the library, and provide measurements for: flow, pressure, temperature, and volume. The sensors must be properly scaled using the "Configuration" option.

Scaling up is done by giving minimum and maximum values of the taken measures in the circuit (for example, P_{min} = 1 bar and P_{max} = 120 bar) and the minimum and maximum of the output signal (for example, $Signal_{min}$ = -10 bar and $Signal_{max}$ = +10).

If the measure is outside the extreme values then the output signal will take the extreme value (for example, if the measurement is 155 bar, then the signal value will be +10).

4.5.4.2 Generic Stand-alone Transducer

There is a generic sensor whose output value is defined manually for the simulation by clicking on the symbol. You can set the desired extreme values in the properties dialog box.

Figure 4–118: Generic Stand-alone Transducer

Any variable created with the variable manager and used in a SFC or a logic controller is considered a sensor, digital or analog.

4.5.4.3 Mechanical Sensors

These sensors are part of the actuators and they must be activated using the cylinders' or motors' "Component properties" Dialog Box. For cylinders, the sensor's activation is found in the "Builder" by selecting the "Sensor" button. For motors, the sensor's activation is found in the option "Sensors".

Three sensors are available with the linear and rotary actuators:

- Position (linear or angular);
- Speed (linear or angular);
- External Efforts (Force or torque).

When a sensor is activated (when the box is checked), a variable is made available in the "Variable Manager". Each variable can be read by a control device (proportional hydraulic), an internal logic variable (SFC workshop, Ladder Programming) or specific component controls (Valves Controls).

For each sensor, the output signal range (minimum and maximum values) must be properly set. For the different sensors the possible range is:

1. Position: the range is from 0 to maximum extension;
2. Speed: between +/- maximum limit (advanced data);
3. Efforts: between +/- maximum limit (advanced data).

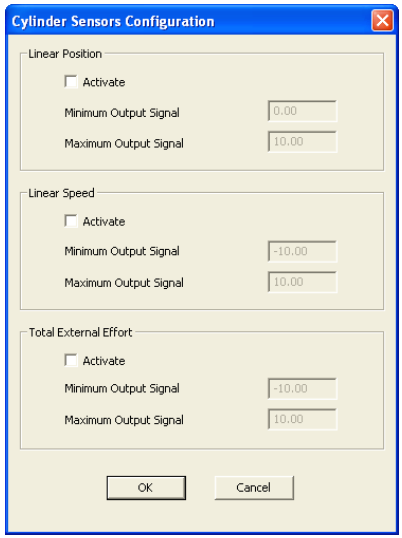


Figure 4–119: "Cylinder Sensors Configuration" Dialog Box

4.5.4.4 Directional Valve Sensors

When a directional valve is inserted on a diagram, a percentage variable type without name is created in the "Variable Manager". Modify it in the manager by giving it a name and it could be used in a variable assignment.

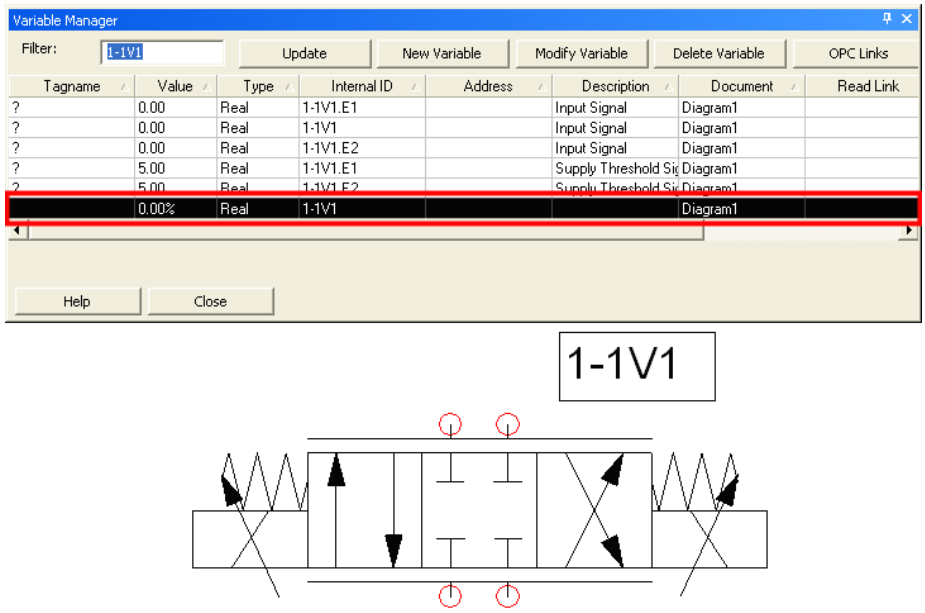


Figure 4-120: Proportional Valve with LVDT (%) Visible in the Variable Manager

For On/Off directional valves, you must use the "Position Sensor" normally opened or closed.

4.5.5 Set Point Devices

4.5.5.1 Analog Devices

The following components generate analog signals to control electromagnets used on proportional components such a directional valves, pressure and flow valves, pumps, etc. These devices can be controlled by linking the set point device output variable to a proportional component.

When in simulation mode, click on the set point device to activate it.

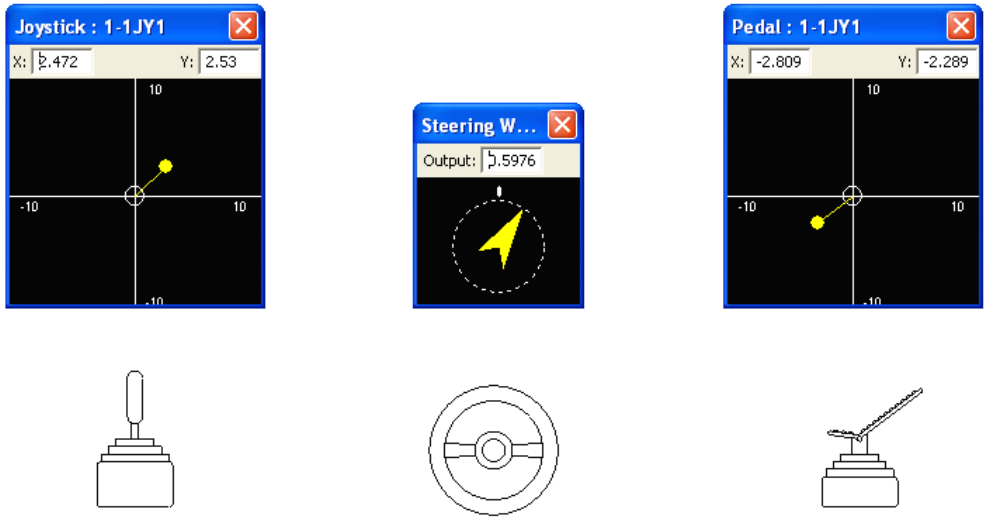


Figure 4–121: Set Point Device in Simulation Mode

In order to control multiple devices simultaneously, it is possible to send the real command of devices connected via USB ports to the computer. If these devices are compatible with *Windows*, they will be compatible with Automation Studio™.

It is then possible to assign a hardware interface to joysticks, steering wheels and pedals included in the proportional (hydraulic and pneumatic) workshops. It is also possible to control, for example, a hydraulic system via electronic controls available commercially. This can be configured by a drop-down list in the configuration branch.

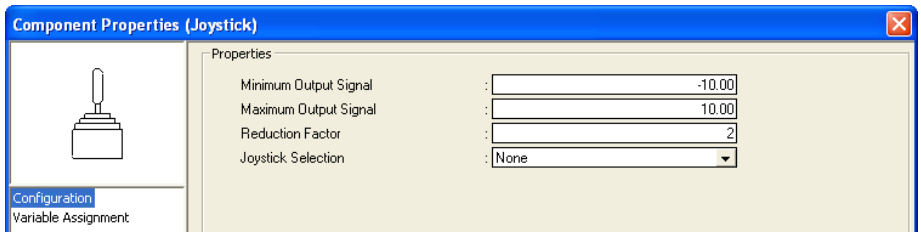


Figure 4-122: Hardware Interface Assignments

4.5.5.2 Graph Generators

For each generator defined by a curve, it is possible to assign a Boolean variable that allows triggering the curve reading with this Boolean variable. This variable is created only if the

"Controlled by variable" box is checked in the component configuration dialog box.

The manual control is preserved and allows you to change the variable's state.

Graph Variable Generator

The command generated by the following component comes from time curve configurable in the component properties dialog box.

Figure 4–123: Graph Variable Generator

This component generates a signal based on the curve shown in its properties ("Set Point Curve"). At each moment this signal is stored in a variable.

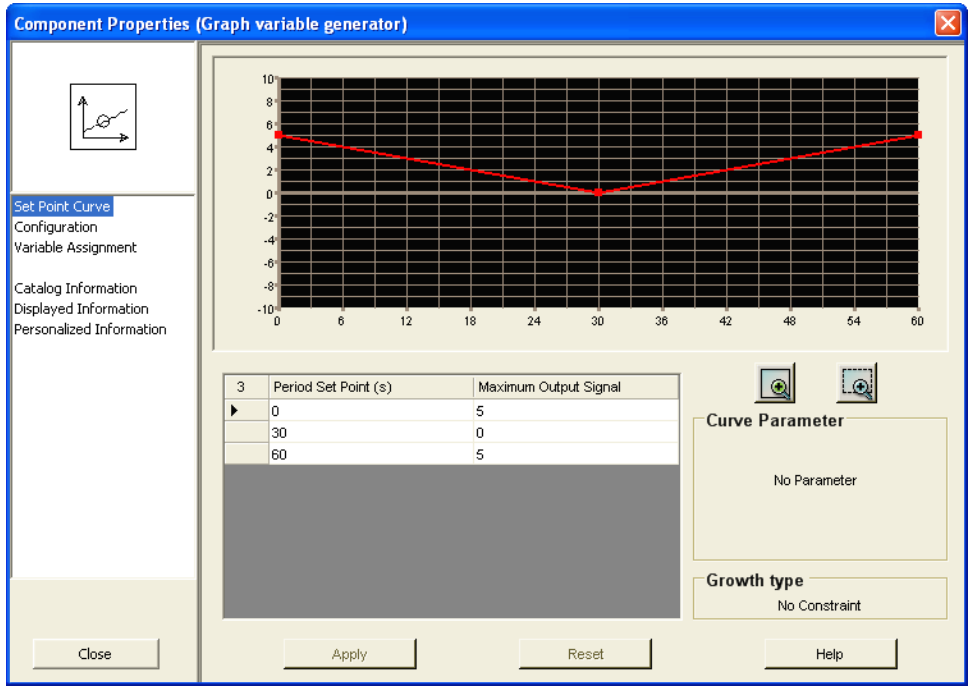


Figure 4–124: Graph Variable Generator Curve

The signal generation can start with the simulation, or with a manual signal. The signal can be

generated in a loop mode or not.

Lookup Table

This component allows creating a function Y(X), whose curve values are contained in a spreadsheet.

Figure 4-125: Component Graph Y(X)

A Boolean variable allows activating the Y reading as an X function. This component increases the number of possibilities to manage the variable.

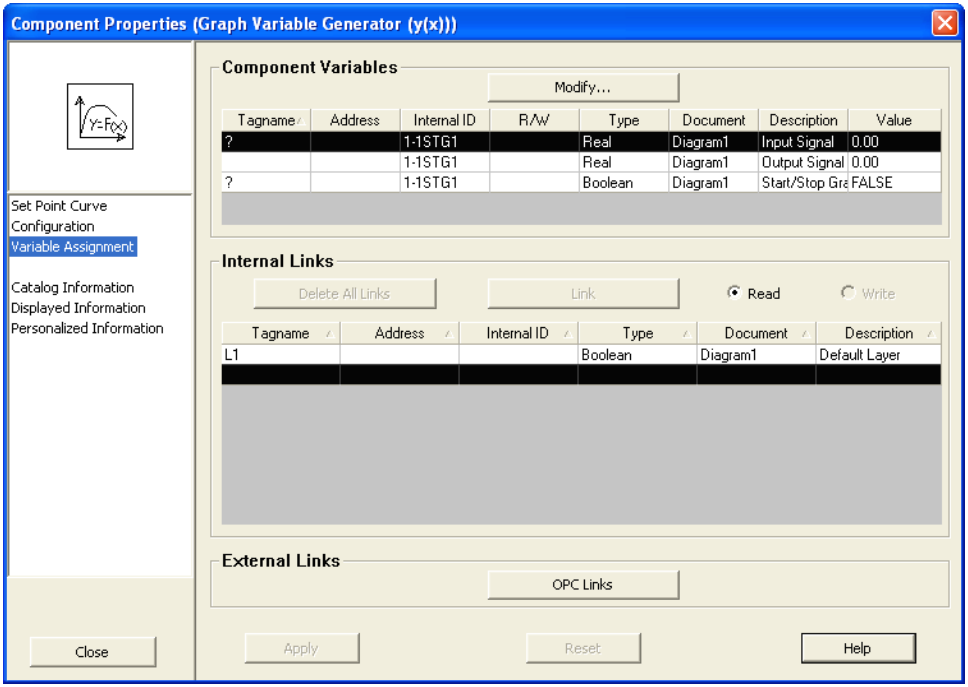


Figure 4-126: Lookup Table Variable

The Look-Up Table allows assigning a speed operation curve to a thermal motor as a torque function.

4.5.6 Control Device

The control device is an important component to build an electronic control for servo valves. This component outputs a variable made of a combination of variables supplied by other components or generated.

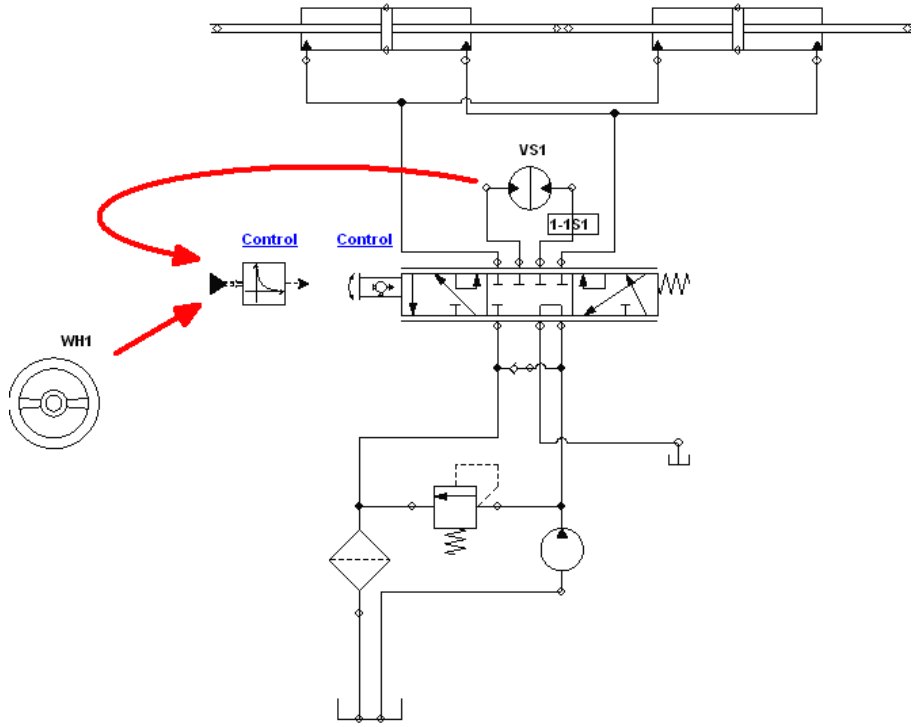


Figure 4–127: Circuit with Control Device

In the above example, the controller compares, using the equation *Error = 1-1WH1.WH1 - 1-1S1.VS1,* the wheel output to the metering unit output to evaluate the cylinder's position of this power steering. In this case, the control device output will take the value shown in the following figure.

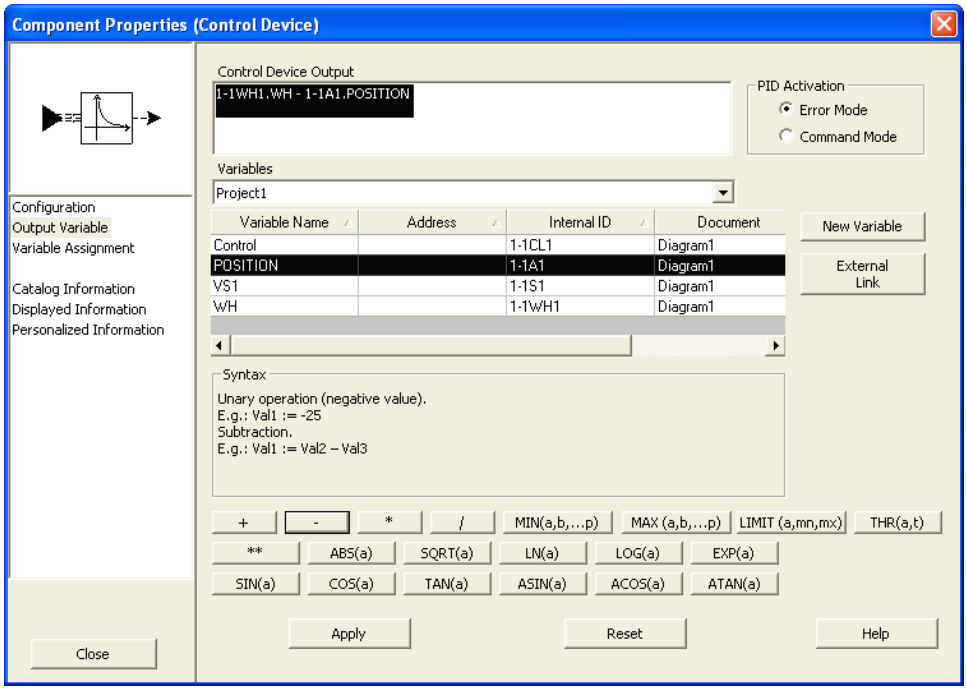


Figure 4–128: Example of Computing the Output of a Controller

The error mode selected in the dialog box implies that the controller uses a PID for its output. In simulation mode, you can access the proportional, integral and derivative gain by clicking on the controller.

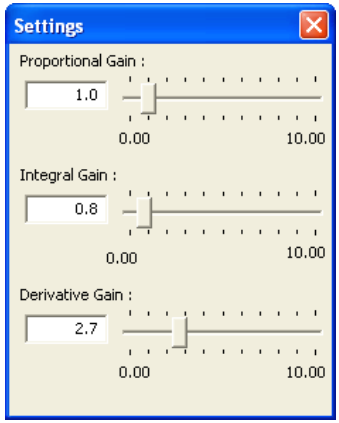


Figure 4–129: PID Settings

When in error mode, the output signal is the difference between a requested set point and the actual value. This error can be a simple difference (Set point – Actual value) or a more complex equation including many variables.

If the command mode is selected, the value is directly sent to the receiver without passing through the PID.

4.6 Special Simulation Considerations

4.6.1 Customized Reservoir

A pump can drive (or suck) oil out of a connector positioned higher than the reservoir volume.

When a customized reservoir is filled at full capacity (depending on its technical characteristics) it is possible to introduce 25% more fluid.

4.6.2 Directional Valves

The directional valves generate instability when the "Bulk Modulus Simulation" option is checked and the involved oil volume is minimal. This behaviour depends on the oil used.

When the fluid volume is minimal, check valves used in the directional valves generate higher compressibility than when they are used alone.

4.6.3 Fluid Lines

A very short dynamic line can cause simulation mistakes.

5 Component Builders

Builders are available for the following components:

- Pneumatic and hydraulic cylinders;
- Pneumatic and hydraulic directional valves;
- Hydraulic proportional directional valves;
- Pneumatic sequencers;
- Spool position for directional valves.

5.1 Cylinder Builder

This builder only modifies the cylinder visual aspect; it does not affect the cylinder's behaviour in simulation, except for brakes and springs.

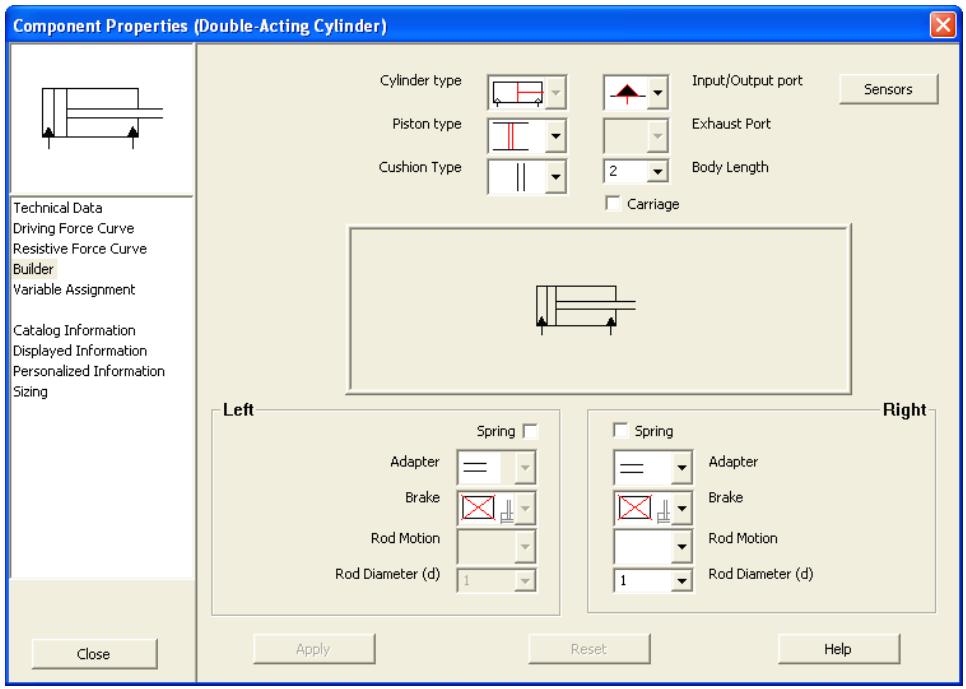


Figure 5–1: Cylinder "Builder» Branch

Zone	Description
Type of Cylinder	Allows you to select one of the following types of cylinders: Single Acting (Retraction), Single Acting (Extension), Double effect, Rodless Double effect, Double effect Double Rod, and Rodless. However, once it has been selected it is no longer possible to change the type of cylinder. In order to modify the type of cylinder, another cylinder from the main library must be selected. The cylinder's categories are sorted by type.
Piston Type	Allows you to select one of the following types of pistons: Magnetic or Non-Magnetic.
Cushion Type	Allows you to select one of the following types of cushions: None, Left, Right, Left & Right, Left variable, Right variable, Left & right variables, Left & right variable, Left variable & right, Left & right variables.
Carriage	Allows you to indicate the presence or the absence of a carriage.
Input/Output port	Allows you to select one of the following input/output ports: exhaust or input port.
Exhaust port	Allows you to select an exhaust port.
Adapter	Allows you to select one of the 10 adapters.
Brake	Allows you to select one of the following brakes. Four options: None, Double Effect, NO and NC
Rod Motion	Allows you to select one of the following rod motions. Four options: Rotation, No Rotation). Symbolic only.
Spring	Allows you to include a spring, or not.
Rod Diameter	Allows you to select one of the following rod diameters: an integer value 0, 1, 2 or 3.

Zone	Description
Body Length	Allows you to determine the body length: an integer value between 2 and 15.
Sensors	Opens the cylinder sensors dialog box. Three sensors are available: position, speed, and force.

Brakes can be configured. The frequently asked questions for this topic are:

- What is the applied resistive force on the cylinder rod when the brake is applied?

  *Resistive force(kN) = Spring Force (kN) – Brake Pressure (kPa)*0.01(kN/kPa)*

- What is the applied resistive force on the cylinder rod when the brake is off?

  *Resistive force(kN) = Brake Pressure (kPa)*0.01(kN/kPa) - Spring Force (kN)*

- Can the spring force be changed?

  The spring force cannot be modified.

The hydraulic and pneumatic springs forces are:

Pneumatic spring force	Hydraulic spring force
Normally applied: 10.0 kN	Normally applied: 100.0 kN
Normally released: 1.0 kN	Normally released: 10.0 kN

- What is the required pressure to fully open a pneumatic brake normally applied?

  It requires 1000.0 kPa, which means 1 bar or 14.7 psi.

- What is the required pressure to fully open a hydraulic brake normally applied?

  It requires 10000 kPa, which means 100 bar or 1470 psi.

- What is the required pressure to reach the resistive force threshold of the hydraulic brake normally released?

  It requires 1000.0 kPa, which means 10 bar or 147 psi.

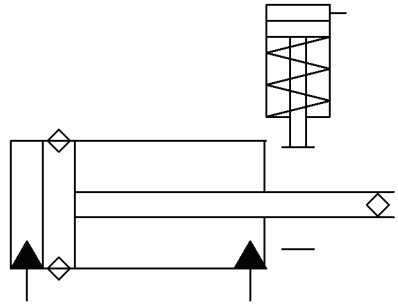


Figure 5–2: Cylinder with Brake

5.2 Directional Valves Builder

5.2.1 Directional Valves Conversion

In Automation Studio™ version 5.4, builders are totally new.

☞ It is POSSIBLE but NOT MANDATORY to update your directional valves. If you are satisfied with your current diagrams designed with a previous Automation Studio™ version, we recommend keeping them as is.

Indeed, the directional valves update redraws only the positions and the unitary commands (such as spring, pilot, solenoid). Then the mixed commands must be configured (such as pilot+directional valve) and also the positions with their internal channels. On the other hand, if you begin a new project, then use the directional valves from the main library.

How to differentiate an “old generation” from a “new generation”?

Method 1: When forcing the valve position, the red check mark, identifying that the valve was forced, is different, as shown in the following figure.

Figure 5–3: Position Forcing Old (Left) and New (Right) Generation

Method 2: In the directional valve properties dialog box, if the valve is of the old generation, the "Upgrade" option is available in the side menu.

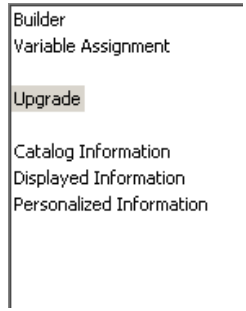


Figure 5–4: "Old Generation" Dialog Box Menu Option

For all "old generation" directional valves, a warning message informs you that an upgrade will improve the component's performance.

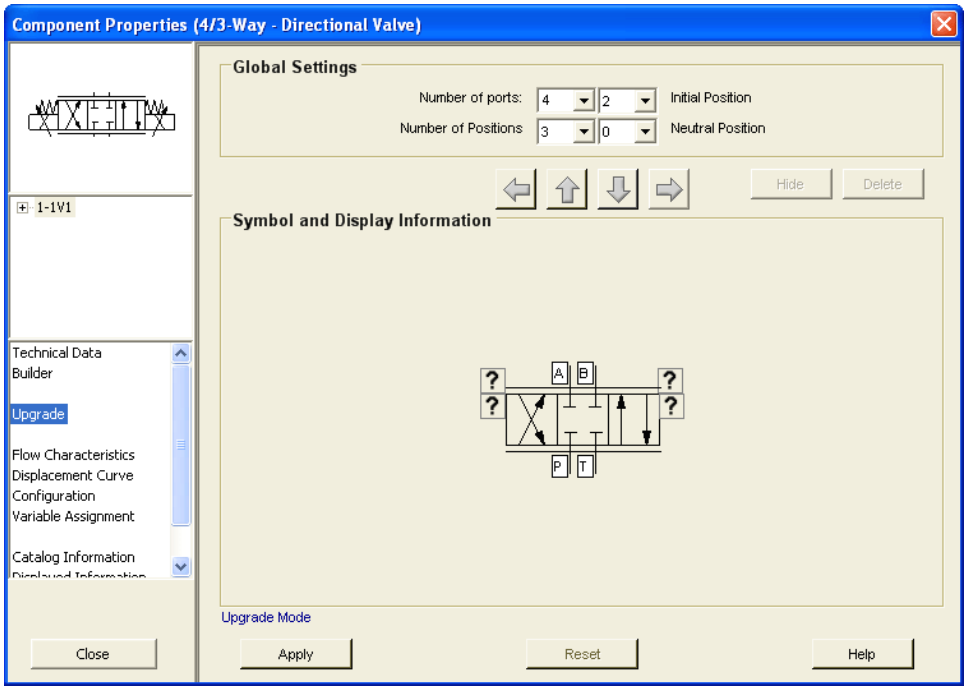


Figure 5–5: Old Generation Directional Valve "Component properties" Dialog Box

If you select the "Upgrade" option, the new directional valve is displayed on the right side of the window. The On/Off directional valves are completely redone.

☞ Upgrading a directional valve is not reversible.

With proportional directional valves, only the control needs to be reviewed, since previously controls did not have a realistic simulation.

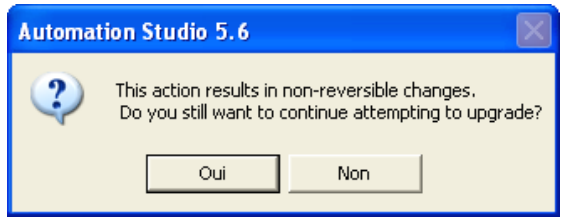


Figure 5–6: Warning Message Before Upgrade

Accessing the Position Sub-Component Properties

In proportional mode, using the previous builder, the pressure drop could be linked to be specific flow.

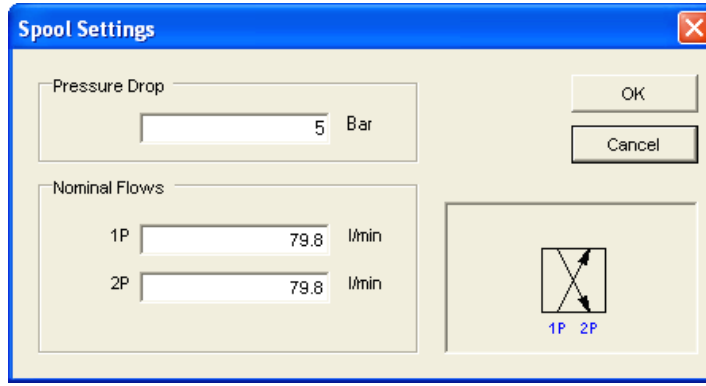


Figure 5–7: Previous Method to Adjust the Internal Channel Properties

With the "new generation" builder, the sub-component's properties can be accessed using the tree structure of the component or by zooming on the component zone (zooming is performed by selecting the component's representation and rolling the mouse scroll wheel) and selecting the sub-component with a click.

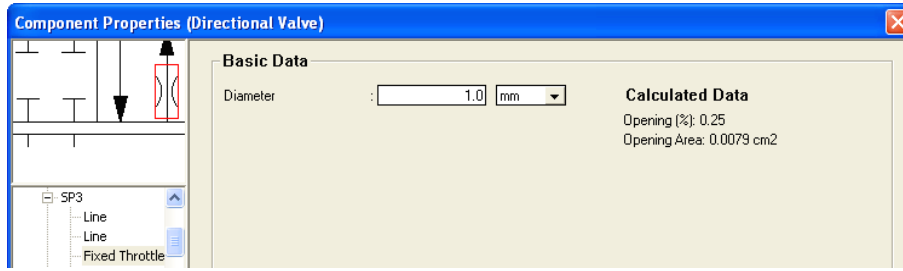


Figure 5–8: New Method to Access Internal Channel Properties

5.2.2 ON/OFF Directional Valves Builder

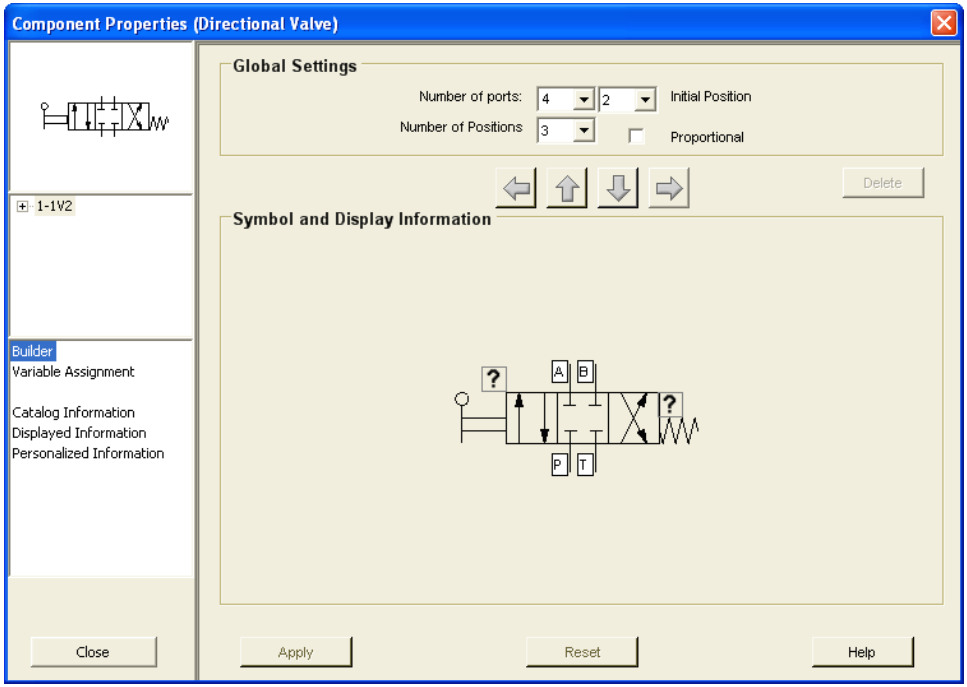


Figure 5–9: Directional Valve "Builder" Branch

Zone	Description
Number of ports	Field to assign number of connectors of the valve. The connector position is assigned upon selection of the first spool. Changing the number of ports re-initializes the entire directional valve. The valve present before will no longer be part of Automation Studio™. Refer to section *5.8 Position Editor* for the position creation procedures.
Number of positions	Field to assign the number of positions for the valve. Changing the number of positions re-initializes the entire directional valve.
Initial Position	Field to select the valve initial position. The initial position is the position where the connector's configuration is represented.

Zone	Description
Proportional	Check box to identify the valve as being proportional. This option is only symbolic, the valve will be simulated as a non-proportional valve. To get a true proportional component, they must be inserted directly from the proportional hydraulic library.
Arrows	Controls to move the position of the valve controls upwards, downwards, to the left or to the right.
Delete	Deletes the selected control.
Apply	Applies the changes.
Reset	Resets the symbol to its initial state.
Help	Button to access the online help dialog box.
Symbol	Representation of the directional valve with the modifications.

Initially, a new directional valve does not have controls, question marks are present on each side of the spool, and can be replaced by other controls in series or parallel to the existing ones.

To reset a directional valve, just change the number of ports or positions.

You can select a specific spool position from a wide selection. There are as many representations as positions.

An editable field containing a default port number is created for each connection ports.

Distributor Configuration

To configure a directional valve:

1. Assign the number of ports;

   Once the valve is configured, if the number of positions of ports or the initial port number is changed, the valve is reset.

2. Assign the number of positions;

3. Assign the initial position number;

4. Check the proportional box, if required;

5. Double-click on the first spool position represented by a question mark.

The "spool selection" dialog box opens. Select the appropriate tabs that correspond to the desired connector positions. If you cannot find a specific spool configuration, you can move to a custom selection in a personalized library, provided that you have the "Position Editor" license. Refer to the section *5.8 Position Editor* for more information about how to create a personalized spool configuration.

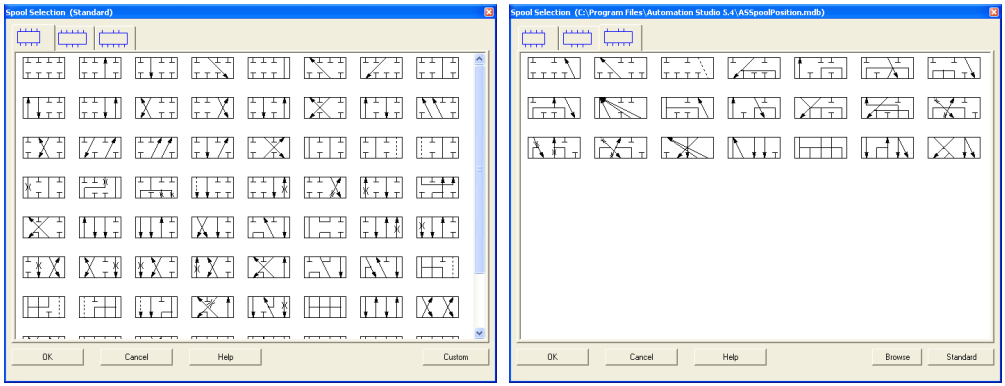


Figure 5–10: Standard and Custom Spool Selection Dialog Box

A custom spool position can be part of various databases. You can access this database by clicking on the "Browse" Button.

1. Select the desired position by double-clicking on it or by clicking on the desired position followed by one click on the "OK" button. Repeat the same procedure for each position;

2. Select a valve control or a control question mark by double-clicking on it;

An existing control will position another control in series or parallel if a question mark is already present in a serial position or parallel position.

3. Repeat the same procedure on the other side.

Once the directional valve is configured, you can move the controls one by one or several at a time. To select more than one control, select the first one and then hold the CTRL key down and click on the other controls to be selected.

A selected control can be deleted using the "Delete" button

5.2.3 Proportional Directional Valve Builder

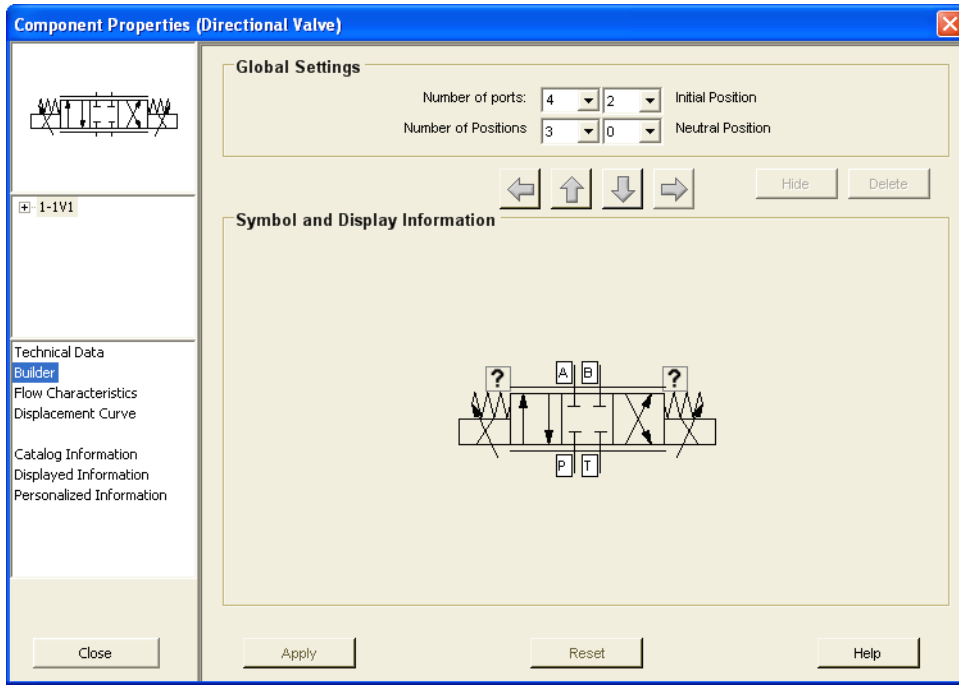


Figure 5–11: Proportional Directional Valve "Builder" Branch

Zone	Description
Number of ports	Field to assign the number of ports of the valve. Changing the

Zone	Description
	number of ports resets the valve configuration. The number of port is displayed in color if Automation Studio™ does not support it. Refer to section *5.8 Position Editor* for the position creation procedures.
Number of positions	Field to assign the number of positions for the valve. Changing the number of ports resets the present valve configuration.
Initial Position	Field to assign the initial valve position. The port number will be attached to this position. However, the control must be configured properly for the initial position both in edition and simulation.
Neutral Position	Field to assign the neutral position. The valve moves to this position when the electrical control is defective (variable value below threshold).
	Controls to move the position of the valve controls upwards, downwards, to the left or to the right.
Delete	Deletes the selected control.
Apply	Applies the changes.
Reset	Resets the symbol to its initial state.
Help	Button to access the online help dialog box.
Symbol	Representation of the directional valve with the modifications.
Hide	Button to hide a spool position. This function is made available to simulate different flow rate for a same position.

Initially, a new directional valve does not have controls, question marks are present on each side of the spool, and can be replaced by other controls in series or parallel to the existing ones.

To reset a directional valve, just change the number of ports or positions.

You can select a specific spool position from a wide selection. There are as many representations as positions.

An editable field containing a default port number is created for each connection ports.

Proportional Distributor Configuration

To configure a directional valve:

1. Assign the number of ports;

Once the valve is configured, the valve is reset when you change the number of position, the number of ports or the initial port number.

2. Assign the number of positions;

3. Assign the initial position number;

4. Check the proportional box, if required;

5. Double-click on the first spool position represented by a question mark;

The "Spool Selection" dialog box opens. Select the appropriate tabs that correspond to the desired connector positions. If you cannot find a specific spool configuration, you can move to a custom selection in a personalized library, provided that you have the "Position Editor" license. Refer to the section *5.8 Position Editor* for more information on how to create a personalized spool configuration.

A custom spool position can be part of various databases. You can access this database by clicking on the "Browse" Button.

6. Select the desired position by double-clicking on it or by clicking on the desired position followed by one click on the "OK" button. Repeat the same procedure for each position;

7. Select a valve control or a control question mark by double-clicking on it;

   The control selection dialog box opens. You can move controls by a simple drag-and-drop.

8. Repeat the same procedure on the other side;

Once the directional valve is configured, you can move the controls one by one or several at a time. To select more than one control, select the first one and then hold the CTRL key down and click on the other controls to be selected.

A selected control can be deleted using the "Delete" button.

9. Once the symbol is configured, you can change the spool position configuration, the displacement curves and flow characteristics of the directional valve.

Hide Property Example

The "Hide" property can be used to create a pressure control valve from a directional valve. This function gives a continuous displacement to an On/Off valve. In the following example, the pressure control valve is created with a hidden spool position (channel with check valve); the hidden position is marked with an "H". Notice that the visible position actually has plugs (drawn in white), and the arrow is actually a drawing. This position is customized.

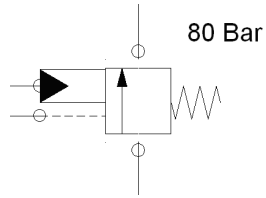


Figure 5–12: Pressure Controlled Valves Built Using a Proportional Directional Valve

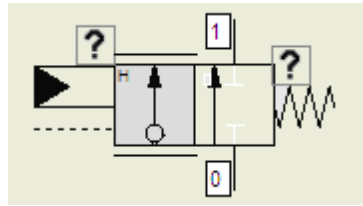


Figure 5–13: Builder Representation of the Above Valve

If for a given position, you want different flow configuration, all you have to do is to copy this position in the builder.

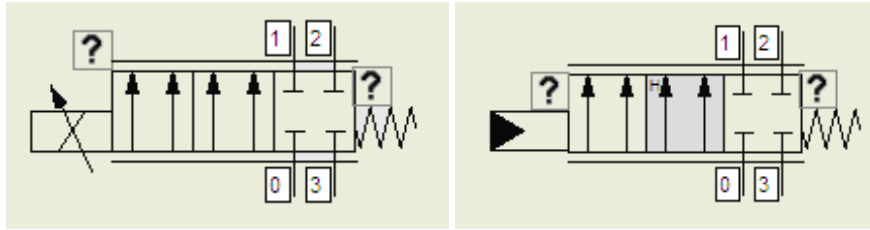


Figure 5–14: Copying a Position

This position is created with two channels (left and right). You must now set the flow rate for position 1 and position 2 as needed.

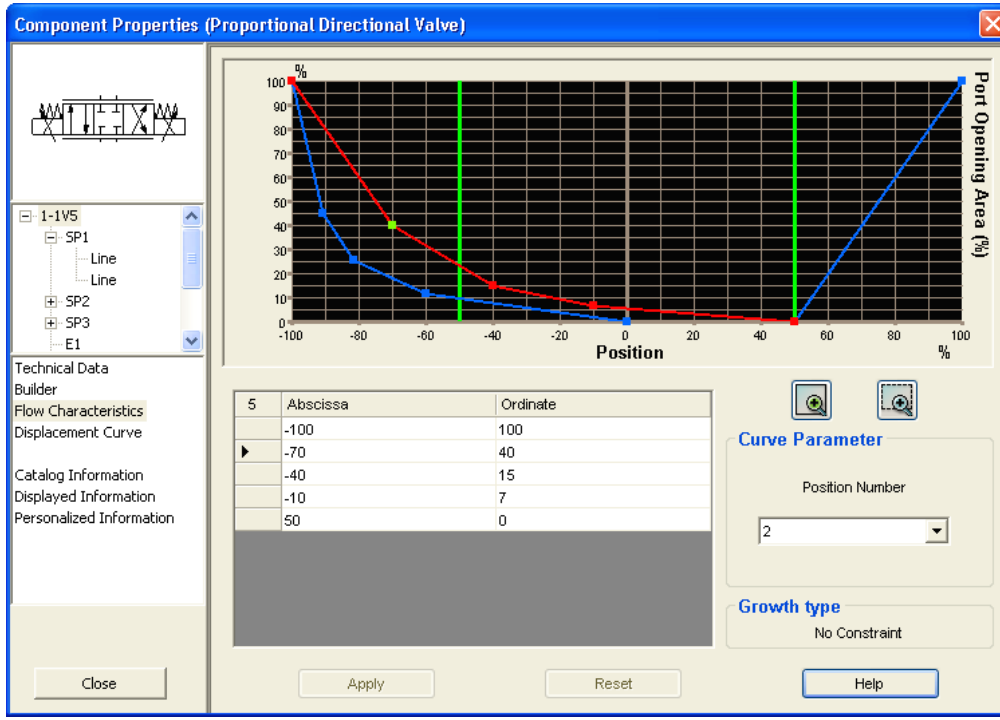


Figure 5–15: Connector Opening Behaviour for Positions 1 and 2

For each of the identical positions, one channel must be deleted. The left position keeps the left channel open; the right position keeps the right position open. To close a channel, assign a nominal flow of 0 l/min, and no pressure drop.

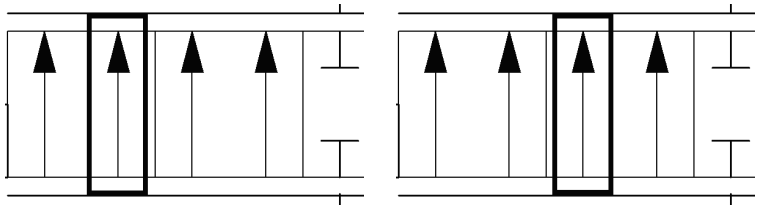


Figure 5–16: Lines to be Closed

The only thing that remains to be done is to hide one of the duplicated positions. One of the spool positions is greyed out.

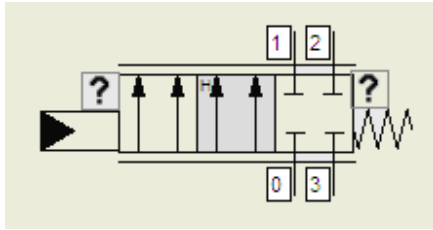


Figure 5–17: Hiding the Duplicated Positions

Following is an example of such a configuration. There is different flow for the different internal channels within the same spool position.

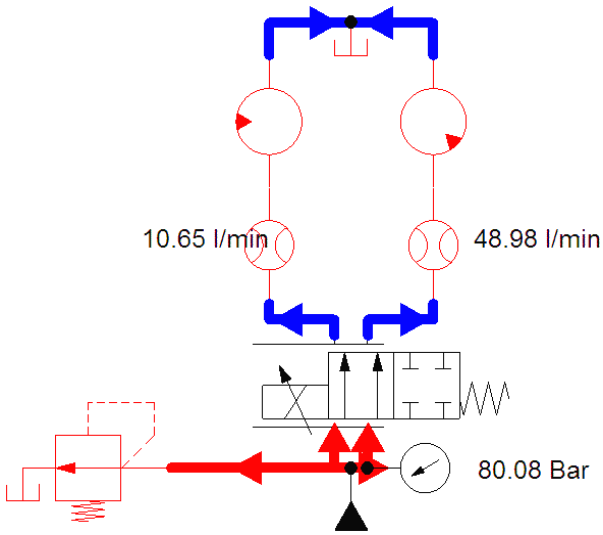


Figure 5–18: Hidden Properties

5.3 Sequencer Builder

Sequencers are pneumatic components controlled by sequential automation. Once it has received a signal and that all requirements are met, the sequencer moves to the next step. The sequencer cells are side by side and each cell corresponds to a step. The sequence can be of two types: PAC (Programmable Air Logic) or Shift Register. The number of cells can be from 1 to 10, the default quantity is 2.

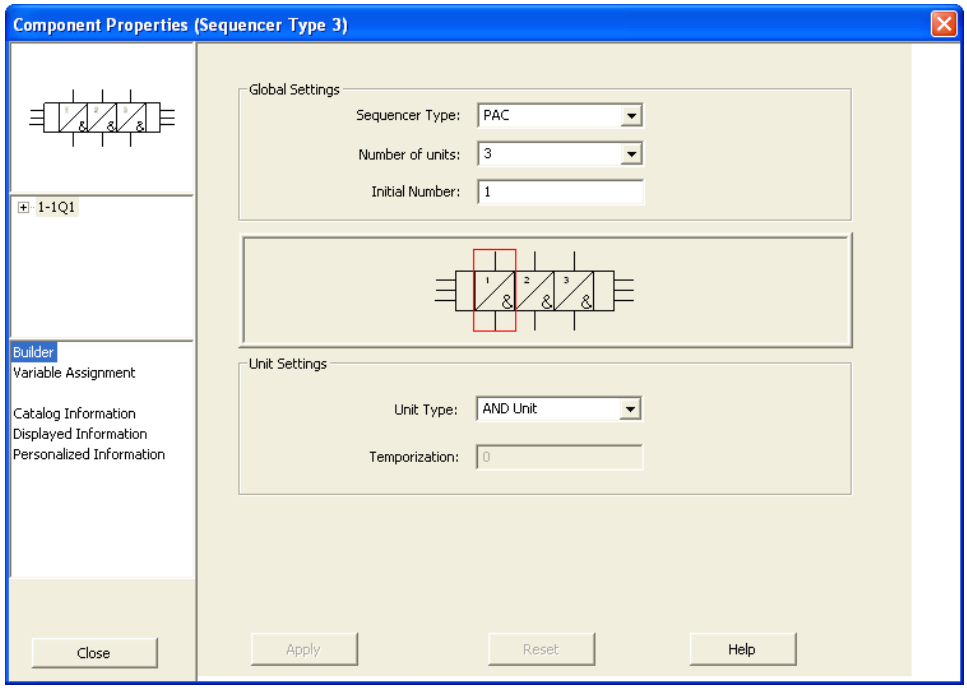


Figure 5–19: Sequencer "Builder" Branch

Zone	Description
Sequencer Type	Drop-down list to select one of the following sequencers: Type I, Type II, PAC, Shift Register.
Number of units	Drop-down list to select the number of units in the sequencer (0 – 10).
Initial Number	Field to assign the initial number for the first cell.

Zone	Description
Cell Types	For each cell, you can select the following functions: Amplifier, AND Unit, NO Unit, YES Unit, Solenoid NC, Bleed Sensor Relay and Timer.
Temporization	Active only for the “Timer” type, to specify the delay time.
Symbol	Unit representation.

Sequencer configuration

To configure a sequencer:

1. Select a type of sequencer;

2. Select the number of unit;

3. Select the initial number.

Based on the type of sequencer (PAC or Shift Register), select a cell on the sequencer then the unit type.

Type I Sequencer

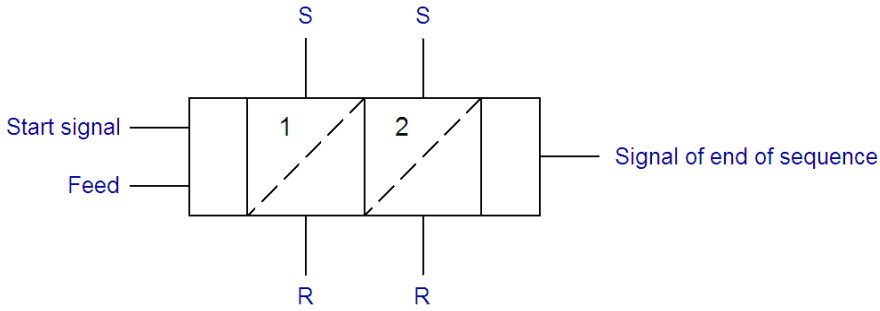


Figure 5–20: Type I sequencer connections

The type 1 sequencer works if the start signal is under pressure. Port S1 is pressurized and the components connected to this port are activated. When this step is complete, the return signal

R1 is pressurized and step 2 is pressurized. This action affects S1 which is now depressurized and starts the S2 step. This cycle will repeat itself for a number of times corresponding to the number of units. At the end of the sequence, the exit stage signal is under pressure and the sequencer is ready for another cycle.

The start and exit signals must be activated simultaneously to start the cycle. If the pressure supply feed is not under pressure, the sequencer will not react.

Type II Sequencer

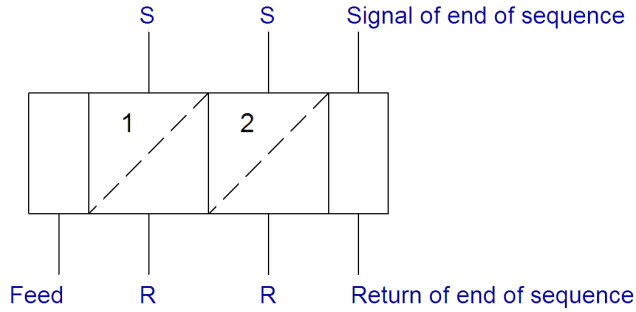


Figure 5–21: Type II Sequencer Connections

The type 2 sequencer works in the same manner as the type 1 sequencer. Only the connections vary.

PAC Type Sequencer

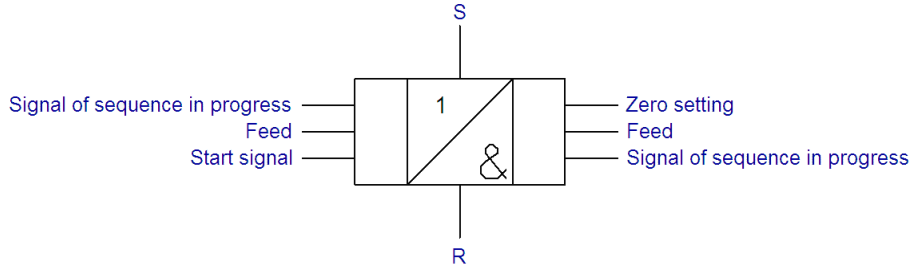


Figure 5–22: PAC Type Sequencer Connections

Each unit of a PAC sequencer has three built-in logical functions:

- A memory function used for supplying the output. The pressure also resets to zero the

preceding units and triggers the next unit;

- An OR function resets to zero the preceding units when the output is under pressure or when a reset signal comes from the next unit;

- A function that allows the transition to the next step. This function can be of various types depending on the desired effect or the nature of the input signals. This function has the following types:
    i) Amplifier;
    ii) AND;
    iii) YES;
    iv) NO;
    v) Electro-valve NC;
    vi) Bleed sensor relay (positive);
    vii) Timers.

Shift Register Sequencer

Each unit of a Shift Register sequencer has two built-in logical functions:

- A memory function used for supplying the output. The pressure also resets to zero the preceding units and triggers the next unit.
- A function that allows the transition to the next step. This function can be of various types depending on the desired effect or the nature of the input signals. This function has the following types:
    i) Amplifier;
    ii) AND;
    iii) YES;
    iv) NO;
    v) Electro-valve NC;
    vi) Bleed sensor relay (positive);
    vii) Timers.

5.4 Pump Builder

Some components, for which Famic Technologies Inc. does not have access to the internal technical data, are displayed as concept diagrams. In Automation Studio™, these concept

diagrams are found in a custom library. The concept diagram is modeled based on manufacturers' specifications. During simulation, the diagram's representation is not animated but the parameters can be tuned.

Each pump or motor found in a personalized library represents only one model of a specific type. To limit the number of symbols in the library, some pumps can be configured. A builder for a specific type will provide all models.

The technical data are as specified by the manufacturers when you select a pump. However, the technical data can be changed as for all other components.

Only the major connectors were modeled, none of the measuring ports are accessible

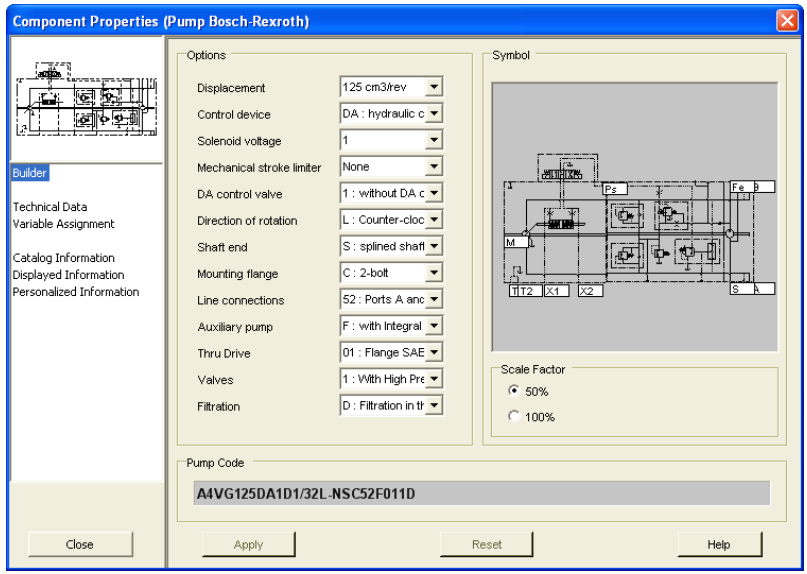


Figure 5–23: Pump "Builder" Branch

Options

Manufacturers' specifications required to build the pump's code and the pump's appropriate symbol. Depending on the content of some fields, other fields may appear or disappear according to the selected pump model. Some of the "Options" fields are passed to the "Technical Data" but cannot be changed.

The displacement of configurable pumps can be edited in the pump builder. The catalog part number will be deleted.

Symbol

Symbol of the selected model. The symbol is automatically adjusted according to the selected model (principally, the "control mechanism"). The final symbol is displayed in the "Symbol" zone. The ports connections can be directly edited in the zone.

The symbol size is 50% of the ISO size. You can get the full size by choosing so in the configuration dialog box.

Pump Code

Manufacturers pump model code. The code is built automatically according to the content of the various "Option" fields offered. This code is also part of report and BOM.

5.4.1 Accessing to Internal Components

It is possible to discretize a configured pump inserted in a circuit. This gives access to all subcomponents belonging to the basic schematic.

To access internal components:

1. With a right click, open the shortcut menu of the configurable pump;

2. Choose the "Access to Internal Components" option.

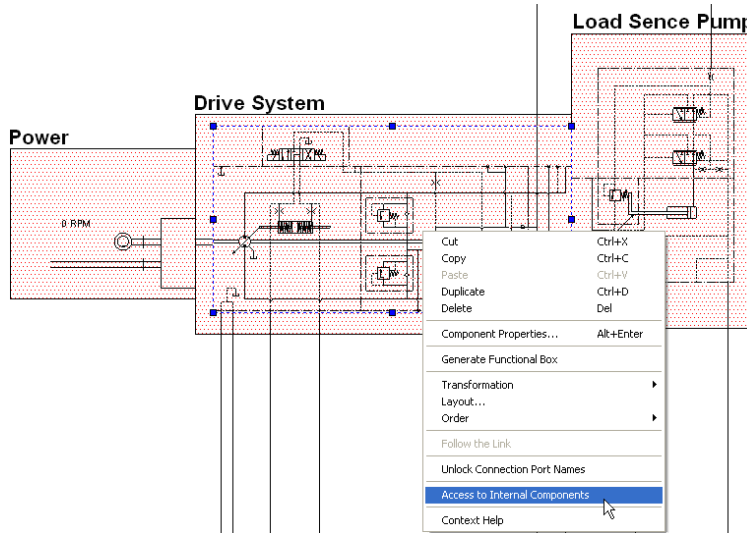


Figure 5-24: "Access to Internal Componenst" Feature

The pump is discretized.

The advantage of converting a pump into a discrete pump is in the possibility to visualize flows and to modify internal component parameters illustrating the main functioning of the pump.

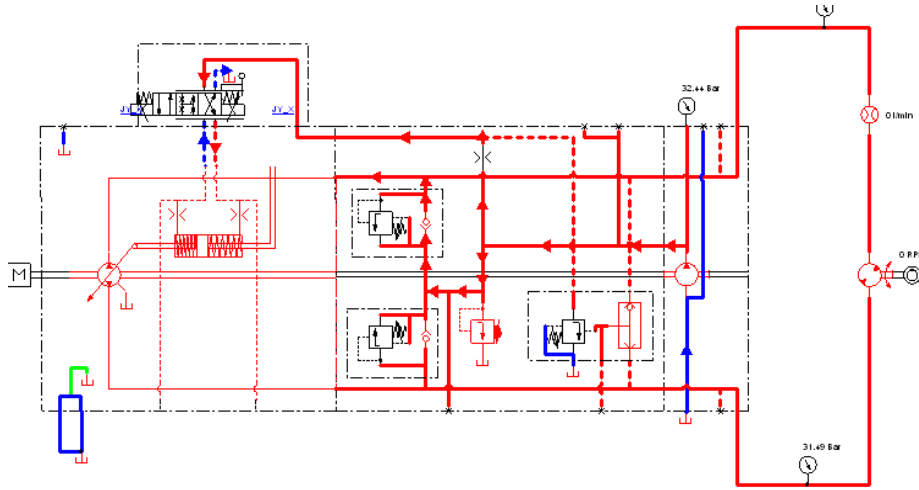


Figure 5-25: Discrete Pump Simulation

Finally, swatch plate symbols are available, in order to allow creating of personalized pumps.

5.5 Reservoir Builder

The customized reservoir can transfer fluid by gravity, from one of its connectors to another lower reservoir. A pressure greater than the hydrostatic pressure can reverse the flow. The reservoir can be configured in the "Builder" branch in the "Component Properties" dialog box.

You can choose the number of ports, their types, their positions on the reservoir's edge, their heights and the flow coefficients at their inputs and outputs. Pressure, temperature or level sensors can also be used. In a model that uses a gravitational field, a reservoir with many ports and sensors can better illustrate (as compared with a tank) draining concepts from one reservoir to another.

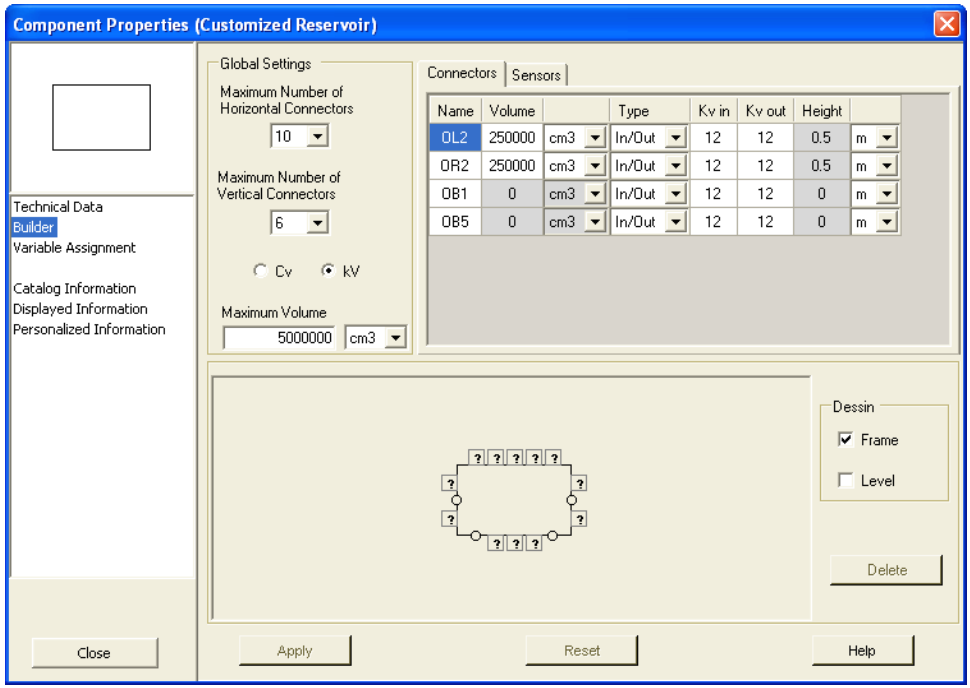


Figure 5-26: "Builder" Branch for Reservoirs, "Connectors" Tab

Zone	Description
Maximal Number of Horizontal Connectors	Specifies the maximum number of upper and lower horizontal connectors.
Maximal Number of Vertical Connectors	Specifies the maximum number of left and right vertical connectors.
Connectors tab	Specifies, for each connector, its name, height, type, flow coefficient *Kv* (in/out), and, whether is has a check valve.
Sensors tab	Specifies, for each sensor, its name, what it measures and in which unit, its connector and its height.
Frame	Specifies whether to show the reservoir's outline.
Level	Specifies whether or not to show the horizontal level (this line is not updated in the simulation mode).
Delete	Deletes the selected connector.

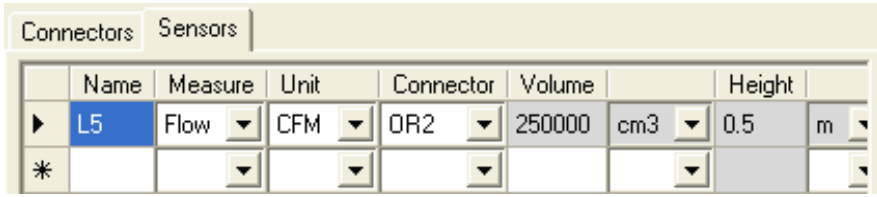


Figure 5-27: "Builder» Branch for Reservoirs, "Sensors" Tab

5.6 Gearbox Builder

A gearbox transfers mechanical power from a power source (electric motor, combustion engine) to one or more targets (pump, compressor). This component distributes the source's mechanical power through its angular velocity and torque. The transmission ratio together with the input velocity and torque will determine the outputs speeds and torques. The power at the input is equal to the sum of the power at the outputs modulo the transmission efficiency.

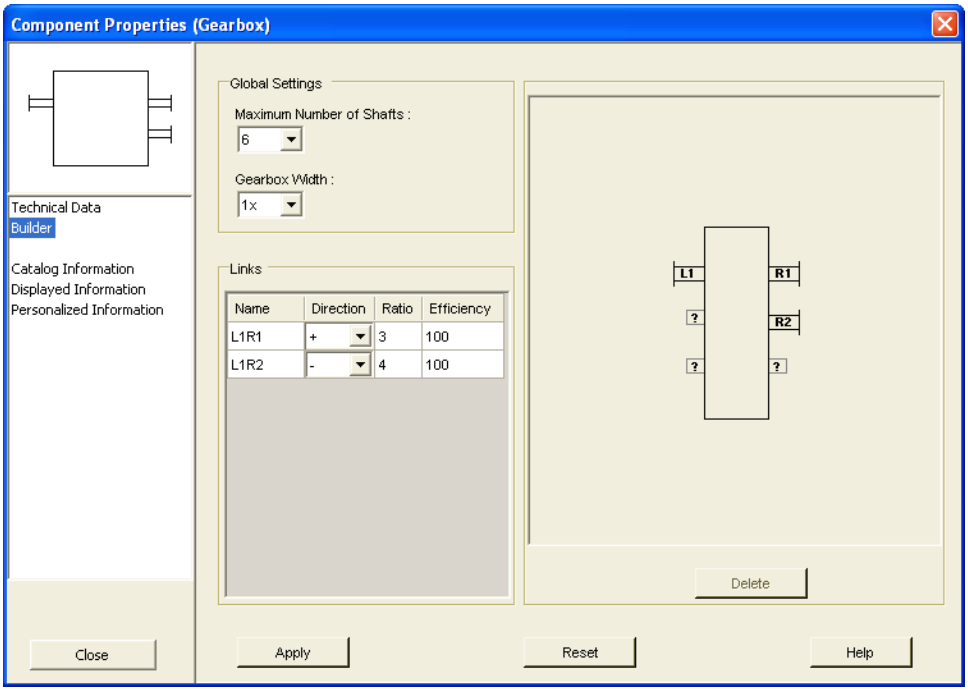


Figure 5-28: "Builder" Branch for a Gearbox

Zone	Description
Maximum Number of Shafts	Specifies the maximum number of shafts with an output connector. Internal details of the gearbox are not considered.
Gearbox Width	Specifies the symbol width on the document.
Links	Specifies the transmission ratio for different pairs of shafts.
Delete	Deletes the selected shaft.

5.7 Pulse Counter Builder

This component allows you to count pneumatic impulses and send a signal when the counter

reaches its activation value. Many options are available.

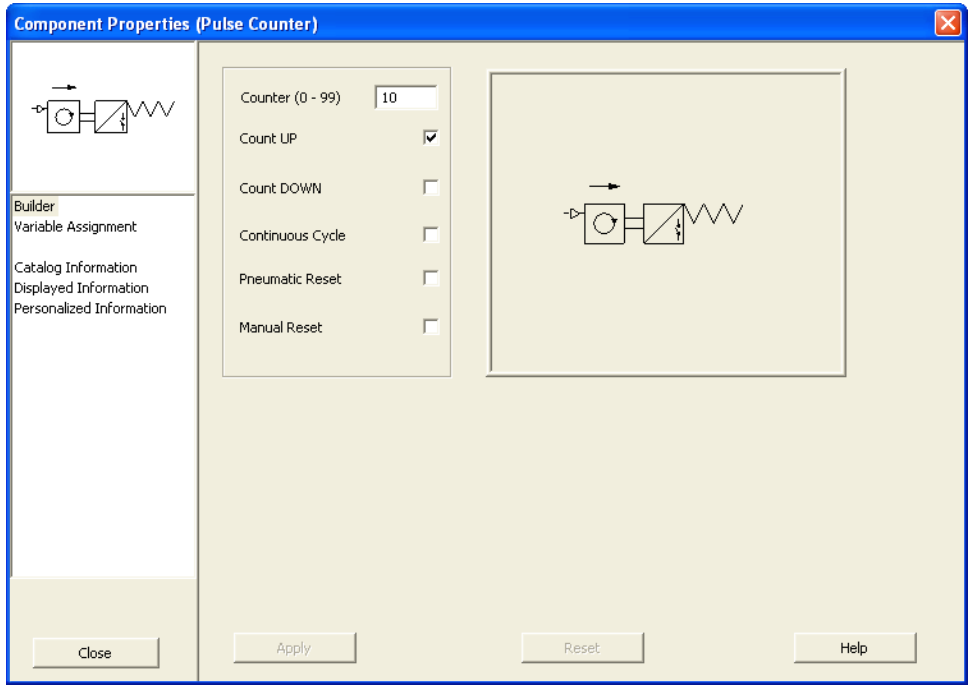


Figure 5-29: "Builder" Branch for a Pulse Counter

Zone	Description
Counter (0-99)	Allows you to define how many impulses are needed to activate component variable.
Count UP	Allows you to increase number of counted impulses.
Count DOWN	Allows you to decrease number of counted impulses.
Continuous Cycle	Allows you to automatically reset the counter when the counter value exceeds the activation value.
Pneumatic Reset	Allows you to reset the counter with an external pneumatic signal.

Zone	Description
Manual Reset	Allows you to reset the counter by pressing the reset button in simulation mode.

5.8 Position Editor

This workshop is available in the Professional version only.

5.8.1 Introduction

An optional function is available to create and generate the configuration of your own spool positions. Even if the function uses the common editor, its result is applied only to the position library, hence the name Position Editor.

5.8.2 Spool Position Generation and Operation Scenario

Generating a Position

These steps show how to create a spool position. Depending on your license, you will be able to create, modify and simulate spool positions for hydraulic and/or pneumatic directional valves, “On/Off” or “Proportional”.

Our example will provide the steps to create a spool position with eight connectors

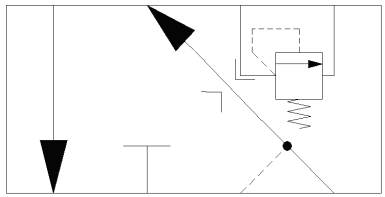


Figure 5–30:Custom Position Example

The steps are as follows:

- Configure the editor;
- Draw the position;
- Generate the position;

- Adjust properties in sub-components;
- Save the customized spool position.

Editor Configuration

Preferably, the grid must be displayed to properly align the components, and to avoid visual errors when zooming and placing the connectors. The "View" toolbar must contain the following active buttons.

Figure 5–31: Active Grid Functions

We recommend that you use your favourite line jumps setting. The default line jump setting is as recommended by the ISO 1219-1 standard. This option is accessible through the menu "View" → "Line's Jump Setting" or from the tool bar button .

Position Editor Symbols

Spool position are drawn using available components in the "Valve Spool Designer" directory. These components can only be linked to each other. It is not possible to link one of these components to other components from the library.

To help designing precisely, when in edition mode, the components are displayed in scale 4. When generated, the symbol is reduced to scale 1.

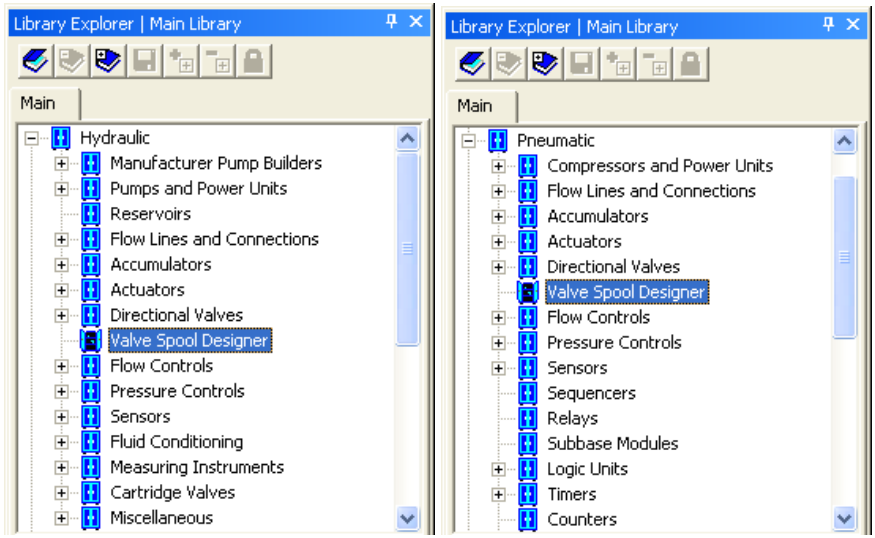


Figure 5–32: Spool Position Symbols Libraries

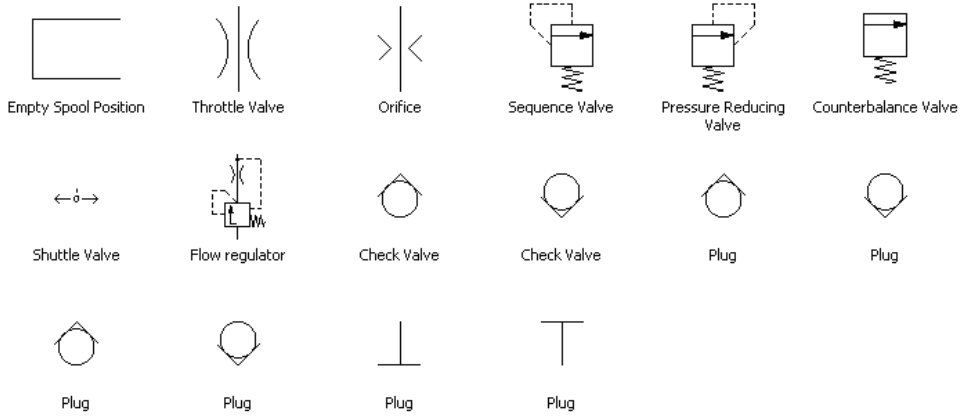


Figure 5–33: Spool Position Symbols

1. Open the library branch “Spool Symbol Position”;

2. Insert the “Empty Spool Position” on the diagram.

Empty Spool Position

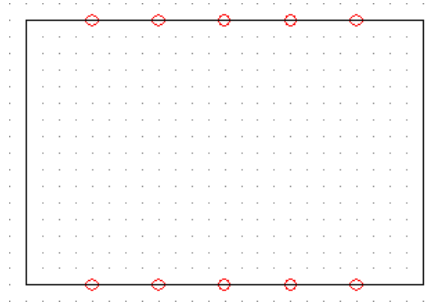


Figure 5–34: Empty Spool Position with 5 Ports

The connectors are positioned closer to each other than existing spool positions. This is done to enable the addition of pressure detection lines "LS" between 2 standard ports.

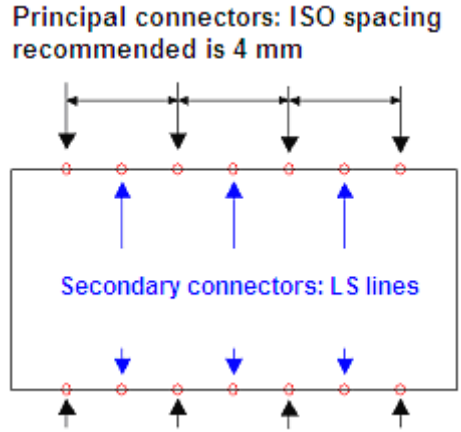


Figure 5–35: Ports Recommended Utilization

For our use, we need four connectors on top following the ISO standard; therefore there will be three secondary connectors. In the Empty Spool Position Builder, you can see examples of numbers (#) of connectors to be selected to achieve recommended ISO configurations. (ISO 1219-1 recommendations for 4 mm between two ports).

By selecting this standard, if you required a valve with four connectors on top and five on the bottom, you must set the spool position with *2 X 5 – 1 = 9* connectors. The additional ports are secondary ports for LS lines or others that will be deleted if nothing is connected to them.

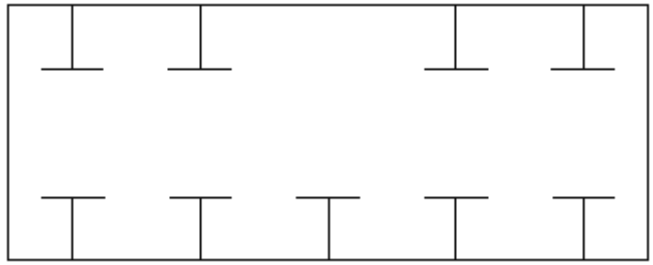


Figure 5–36: ISO Standard: Four Ports on Top and Five at the Bottom

Rules to follow: find out the number of standard ports required on top and bottom; if N is the higher number then select a configuration of (*2 X N – 1*) real ports in the Empty Spool Position dialog box.

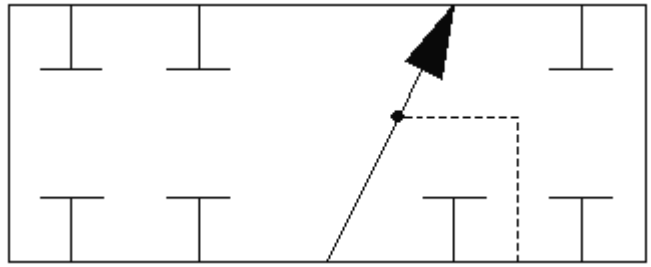


Figure 5–37: Same Configuration but with an LS Line

The Empty Spool Position dialog box opens by double-clicking on the component. By default, this component is configured with five ports on top or on at the bottom.

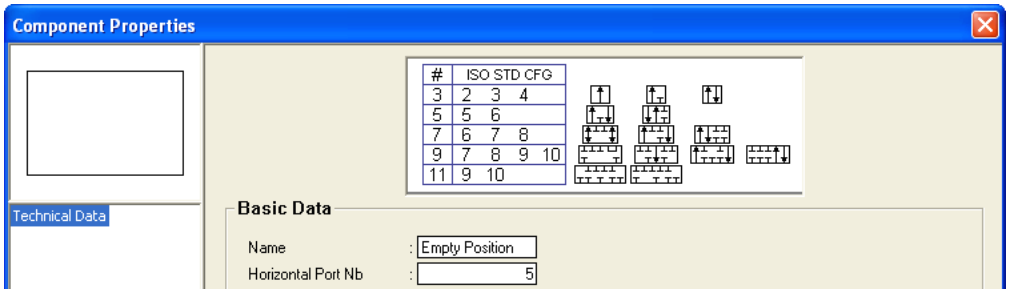


Figure 5–38: Empty Spool Position Dialog Box

For our example, we select seven horizontal ports in order to keep the Automation Studio™ standard of two steps on the grid between connectors.

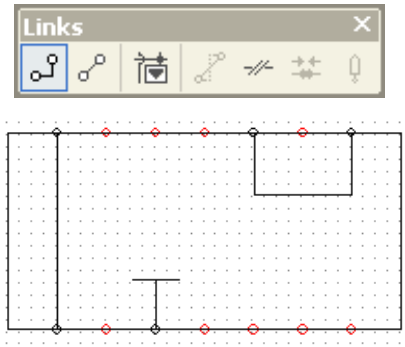
You can name the component. The name will be present in the custom positions windows when selecting a spool position configuration.

Drawing Methodology

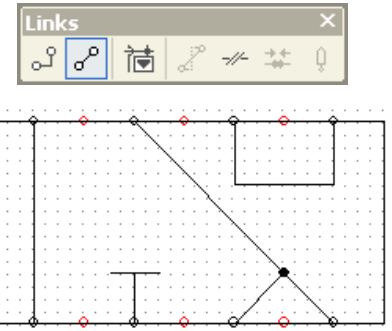
When you create the diagram, you must consider the empty positions as the position circuit limit. Insert lines between the components. We will use the following definition; external lines are lines between a component connector or lines and an empty spool position connector. An internal line is a line between two components connectors.

Draw the position in the following sequence:

1. Insert links and component between connectors;



2. Insert direct links;



All lines and components are configured during the diagram conversion to a spool position. Therefore, do not try to create channel with three lines; keep the diagram as simple as

possible.

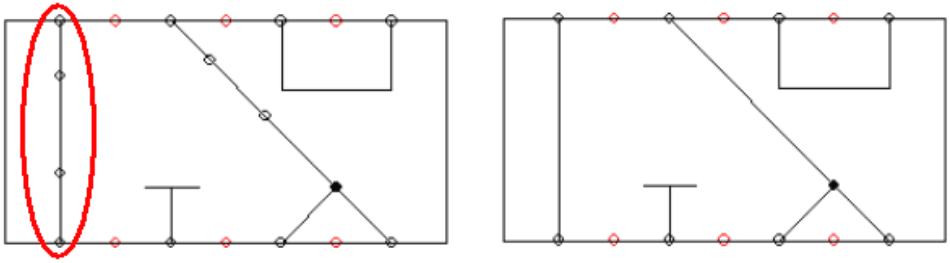


Figure 5–39: Diagram on the Left is Simplified Compared to the Diagram on the Right

3. Insert the components directly on the lines. Use all direct link functions, the automatic insertion benefits, the layout functions and the editor zooming capability to achieve the best possible quality in your drawing;

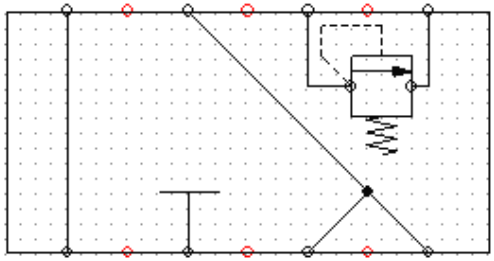


Figure 5–40: Component Insertion on a Line

4. Insert components on diagonal direct links. This can be done in the same way as for normal links (horizontal or vertical) except that prior to insertion, the component must be aligned with the lines; then keep the SHIFT key down, and drag the component on the line. Both component connectors must be on the lines;

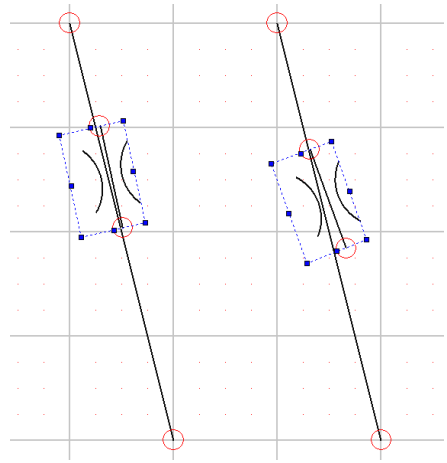


Figure 5–41: Automatic Insertion (good and bad)

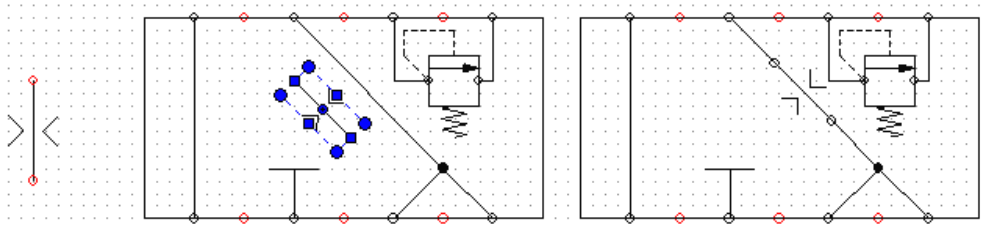


Figure 5–42: Insertion on Direct Links

5. All is left to do is to add arrows on lines to indicate the flow direction. This can be done simultaneously by selecting all lines and by adjusting the "visual parameters" of the lines.

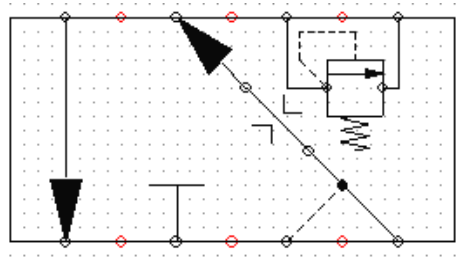


Figure 5–43: Finished Spool Image

If you have external connections to more than two connectors, the lines intersection must be done as follows in order to avoid any major problems when using the directional valve builder. The black dots are added symbols.

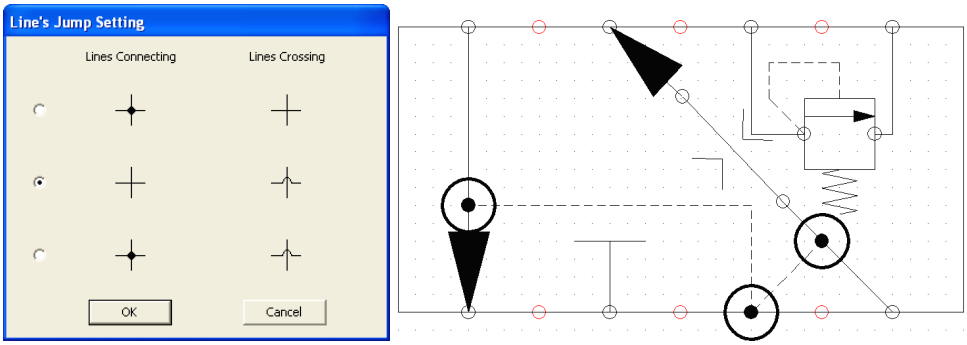


Figure 5–44: External Connections with more than Two Connectors

You can use the standard drawing tools to add details to your spool position design.

☞ Never group components dedicated to spool position creation.

☞ When designing a spool position, the "hide" and layer functions are not available when converting the drawing to a spool position. The grouping function must not be used.

Spool Position Generation

1. Once your design is finished, select all components, including the empty positions;

2. From the shortcut menu (right click on the mouse), select the function "Generate Spool Position".

This function will be offered only if you have exactly one "empty position" in the selection.

Automatic Validation (Error Checking)

Before generating a final product, error messages are displayed if any conceptual errors are detected. Validation is done to ensure that there is no inconsistency.

Possible types of error:

- Elements of the spool are not component from the dedicated spool position component libraries;
- Some connectors are free;

- Some components inside the empty position have not been not selected;
- A close circuit is not directly connected to the empty position.

Errors must be corrected before proceeding.

Position Preview

Once the validation phase is done, a preview dialog box opens containing a representation of the spool position and a component summary.

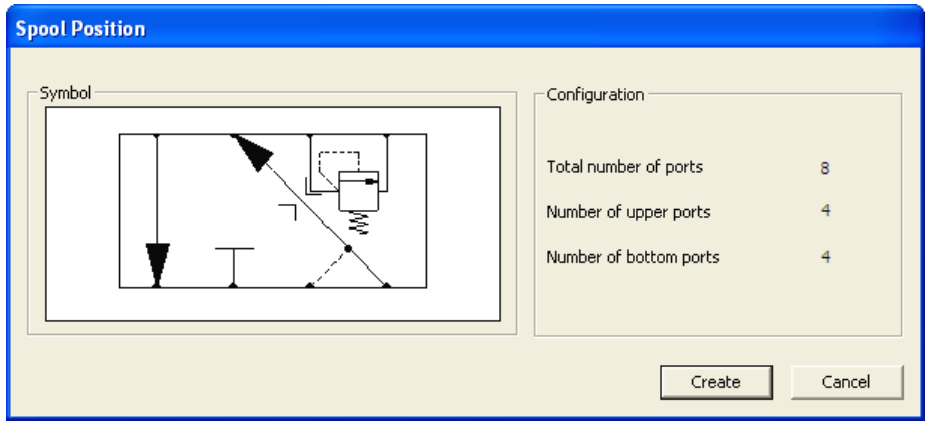


Figure 5–45: Spool Position Preview Summary

There are two possibilities while in the preview:

- Create the spool position: the position diagram changes into a group with a gray background;
- Cancel to go back to the spool position edition.

To create the spool position:

1. Press the “Create” button.

The figure below illustrates the steps to create a spool position.

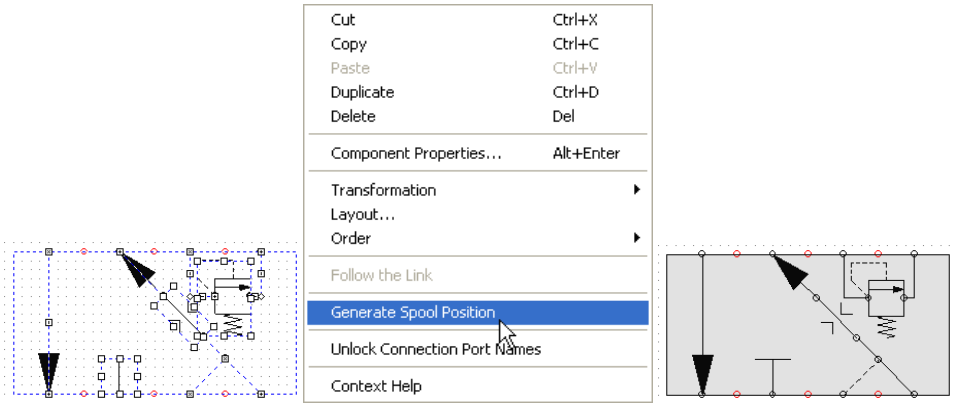


Figure 5–46: Position Creation

Component Configuration

The component properties can be changed during the spool position creation or after, except for channels or lines. In edition mode, the lines are standard lines with properties (diameter, length, roughness and material) which are not proper for position internal channel. Internal lines are converted to channels with adequate properties only once the generation is done. (Nominal flow for a given differential pressure).

The spool position can always be changed through the directional valve builder.

Saving the Spool Position

Now that the spool position is generated, it can be saved in a database at any preferred location.

1. Select “Save Spool Position” in the spool position shortcut menu (right click at the mouse)

Two types of position are available according to your license: On/Off or proportional.

- On/Off Position: once created the position will be part of the On/Off Builder;
- Proportional Position: once created the position will be part of the proportional builder;
- Both: once created the position will be part of both the On/Off and proportional builders.

2. Select the type of spool position that you have generated.

The spool position is now available in the suitable builder (Proportional and/or On/Off).

You can save the personalized spool position in the database provide by Automation Studio™ (ASSpoolPosition.mdb). It is recommended to keep the source file of the spool position, to use the same design for other technologies or to use it as a starting point for other design.

5.8.3 Sharing the Personalized Spool Position

You can share your spool position design with other users as long as they also have the "Spool Position Editor" license. All you have to do is to provide them with the position database. The database is found in the "valve builder" directory directly under the "Automation Studio™" directory, with the extension .mdb.

5.9 Functional Box

A functional box is a configurable component to be used in standard diagrams. This component allows the creation of complex components that can be used in Automation Studio™. The user can configure the component layout, connection ports and behaviour.

The programming required to assign a specific behaviour to a functional box must be done in SFC. HMI components can also be included in the functional box as well as basic display elements, which can be used to design a front representation.

This component gives, among others, endless possibilities to enhance didactic material.

The following figure shows an example of a functional box of a basic automated system.

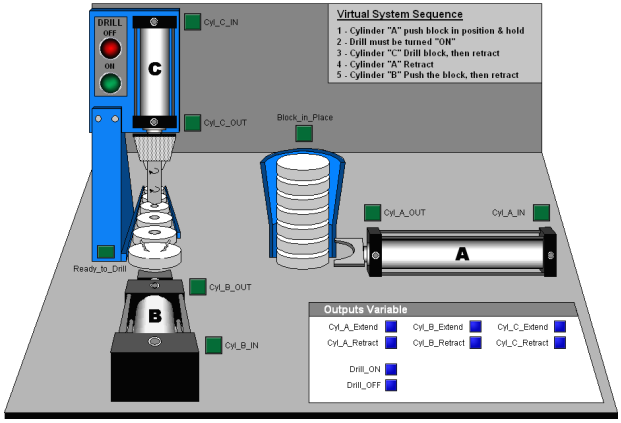


Figure 5-47: Example of "Functional Box" for Training Needs

5.9.1 Creating a Functional Box

5.9.1.1 Editing Functional Box

In this section, a "Hydraulic Drill" component is edited. Once the circuit and SFC edition are completed, it will be shown how to generate and use the final component.

For the demonstration purpose that follows, "Hydraulic Drill" component will be set as a component containing a vertical cylinder to drill a hole by pressing a push-button.

It is recommended to use different layers to facilitate elements edition of the functional box. Using three layers is usually indicated. The recommended layers should contain: the framework, HMI components and finally the graphic elements. We also suggest unlocking only the layer in edition mode, which will greatly facilitate components selection and manipulation.

Here is the procedure to create "Hydraulic Drill" component:

1. Create a standard diagram;

2. Insert a "Functional Box" which can be found in the library under "Hydraulic" or "Pneumatic" → "Miscellaneous" → "Others";

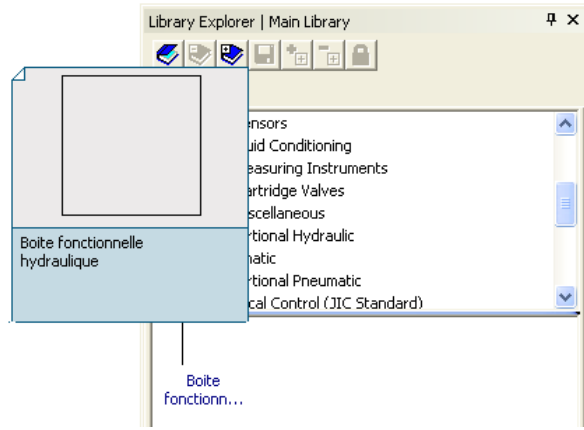


Figure 5-48: Inserting a "Functional Box"

3. Insert the hydraulic components of the circuit. Include proximity switches and enter tagnames **C_IN** et **C_OUT** , as shown;

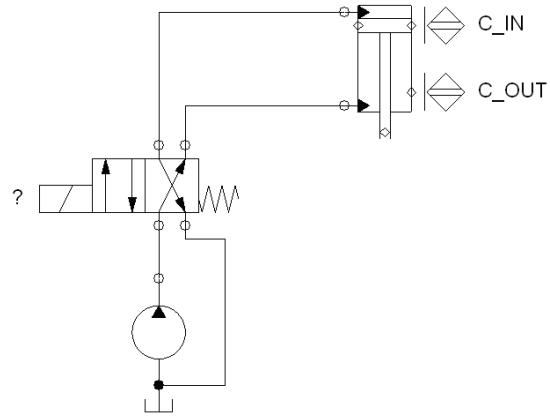


Figure 5-49: Inserting Hydraulic Components

4. Insert a push-button HMI component from the main library "HMI and Control Panels" → "Controls & Commands" → "Buttons" branch and type **START** as the "Tag name" in the pop-up dialog box;

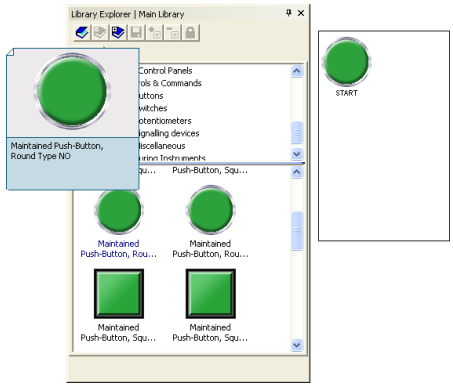


Figure 5-50: Inserting and Configurating Pushbutton

5. Draw, using graphic elements, a double-acting cylinder;

6. Resize the "Functional Box" as needed and add as much HMI elements as you want to improve the visual aspect of your "Functional Box". Here, indicator light are used;

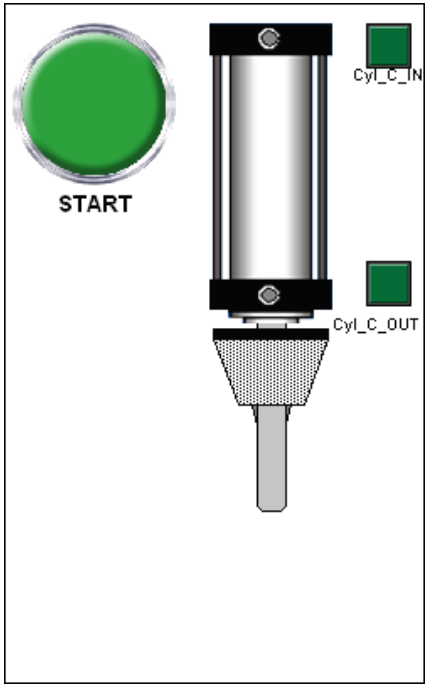


Figure 5-51: Adding Graphic Elements

7. Create a new SFC diagram in the "Project Explorer" root location. Insert an initial step followed by a transition, a normal step and a transition, then link the end to the beginning to allow looping;

8. Create Boolean variables named **DRILLING, Cyl_C_IN** and **Cyl_C_Out**;

☞ When creating variables, it is essential to associate these variables to the corresponding SFC document, as shown in the following figure.

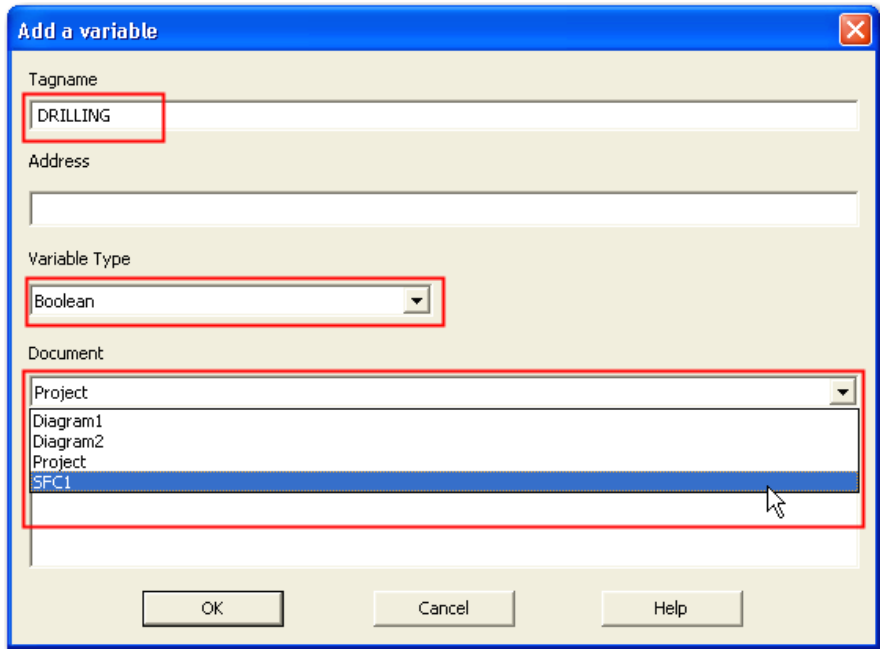


Figure 5-52: Creating DRILLING Variable

9. Configure the SFC logic as followed:

- Step1: **DRILLING :=0**;
- Transition 1: **Cyl_C_IN AND 1-1SS1.START**;
- Step 2: **DRILLING := 1**;
- Transition 2: **Cyl_C_OUT.**

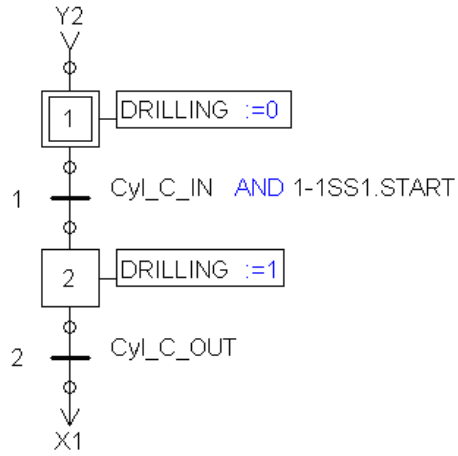


Figure 5-53: Editing Steps and Transitions

10. Assign the directional valve solenoid to the "DRILLNG" variable;

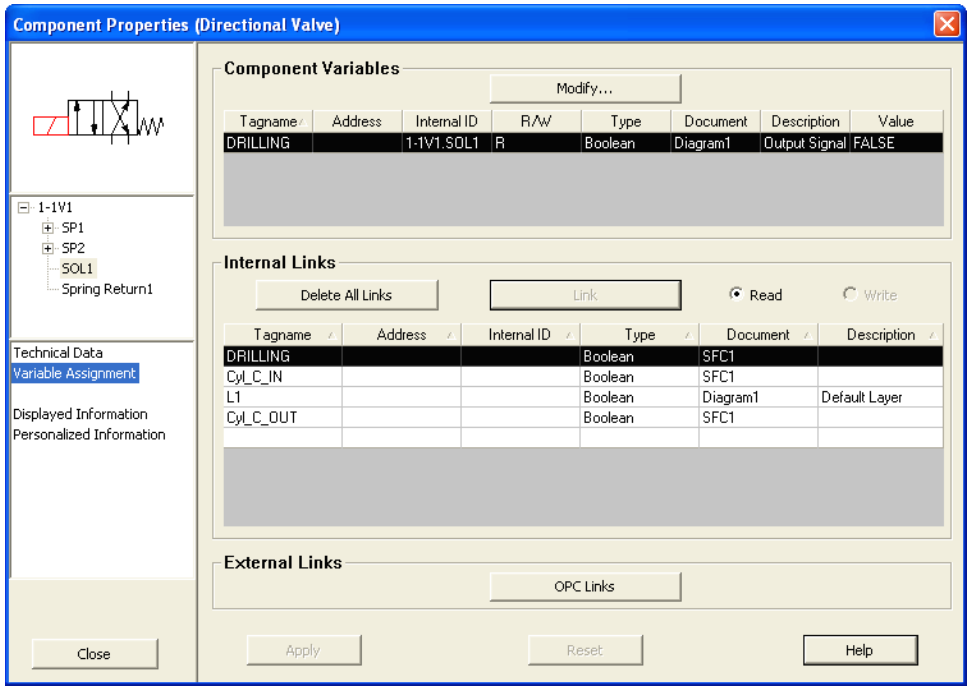


Figure 5-54: Assigning "DRILLING" Variable

11. Configure the cylinder animation to emulate its up and down movements by associating cylinder positions to the vertical translations of the animation.

☞ Refer to *HMI & Control Panels Workshop User's Guide* for details about primitives' animation.

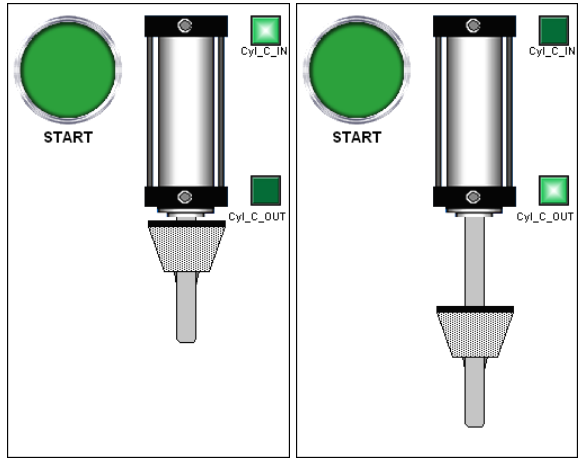


Figure 5-55: Animating the "Hydraulic Drill"

This completes the "Hydraulic Drill" component editing. In the next section, you will learn how to generate the final component.

5.9.1.2 Generating the Functional Box

Functional box generation is quite simple.

1. Select all elements to include in the functional box;

2. Right click on these components and select the "Generate Functional Box" function in the shortcut menu;

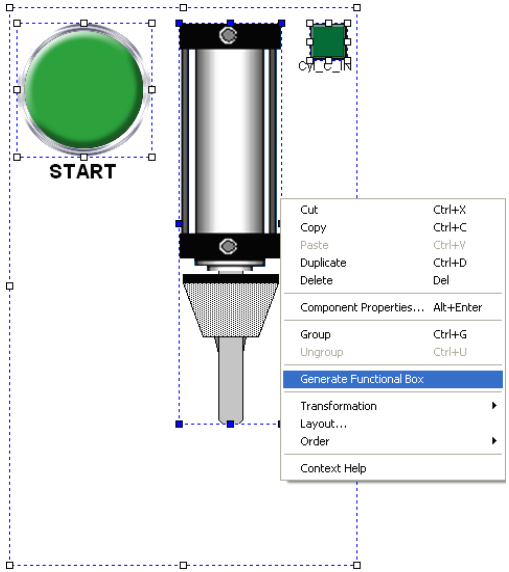


Figure 5-56: "Generate Functional Box" function

Thereafter a dialog box appears allowing the selection of SFC(s) associated with the functional box.

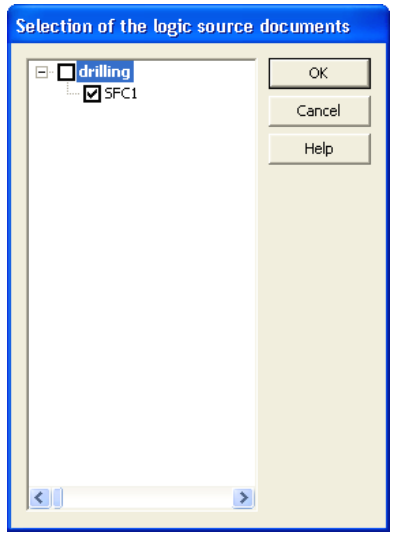


Figure 5-57: Selecting Source Documents

3. Select SFC documents that contain functional box logic;

☞ SFC diagrams should be at the "Project Explorer" root location. The fictional box performance can be compromised if the diagrams are placed in a folder.

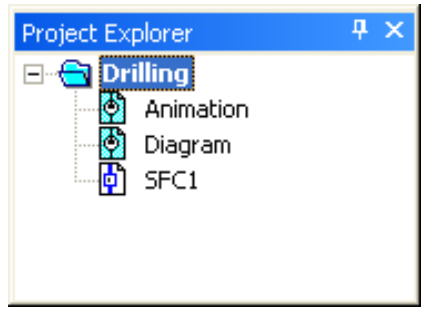


Figure 5-58: SFC Diagram at the "Project Explorer" Root

The dialogue box "Options of Functional box generation" appears.

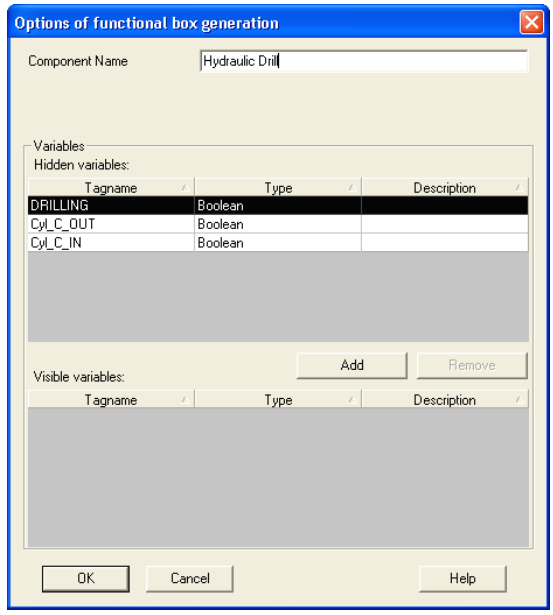


Figure 5-59: "Options of Functional box generation" Dialog Box

Here are the element descriptions of this dialog box.

Zone or Element	Description
Component Name	Appoints the generated component. This name identifies the component in the personalized library.
Hidden variables	Lists global variables associated with the SFC diagram which are internal to the functional box component. These variables will not be visible in the variable manager of a project containing the component functional box.
Visible variables	Lists global variables associated with the SFC diagram that will be visible in the variable manager of a project containing the component functional box.
"Add" button	Moves selected hidden variable to a visible variable.
"Remove" button	Moves selected visible variable to a hidden variable.

4. Fill in the "Options of Functional box generation" fields as followed:

   a) Type **Hydraulic Drill** in the "Component Name" field;

   b) Choose variables you want to make visible in the variable manager. For our example, add all variable in the "Visible Variables" Zone.

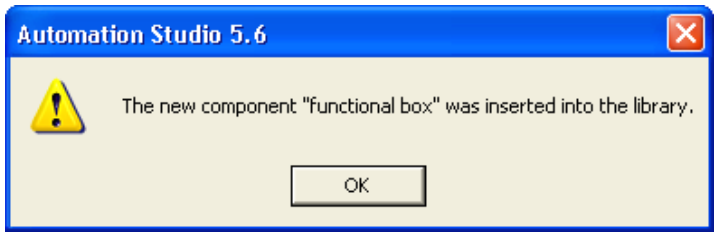


Figure 5-60: Confirmation Message

5. Click on "OK" and a message confirming the fictional box creation is displayed.

5.9.1.3 Using Functional Box

When created, the functional box is automatically inserted in a personalized library called

"Functional Boxes".

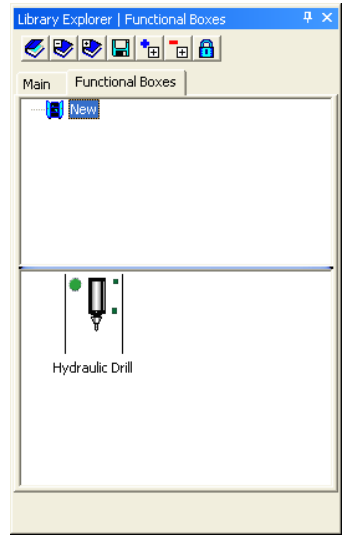


Figure 5-61: "Functional Boxes" Personalised Library

Note that the library really contains the "Hydraulic Drill" component generated in the previous section. This component can be dragged and dropped onto a diagram just like any other component.

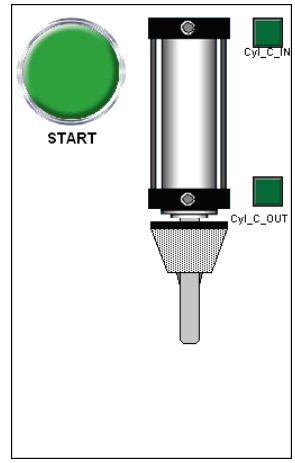


Figure 5-62: "Hydraulic Drill" Component

Save carefully diagrams that allowed you to design your functional box, because you cannot modify a generated component.

A functional box is a standalone component, the logic programming is now part of it and the SFC diagram is no longer required. It is authorized to move the functional box component in a customized library and use it as many times as necessary in standard diagrams. The functional box component is editable and configurable as all the other Automation Studio™ components.

Also, after creating a functional box, you have access to internal variables of the box through the "Component properties" → "Variable Assignment". For variable made visible, it is possible to link read/write all these variables, in order to reassign them to the variables of any project where the functional box is inserted.

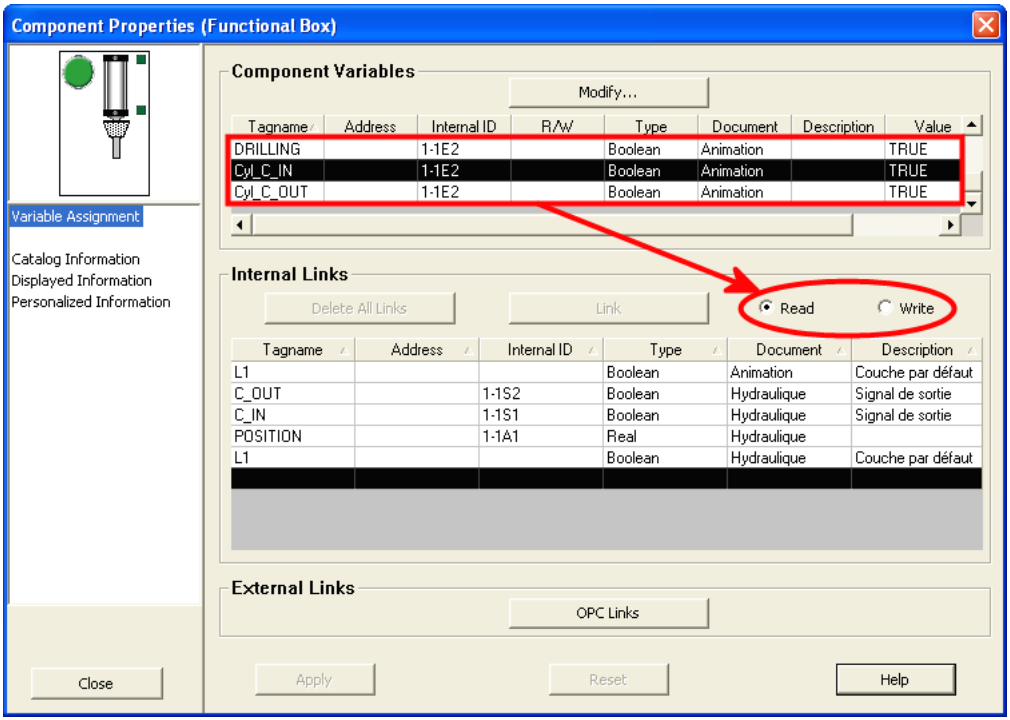


Figure 5-63: Variable assignment of a "Functional Box"

By variable assignment, it will be possible to make your functional box interact properly with any built circuit.

6 Additional Information

6.1 Component Detailed Description

Refer to the component online help, available in the component's shortcut menu or refer to the *Automation Studio™ "User's Guide"* for their simulation properties.

6.2 The Other Simulators

Per analogy, the electrotechnical simulator is designed with the same principles as for the hydraulic simulator. However there is no thermal calculation available. The SFC, Ladder, and Electrical Control workshops' variables can output to or received from a fluid workshop during simulation.

6.3 Defective Components

For most components, a maximum pressure and a maximal temperature can be set. If this component limit is reached, a flashing P for pressure or T for temperature is displayed at one of the connection ports.

Figure 6–1: Flashing Symbol of Reached Pressure or Temperature Limit

This option is activated from the menu "Tools" → "Options…" → "General Simulation" → "Defective Component" and in the "Project Properties" → "Options" → "General Simulation".

6.4 Open Path Detection

In order to optimize circuit design and connection understanding, a diagnostic for open paths was added. By simple clicking on a path, all the fluid lines linked continually to the initial one are underlined, including the initial fluid line. All the other fluid lines and components are displayed in dark gray.

In simulation mode, a new button is added to the “Simulation” toolbar.

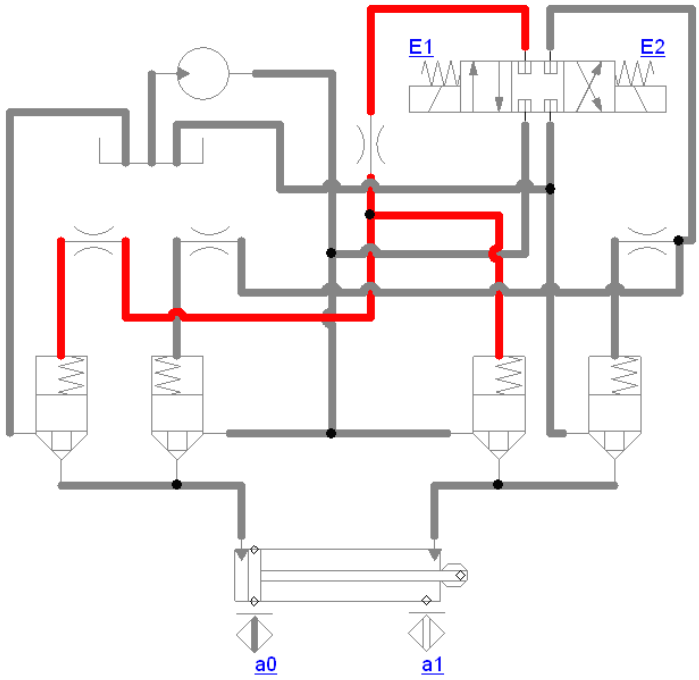


Figure 6-2: “Open Path Detection” Option

This button remains active in simulation mode. When this function is selected, the cursor becomes a coloured conduct and the simulation is paused.

To use “Open Path Detection” diagnostic:

1. Start a desired simulation;

2. Click on the button in the “Simulation” toolbar;

   The cursor takes this shape .

3. Click on a line for which you want to detect an open path.

The cursor takes this shape .

The open paths, allowing the fluid to flow, are highlighted in red. This state is held until a selection is made, the option is disabled or the simulation is restarted.

This open path detection option creates temporary simulation pause when a line is selected. So you can restart a simulation exactly where it was stopped, when using the function.

After using "Open Path Detection", you cannot force directional valve position.

6.5 Error detection in diagrams in simulation mode

The fluid simulator can calculate if a connection port is open. If so, a message appears when the simulation is launched and unconnected components turn fuchsia. The list of affected components can be found in the "Messages" window.

This function can be launched manually in the "Tools" menu → "Verify Connections".

6.6 Symbol in Simulation Mode

Components Animation

In simulation mode, if a component is red (default color), an animation window can be opened from the component's shortcut menu. This command is available for fluid components in simulation mode. It opens a window showing a cross-section view of the animated component according to the component status.

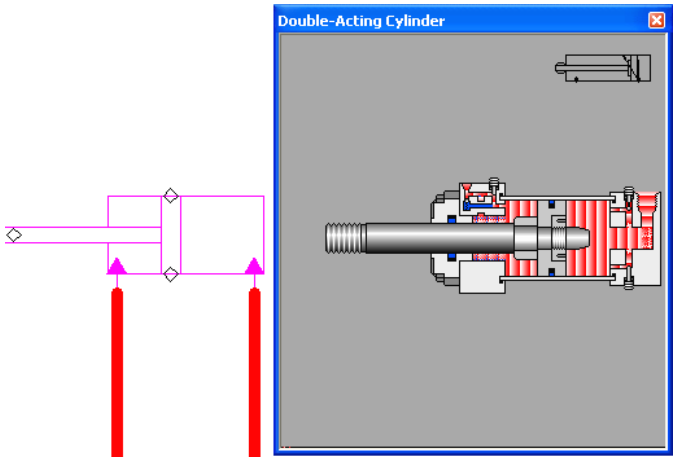


Figure 6-3: Animating Symbol with Cross-Section View

The following list contains the component offering a cross-section animated view:

- Accumulators;
- Check Valves;
- Flow meters;
- Flow Regulators, with check valve or compensated;
- Filters;
- Pressure Regulators;
- Bidirectional motors;
- Unidirectional motors;
- Pumps;
- Pressure Compensated Variable Pumps;
- Pressure Reducers;
- Return Reservoirs;
- Control valves with check valves;
- Sequence valves;
- Sequence valves and check valves;
- Differential Cylinders;
- Double-acting Cylinders with absorber.

All components can be found in the hydraulic or pneumatic section “Cross-Section Symbols”.

Lines Animation

In order to meet training needs, we have introduced an animated view of the lines. However, because this feature is very computer time demanding, depending on the size of the project to simulate, we decided to put it as an option. Therefore, two project options are added to the hydraulic and pneumatic simulation options. They are disabled by default and can be accessed by choosing "File" → "Project Properties" → "Options" and by checking "Animated Flow" box for the desired technology.

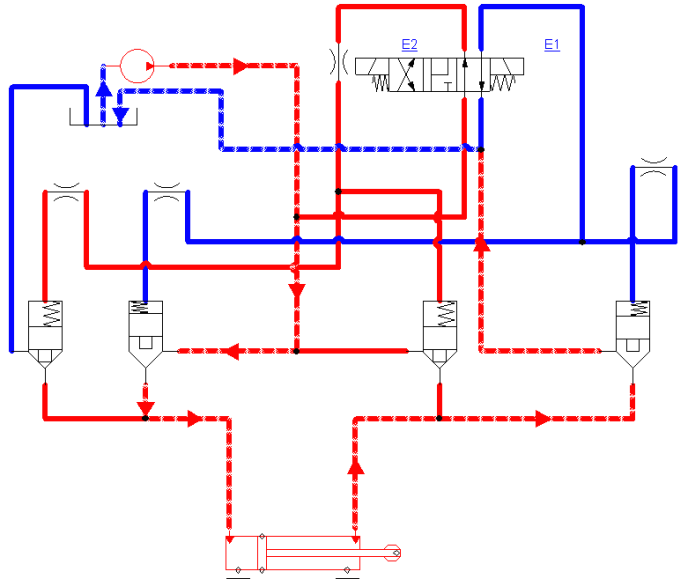


Figure 6-4: Lines Animation

Tuning

When in simulation mode, a click on specific components allows to adjust its operational parameters (cracking pressure of a valve) or to control its position or state (directional valve, isolation valve, etc.).

The cursor turns into a hand when placed over a component which can be set during simulation.

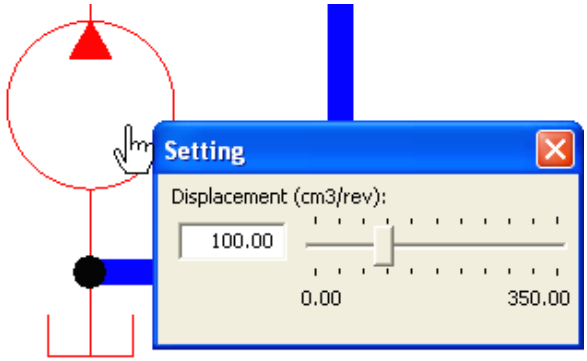


Figure 6-5: Component Dynamic Setting in Simulation Mode

Forcing

Clicking on a valve position will force this valve to take this position and overrule the simulation's calculation for this valve.

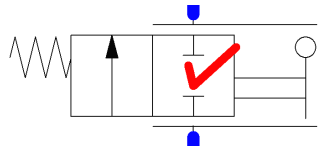


Figure 6–6: Forcing a Valve Position

6.7 Tools

Plotter

For most hydraulic and pneumatic regulating components, you can plot input and/or output pressures, or flow passing through them. Except for directional valves with different flow rate for each valve position. Measuring devices are then used to measure the performance of these valves which output value can be plotted.

Sizing Sheet

Sizing Sheets are static calculation tools. Therefore, some dynamic parameters such as the (input and/or output) pressure or flow cannot be known to the sizing sheets; you must enter them onto the sheets.

Also, even if component models are always improved in every Automation Studio™ version, some parameters are not used for modelization, therefore the simulator does not include them in its calculation (motor volumetric or mechanical efficiency).

☞ Calculation sheets use only numerical dates. They don't use technical data such as curves, or variable assignation (for instance, the driving and resistive force curves for a cylinder are not used by the calculation sheet).

6.8 Demo Files

The following figure shows where Automation Studio™ demo files can be found. They are provided to complement the information found in this manual.

These files can be opened directly from the component online help.

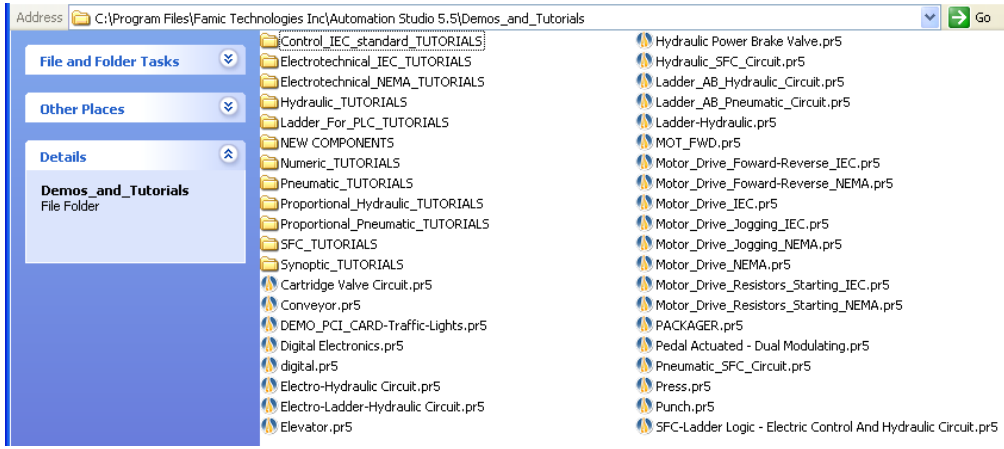


Figure 6–7: Demo Files

6.9 Options List

For a workshop options to be available, at least one drawing of this workshop must be opened with at least one component inserted on the corresponding workshop diagram.

6.9.1 Simulation Options

Hydraulic and Pneumatic

Option	Function
Link Width in Simulation	Specifies the pressure lines width during simulation.
Flow Arrow Width	Specifies the flow arrowhead width during simulation.
Flow Arrow Height	Specifies the flow arrowhead height during simulation.
Minimum Pressure	Specifies the circuit minimum pressure.
Low Pressure Limit	Specifies the pressure at which lines are considered low pressure.
High Pressure Limit	Specifies the pressure at which lines are considered high pressure.
Maximum Pressure	Specifies the circuit maximum pressure during simulation.
High Pressure Line	Specifies the high pressure line color during simulation.
Low Pressure Line	Specifies the low pressure line color during simulation.
Vacuum Pressure Line	Specifies the vacuum pressure line color during simulation.
High Pilot Line	Specifies the high pressure pilot line color during simulation.
Low Pilot Line	Specifies the low pressure pilot line color during simulation.
Vacuum Pilot Line	Specifies the vacuum pilot line color during simulation.
Drain Line Color (Hydraulic only)	Specifies the drain line color during simulation.

Option	Function
High Load Sense Line	Specifies the high load sense line color during simulation.
Low Load Sense Line	Specifies the low load sense line color during simulation.
Vacuum Load Sense Line	Specifies the vacuum load sense line color during simulation.
High Return Line	Specifies the high return line color during simulation.
Low Return Line	Specifies the low return line color during simulation.
Vacuum Return Line	Specifies the vacuum return line color during simulation.
High Custom Line	Specifies the high custom line color during simulation.
Low Custom Line	Specifies the low custom line color during simulation.
Vacuum Custom Line	Specifies the vacuum custom line color during simulation.

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